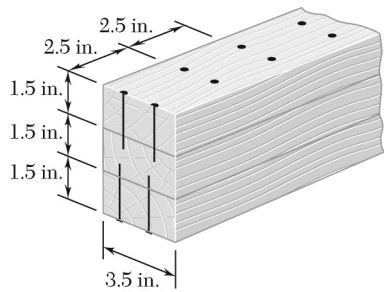


CHAPTER 6



PROBLEM 6.1

Three boards, each of 1.5×3.5 -in. rectangular cross section, are nailed together to form a beam that is subjected to a vertical shear of 250 lb. Knowing that the spacing between each pair of nails is 2.5 in., determine the shearing force in each nail.

SOLUTION

$$I = \frac{1}{12}bh^3 = \frac{1}{12}(3.5)(4.5)^3 = 26.578 \text{ in}^4$$

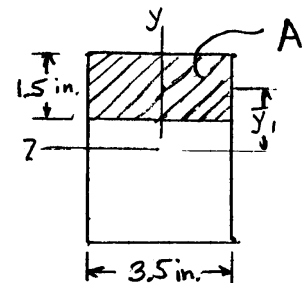
$$A = (3.5)(1.5) = 5.25 \text{ in}^2$$

$$\bar{y}_1 = 1.5 \text{ in.}$$

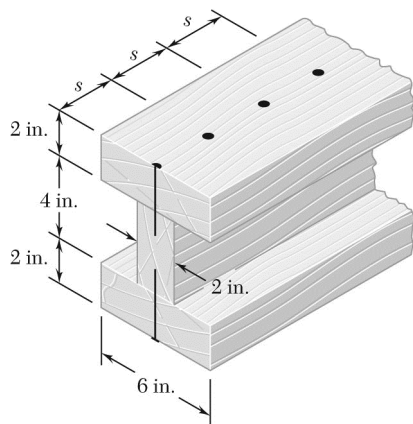
$$Q = Ay_1 = 7.875 \text{ in}^3$$

$$q = \frac{VQ}{I} = \frac{(250)(7.875)}{26.578} = 74.074 \text{ lb/in}$$

$$qs = 2F_{\text{nail}} \quad F_{\text{nail}} = \frac{qs}{2} = \frac{(74.074)(2.5)}{2}$$



$$F_{\text{nail}} = 92.6 \text{ lb} \quad \blacktriangleleft$$



PROBLEM 6.2

Three boards, each 2 in. thick, are nailed together to form a beam that is subjected to a vertical shear. Knowing that the allowable shearing force in each nail is 150 lb, determine the allowable shear if the spacing s between the nails is 3 in.

SOLUTION

$$I_1 = \frac{1}{12}bh^3 + Ad^2$$

$$= \frac{1}{12}(6)(2)^3 + (6)(2)(3)^2 = 112 \text{ in}^4$$

$$I_2 = \frac{1}{12}bh^3 = \frac{1}{12}(2)(4)^3 = 10.667 \text{ in}^4$$

$$I_3 = I_1 = 112 \text{ in}^4$$

$$I = I_1 + I_2 + I_3 = 234.667 \text{ in}^4$$

$$Q = A_1\bar{y}_1 = (6)(2)(3) = 36 \text{ in}^3$$

$$qs = F_{\text{nail}} \quad (1)$$

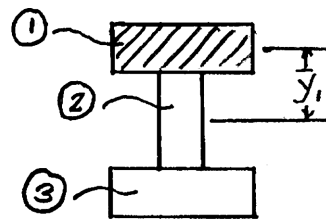
$$q = \frac{VQ}{I} \quad (2)$$

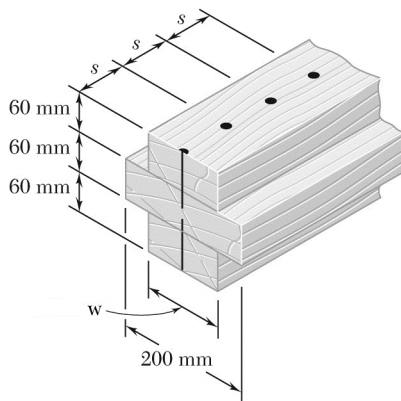
Dividing Eq. (2) by Eq. (1),

$$\frac{1}{s} = \frac{VQ}{F_{\text{nail}}I}$$

$$V = \frac{F_{\text{nail}}I}{Qs} = \frac{(150)(234.667)}{(36)(3)}$$

$$V = 326 \text{ lb} \quad \blacktriangleleft$$



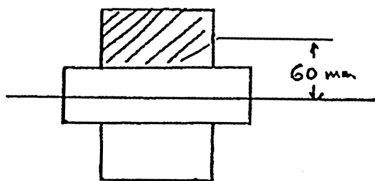


PROBLEM 6.3

Three boards are nailed together to form a beam shown, which is subjected to a vertical shear. Knowing that the spacing between the nails is $s = 75 \text{ mm}$ and that the allowable shearing force in each nail is 400 N , determine the allowable shear when $w = 120 \text{ mm}$.

SOLUTION

Part	$A(\text{mm}^2)$	$d(\text{mm})$	$Ad^2(10^6 \text{mm}^4)$	$\bar{I}(10^6 \text{mm}^4)$
Top plank	7200	60	25.92	2.16
Middle Plank	12,000	0	0	3.60
Bottom Plank	7200	60	25.92	2.16
Σ			51.84	7.92



$$I = \Sigma Ad^2 + \Sigma \bar{I} = 59.76 \times 10^6 \text{ mm}^4 = 59.76 \times 10^{-6} \text{ m}^4$$

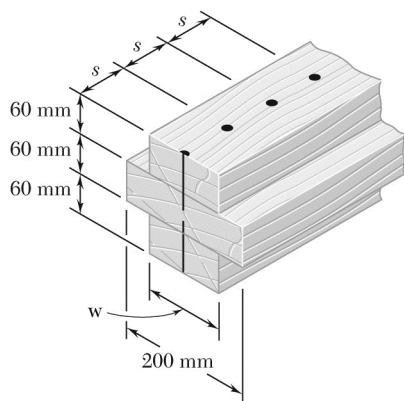
$$Q = (7200)(60) = 432 \times 10^3 \text{ mm}^3 = 432 \times 10^{-6} \text{ m}^3$$

$$q = \frac{VQ}{I} \quad F_{\text{nail}} = qs$$

$$q = \frac{F_{\text{nail}}}{s} \quad V = \frac{Iq}{Q} = \frac{IF_{\text{nail}}}{Qs}$$

$$V = \frac{(59.76 \times 10^{-6})(400)}{(432 \times 10^{-6})(75 \times 10^{-3})}$$

$$V = 738 \text{ N} \quad \blacktriangleleft$$



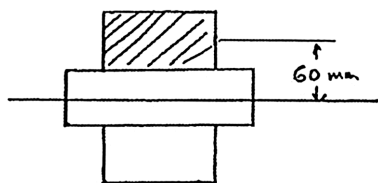
PROBLEM 6.4

Solve Prob. 6.3, assuming that the width of the top and bottom boards is changed to $w = 100 \text{ mm}$.

PROBLEM 6.3 Three boards are nailed together to form a beam shown, which is subjected to a vertical shear. Knowing that the spacing between the nails is $s = 75 \text{ mm}$ and that the allowable shearing force in each nail is 400 N , determine the allowable shear when $w = 120 \text{ mm}$.

SOLUTION

Part	$A(\text{mm}^2)$	$d \text{ (mm)}$	$Ad^2(10^6 \text{ mm}^4)$	$\bar{I}(10^6 \text{ mm}^4)$
Top plank	6000	60	21.6	1.80
Middle Plank	12,000	0	0	3.60
Bottom Plank	6000	60	21.6	1.80
Σ			43.2	7.20



$$I = \sum Ad^2 + \sum \bar{I} = 50.4 \times 10^6 \text{ mm}^4 = 50.4 \times 10^{-6} \text{ m}^4$$

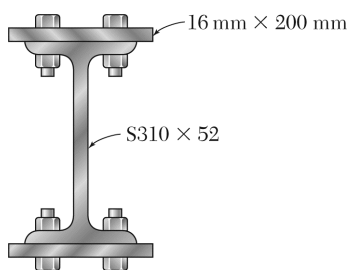
$$Q = (6000)(60) = 360 \times 10^3 \text{ mm}^3 = 360 \times 10^{-6} \text{ m}^3$$

$$q = \frac{VQ}{I} \quad F_{\text{nail}} = qs$$

$$q = \frac{F_{\text{nail}}}{s} \quad V = \frac{Iq}{Q} = \frac{IF_{\text{nail}}}{Qs}$$

$$V = \frac{(50.4 \times 10^{-6})(400)}{(360 \times 10^{-6})(75 \times 10^{-3})}$$

$$V = 747 \text{ N} \quad \blacktriangleleft$$



PROBLEM 6.5

The American Standard rolled-steel beam shown has been reinforced by attaching to it two 16×200 -mm plates, using 18-mm-diameter bolts spaced longitudinally every 120 mm. Knowing that the average allowable shearing stress in the bolts is 90 MPa, determine the largest permissible vertical shearing force.

SOLUTION

Calculate moment of inertia:

Part	$A(\text{mm}^2)$	$d(\text{mm})$	$Ad^2(10^6 \text{ mm}^4)$	$\bar{I}(10^6 \text{ mm}^4)$
Top plate	3200	*160.5	82.43	0.07
S310 x 52	6650	0		95.3
Bot. plate	3200	*160.5	82.43	0.07
Σ			164.86	95.44

$$*d = \frac{305}{2} + \frac{16}{2} = 160.5 \text{ mm}$$

$$I = \Sigma Ad^2 + \Sigma \bar{I} = 260.3 \times 10^6 \text{ mm}^4 = 260.3 \times 10^{-6} \text{ m}^4$$

$$Q = A_{\text{plate}} d_{\text{plate}} = (3200)(160.5) = 513.6 \times 10^3 \text{ mm}^3 = 513.6 \times 10^{-6} \text{ m}^3$$

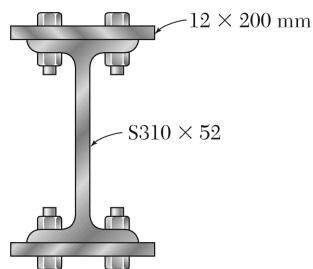
$$A_{\text{bolt}} = \frac{\pi}{4} d_{\text{bolt}}^2 = \frac{\pi}{4} (18 \times 10^{-3})^2 = 254.47 \times 10^{-6} \text{ m}^2$$

$$F_{\text{bolt}} = \tau_{\text{all}} A_{\text{bolt}} = (90 \times 10^6)(254.47 \times 10^{-6}) = 22.90 \times 10^3 \text{ N}$$

$$qs = 2F_{\text{bolt}} \quad q = \frac{2F_{\text{bolt}}}{s} = \frac{(2)(22.90 \times 10^3)}{120 \times 10^{-3}} = 381.7 \times 10^3 \text{ N/m}$$

$$q = \frac{VQ}{I} \quad V = \frac{Iq}{Q} = \frac{(260.3 \times 10^{-6})(381.7 \times 10^3)}{513.6 \times 10^{-6}} = 193.5 \times 10^3 \text{ N}$$

$$V = 193.5 \text{ kN} \quad \blacktriangleleft$$



PROBLEM 6.6

Solve Prob. 6.5, assuming that the reinforcing plates are only 12 mm thick.

PROBLEM 6.5 The American Standard rolled-steel beam shown has been reinforced by attaching to it two 16×200 -mm plates, using 18-mm-diameter bolts spaced longitudinally every 120 mm. Knowing that the average allowable shearing stress in the bolts is 90 MPa, determine the largest permissible vertical shearing force.

SOLUTION

Calculate moment of inertia:

Part	$A(\text{mm}^2)$	$d(\text{mm})$	$Ad^2(10^6 \text{mm}^4)$	$\bar{I}(10^6 \text{mm}^4)$
Top plate	2400	*158.5	60.29	0.03
S310 x 52	6650	0	0	95.3
Bot. plate	2400	*158.5	60.29	0.03
Σ			120.58	95.36

$$*d = \frac{305}{2} + \frac{12}{2} = 158.5 \text{ mm}$$

$$I = \Sigma Ad^2 + \Sigma \bar{I} = 215.94 \times 10^6 \text{mm}^4 = 215.94 \times 10^{-6} \text{m}^4$$

$$Q = A_{\text{plate}} d_{\text{plate}} = (200)(12)(158.5) = 380.4 \times 10^3 \text{mm}^3 = 380.4 \times 10^{-6} \text{m}^3$$

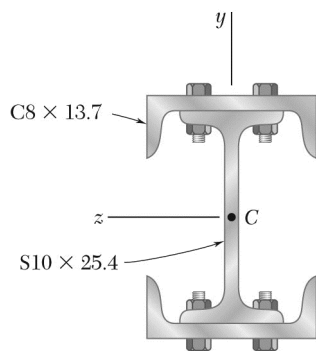
$$A_{\text{bolt}} = \frac{\pi}{4} d_{\text{bolt}}^2 = \frac{\pi}{4} (18 \times 10^{-3})^2 = 254.47 \times 10^{-6} \text{m}^2$$

$$F_{\text{bolt}} = \tau_{\text{all}} A_{\text{bolt}} = (90 \times 10^6)(254.47 \times 10^{-6}) = 22.902 \times 10^3 \text{N}$$

$$qs = 2F_{\text{bolt}} \quad q = \frac{2F_{\text{bolt}}}{s} = \frac{(2)(22.903 \times 10^3)}{120 \times 10^{-3}} = 381.7 \times 10^3 \text{N/m}$$

$$q = \frac{VQ}{I} \quad V = \frac{Iq}{Q} = \frac{(215.94 \times 10^{-6})(381.7 \times 10^3)}{380.4 \times 10^{-6}} = 217 \times 10^3 \text{N}$$

$$V = 217 \text{ kN} \quad \blacktriangleleft$$



PROBLEM 6.7

A column is fabricated by connecting the rolled-steel members shown by bolts of $\frac{3}{4}$ -in. diameter spaced longitudinally every 5 in. Determine the average shearing stress in the bolts caused by a shearing force of 30 kips parallel to the y axis.

SOLUTION

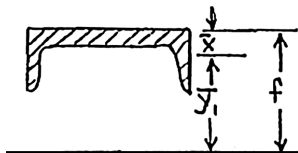
Geometry:

$$f = \left(\frac{d}{2}\right)_s + (t_w)_C$$

$$= \frac{10.0}{2} + 0.303 = 5.303 \text{ in.}$$

$$\bar{x} = 0.534 \text{ in.}$$

$$\bar{y}_1 = f - \bar{x} = 5.303 - 0.534 = 4.769 \text{ in.}$$



Determine moment of inertia.

Part	$A(\text{in}^2)$	$d(\text{in.})$	$Ad^2(\text{in}^4)$	$\bar{I}(\text{in}^4)$
C8 x 13.7	4.04	4.769	91.88	1.52
S10 x 25.4	7.4	0	0	123
C8 x 13.7	4.04	4.769	91.88	1.52
Σ			183.76	126.04

$$I = \Sigma Ad^2 + \Sigma \bar{I} = 183.76 + 126.04 = 309.8 \text{ in}^4$$

$$Q = A \bar{y}_1 = (4.04)(4.769) = 19.267 \text{ in}^3$$

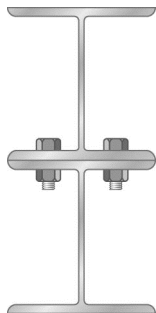
$$q = \frac{VQ}{I} = \frac{(30)(19.267)}{309.8} = 1.8658 \text{ kip/in}$$

$$F_{\text{bolt}} = \frac{1}{2} qs = \left(\frac{1}{2}\right)(1.8658)(5) = 4.664 \text{ kips}$$

$$A_{\text{bolt}} = \frac{\pi}{4} d_{\text{bolt}}^2 = \frac{\pi}{4} \left(\frac{3}{4}\right)^2 = 0.4418 \text{ in}^2$$

$$\tau_{\text{bolt}} = \frac{F_{\text{bolt}}}{A_{\text{bolt}}} = \frac{4.664}{0.4418} = 10.56 \text{ ksi}$$

$$\tau_{\text{bolt}} = 10.56 \text{ ksi} \quad \blacktriangleleft$$



PROBLEM 6.8

The composite beam shown is fabricated by connecting two W6 × 20 rolled-steel members, using bolts of $\frac{5}{8}$ -in. diameter spaced longitudinally every 6 in. Knowing that the average allowable shearing stress in the bolts is 10.5 ksi, determine the largest allowable vertical shear in the beam.

SOLUTION

$$W6 \times 20: \quad A = 5.87 \text{ in}^2, \quad d = 6.20 \text{ in.}, \quad I_x = 41.4 \text{ in}^4$$

$$\bar{y} = \frac{1}{2}d = 3.1 \text{ in.}$$

Composite:

$$I = 2[41.4 + (5.87)(3.1)^2]$$

$$= 195.621 \text{ in}^4$$

$$Q = A\bar{y} = (5.87)(3.1) = 18.197 \text{ in}^3$$

Bolts:

$$d = \frac{5}{8} \text{ in.}, \quad \tau_{\text{all}} = 10.5 \text{ ksi}, \quad s = 6 \text{ in.}$$

$$A_{\text{bolt}} = \frac{\pi}{4} \left(\frac{5}{8} \right)^2 = 0.30680 \text{ in}^2$$

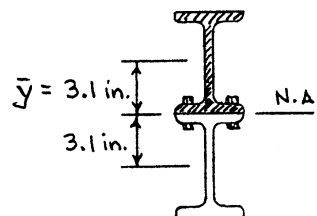
$$F_{\text{bolt}} = \tau_{\text{all}} A_{\text{bolt}} = (10.5)(0.30680) = 3.2214 \text{ kips}$$

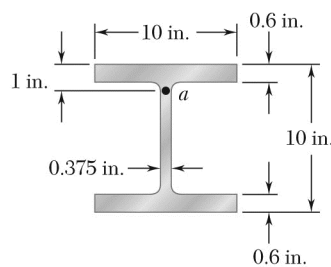
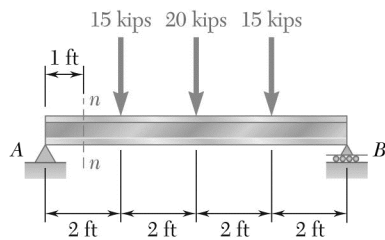
$$q = \frac{2F_{\text{bolt}}}{s} = \frac{(2)(3.2214)}{6} = 1.07380 \text{ kips/in}$$

Shear:

$$q = \frac{VQ}{I} \quad V = \frac{Iq}{Q} = \frac{(195.621)(1.0780)}{18.197}$$

$$V = 11.54 \text{ kips} \quad \blacktriangleleft$$





PROBLEM 6.9

For the beam and loading shown, consider section $n-n$ and determine (a) the largest shearing stress in that section, (b) the shearing stress at point a .

SOLUTION

By symmetry, $R_A = R_B$.

$$+\uparrow \sum F_y = 0:$$

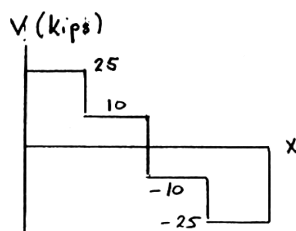
$$R_A + R_B - 15 - 20 - 15 = 0$$

$$R_A = R_B = 25 \text{ kips}$$

From shear diagram,

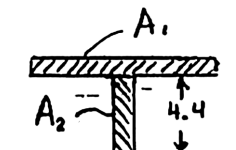
$$V = 30 \text{ kips at } n-n.$$

Determine moment of inertia.



Part	$A(\text{in}^2)$	$d(\text{in.})$	$Ad^2(\text{in}^4)$	$\bar{I}(\text{in}^4)$
Top Flng	6	4.7	132.54	0.18
Web	3.30	0	0	21.30
Bot. Flng	6	4.7	132.54	0.18
Σ			265.08	21.66

(a)



$$I = \sum Ad^2 + \sum \bar{I} = 286.74 \text{ in}^4$$

Part	$A(\text{in}^2)$	$\bar{y}(\text{in.})$	$A\bar{y}(\text{in}^3)$
①	6	4.7	28.2
②	1.65	2.2	3.63
Σ			31.83

$$Q = \sum A\bar{y}$$

$$= 31.83 \text{ in}^3$$

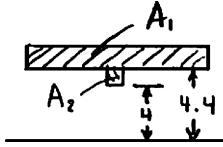
$$t = 0.375 \text{ in.}$$

$$\tau_{\max} = \frac{VQ_{\max}}{It} = \frac{(25)(31.83)}{(286.74)(0.375)}$$

$$\tau_{\max} = 7.40 \text{ ksi} \quad \blacktriangleleft$$

PROBLEM 6.9 (Continued)

(b)



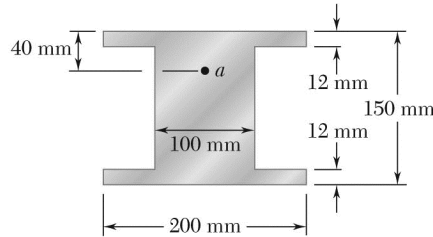
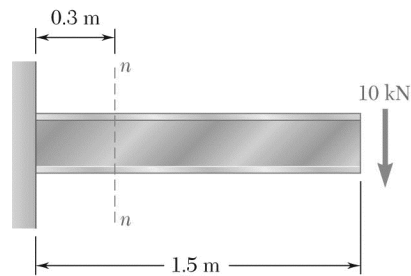
Part	$A(\text{in}^2)$	$\bar{y}(\text{in.})$	$A\bar{y}(\text{in}^3)$
①	6	4.7	28.2
②	0.15	4.2	0.63
Σ			28.83

$$Q = \Sigma A\bar{y} = 28.83 \text{ in}^3$$

$$t = 0.375 \text{ in.}$$

$$\tau = \frac{VQ}{It} = \frac{(23)(28.83)}{(286.74)(0.375)}$$

$$\tau = 6.70 \text{ ksi} \quad \blacktriangleleft$$



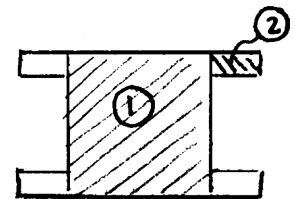
PROBLEM 6.10

For the beam and loading shown, consider section $n-n$ and determine (a) the largest shearing stress in that section, (b) the shearing stress at point a .

SOLUTION

At section $n-n$, $V = 10$ kN.

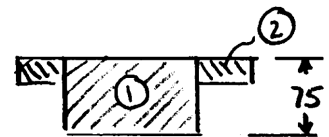
$$\begin{aligned}
 I &= I_1 + 4I_2 \\
 &= \frac{1}{12}b_1h_1^3 + 4\left(\frac{1}{12}b_2h_2^3 + A_2d_2^2\right) \\
 &= \frac{1}{12}(100)(150)^3 + 4\left[\left(\frac{1}{12}\right)(50)(12)^3 + (50)(12)(69)^2\right] \\
 &= 28.125 \times 10^6 + 4[0.0072 \times 10^6 + 2.8566 \times 10^6] \\
 &= 39.58 \times 10^6 \text{ mm}^4 = 39.58 \times 10^{-6} \text{ m}^4
 \end{aligned}$$



$$\begin{aligned}
 (a) \quad Q &= A_1\bar{y}_1 + 2A_2\bar{y}_2 \\
 &= (100)(75)(37.5) + (2)(50)(12)(69) \\
 &= 364.05 \times 10^3 \text{ mm}^3 = 364.05 \times 10^{-6} \text{ m}^3 \\
 t &= 100 \text{ mm} = 0.100 \text{ m}
 \end{aligned}$$

$$\tau_{\max} = \frac{VQ}{It} = \frac{(10 \times 10^3)(364.05 \times 10^{-6})}{(39.58 \times 10^{-6})(0.100)} = 920 \times 10^3 \text{ Pa}$$

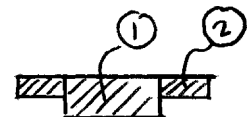
$$\tau_{\max} = 920 \text{ kPa} \quad \blacktriangleleft$$

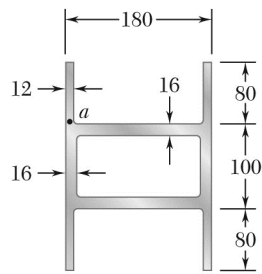


$$\begin{aligned}
 (b) \quad Q &= A_1\bar{y}_1 + 2A_2\bar{y}_2 \\
 &= (100)(40)(55) + (2)(50)(12)(69) \\
 &= 302.8 \times 10^3 \text{ mm}^3 = 302.8 \times 10^{-6} \text{ m}^3 \\
 t &= 100 \text{ mm} = 0.100 \text{ m}
 \end{aligned}$$

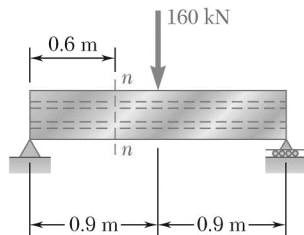
$$\tau_a = \frac{VQ}{It} = \frac{(10 \times 10^3)(302.8 \times 10^{-6})}{(39.58 \times 10^{-6})(0.100)} = 765 \times 10^3 \text{ Pa}$$

$$\tau_a = 765 \text{ kPa} \quad \blacktriangleleft$$





Dimensions in mm



PROBLEM 6.11

For the beam and loading shown, consider section $n-n$ and determine (a) the largest shearing stress in that section, (b) the shearing stress at point a .

SOLUTION

At section $n-n$,

$$V = 80 \text{ kN}$$

Consider cross section as composed of rectangles of types ①, ②, and ③.

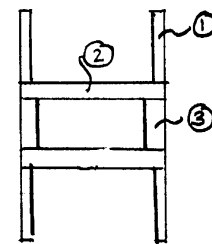
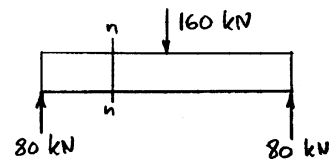
$$I_1 = \frac{1}{12}(12)(80)^3 + (12)(80)(90)^2 = 8.288 \times 10^6 \text{ mm}^4$$

$$I_2 = \frac{1}{12}(180)(16)^3 + (180)(16)(42)^2 = 5.14176 \times 10^6 \text{ mm}^4$$

$$I_3 = \frac{1}{12}(16)(68)^3 = 419.24 \times 10^3 \text{ mm}^4$$

$$I = 4I_1 + 2I_2 + 2I_3 = 44.274 \times 10^6 \text{ mm}^4$$

$$= 44.274 \times 10^{-6} \text{ m}^4$$



(a) Calculate Q at neutral axis.

$$Q_1 = (12)(80)(90) = 86.4 \times 10^3 \text{ mm}^4$$

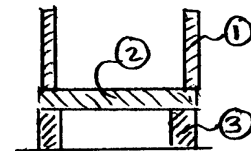
$$Q_2 = (180)(16)(42) = 120.96 \times 10^3 \text{ mm}^4$$

$$Q_3 = (16)(34)(17) = 9.248 \times 10^3 \text{ mm}^4$$

$$Q = 2Q_1 + Q_2 + 2Q_3 = 312.256 \times 10^3 \text{ mm}^3 = 312.256 \times 10^{-6} \text{ m}^3$$

$$\tau = \frac{VQ}{It} = \frac{(80 \times 10^3)(312.256 \times 10^{-6})}{(44.274 \times 10^{-6})(2 \times 16 \times 10^{-3})} = 17.63 \times 10^6 \text{ Pa}$$

$$\tau = 17.63 \text{ MPa} \quad \blacktriangleleft$$

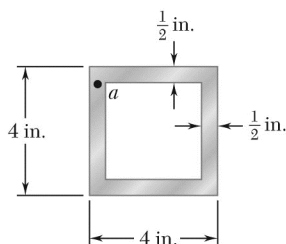
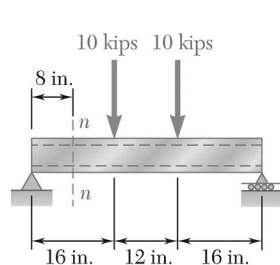


(b) At point a ,

$$Q = Q_1 = 86.4 \times 10^3 \text{ mm}^4 = 86.4 \times 10^{-6} \text{ m}^4$$

$$\tau = \frac{VQ}{It} = \frac{(80 \times 10^3)(86.4 \times 10^{-6})}{(44.274 \times 10^{-6})(12 \times 10^{-3})} = 13.01 \times 10^6 \text{ Pa}$$

$$\tau = 13.01 \text{ MPa} \quad \blacktriangleleft$$



PROBLEM 6.12

For the beam and loading shown, consider section $n-n$ and determine (a) the largest shearing stress in that section, (b) the shearing stress at point a .

SOLUTION

By symmetry, $R_A = R_B$.

$$+\uparrow \Sigma F_y = 0: R_A + R_B - 10 - 10 = 0$$

$$R_A = R_B = 10 \text{ kips}$$

From the shear diagram,

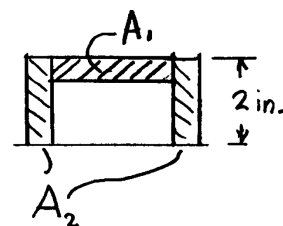
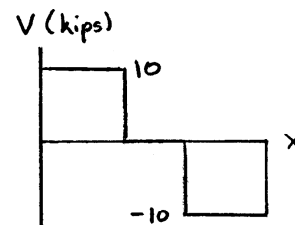
$$V = 10 \text{ kips at } n-n.$$

$$\begin{aligned} I &= \frac{1}{12} b_2 h_2^3 - \frac{1}{12} b_1 h_1^3 \\ &= \frac{1}{12} (4)(4)^3 - \frac{1}{12} (3)(3)^3 = 14.583 \text{ in}^4 \end{aligned}$$

$$(a) \quad Q = A_1 \bar{y}_1 + A_2 \bar{y}_2 = (3) \left(\frac{1}{2} \right) (1.75) + (2) \left(\frac{1}{2} \right) (2)(1) = 4.625 \text{ in}^3$$

$$t = \frac{1}{2} + \frac{1}{2} = 1 \text{ in.}$$

$$\tau_{\max} = \frac{VQ}{It} = \frac{(10)(4.625)}{(14.583)(1)}$$

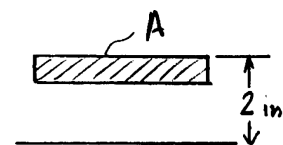


$$\tau_{\max} = 3.17 \text{ ksi} \quad \blacktriangleleft$$

$$(b) \quad Q = A \bar{y} = (4) \left(\frac{1}{2} \right) (1.75) = 3.5 \text{ in}^3$$

$$t = \frac{1}{2} + \frac{1}{2} = 1 \text{ in.}$$

$$\tau = \frac{VQ}{It} = \frac{(10)(3.5)}{(14.583)(1)}$$



$$\tau_a = 2.40 \text{ ksi} \quad \blacktriangleleft$$



For a beam having the cross section shown, determine the largest allowable vertical shear if the shearing stress is not to exceed 60 MPa.

$$I = \frac{1}{12}(50 \text{ mm})(120 \text{ mm})^3 - 2\left[\frac{1}{12}(30 \text{ mm})^4 + (30 \text{ mm} \times 30 \text{ mm})(35 \text{ mm})^2\right]$$
$$I = 7.2 \times 10^6 \text{ mm}^4 - 2[1.170 \times 10^6 \text{ mm}^4] = 4.86 \times 10^6 \text{ mm}^4$$
$$= 4.86 \times 10^{-6} \text{ m}^4$$

Diagram of a T-section beam with dimensions:

- Top flange width: 50
- Top flange thickness: 10
- Web height: 30
- Web thickness: 20
- Total height: 55
- Bottom flange width: 35
- Bottom flange thickness: 10
- Section label: N.A.

$$\begin{aligned} t &= 2(10\text{ mm}) = 0.02\text{ m} \\ Q &= (10\text{ mm} \times 50\text{ mm})(55\text{ mm}) + 2[(10\text{ mm} \times 30\text{ mm})(35\text{ mm})] \\ &= 48.5 \times 10^3\text{ mm}^3 = 48.5 \times 10^{-6}\text{ m}^3 \end{aligned}$$

PROBLEM 6.13 (Continued)

For $\tau_{\text{all}} = 60 \text{ MPa}$,

$$\tau_m = \tau_{\text{all}} = \frac{VQ}{It}$$

$$60 \times 10^6 \text{ Pa} = \frac{V(48.5 \times 10^{-6} \text{ m}^3)}{(4.86 \times 10^{-6} \text{ m}^4)(0.02 \text{ m})} \quad V = 120.3 \text{ kN} \quad \blacktriangleleft$$

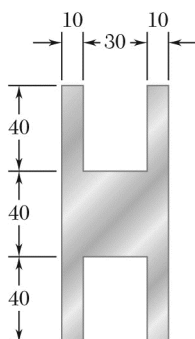
Check τ at neutral axis:

$$t = 50 \text{ mm} = 0.05 \text{ m}$$

$$Q = (50 \times 60)(30) - (30 \times 30)(35) = 58.5 \times 10^3 \text{ mm}^3$$

$$\tau = \frac{VQ}{It} = \frac{(120.3 \text{ kN})(58.5 \times 10^{-6} \text{ m}^3)}{(4.86 \times 10^{-6} \text{ m}^4)(0.05 \text{ m})} = 29.0 \text{ MPa} < 60 \text{ MPa} \quad \underline{\text{OK}}$$

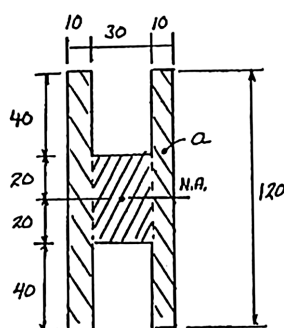
Dimensions in mm



PROBLEM 6.14

For a beam having the cross section shown, determine the largest allowable vertical shear if the shearing stress is not to exceed 60 MPa.

SOLUTION

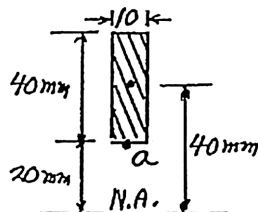


Calculate moment of inertia.

$$\begin{aligned}
 I &= 2 \left[\frac{1}{12} (10 \text{ mm})(120 \text{ mm})^3 \right] + \frac{1}{12} (30 \text{ mm})(40 \text{ mm})^3 \\
 &= 2[1.440 \times 10^6 \text{ mm}^4] + 0.160 \times 10^6 \text{ mm}^4 \\
 &= 3.04 \times 10^6 \text{ mm}^4
 \end{aligned}$$

$$I = 3.04 \times 10^{-6} \text{ m}^4$$

Assume that τ_m occurs at point a .



$$t = 10 \text{ mm} = 0.01 \text{ m}$$

$$\begin{aligned}
 Q &= (10 \text{ mm} \times 40 \text{ mm})(40 \text{ mm}) \\
 &= 16 \times 10^3 \text{ mm}^3 \quad Q = 16 \times 10^{-6} \text{ m}^3
 \end{aligned}$$

For $\tau_{\text{all}} = 60 \text{ MPa}$,

$$\tau_m = \tau_{\text{all}} = \frac{VQ}{It}$$

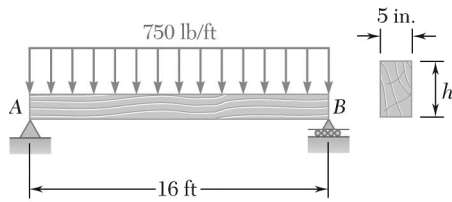
$$60 \times 10^6 \text{ Pa} = \frac{V(16 \times 10^{-6} \text{ m}^3)}{(3.04 \times 10^{-6} \text{ m}^4)(0.01 \text{ m})} \quad V = 114.0 \text{ kN} \quad \blacktriangleleft$$

Check τ at neutral axis:

$$t = 50 \text{ mm} = 0.05 \text{ m}$$

$$Q = 2[(10 \times 60)(30)] + (30 \times 20)(10) = 42 \times 10^3 \text{ mm}^3 = 42 \times 10^{-6} \text{ m}^3$$

$$\tau_{NA} = \frac{VQ}{It} = \frac{(114.0 \text{ kN})(42 \times 10^{-6} \text{ m}^3)}{(3.04 \times 10^{-6} \text{ m}^4)(0.05 \text{ m})} = 31.5 \text{ MPa} < 60 \text{ MPa} \quad \underline{\text{OK}}$$



PROBLEM 6.15

For the beam and loading shown, determine the minimum required depth h , knowing that for the grade of timber used, $\sigma_{\text{all}} = 1750$ psi and $\tau_{\text{all}} = 130$ psi.

SOLUTION

Total load: $(750 \text{ lb/ft})(16 \text{ ft}) = 12 \times 10^3 \text{ lb}$

Reaction at A: $R_A = 6 \times 10^3 \text{ lb} \uparrow$

$$V_{\text{max}} = 6 \times 10^3 \text{ lb}$$

$$M_{\text{max}} = \frac{1}{2}(8 \text{ ft})(6 \times 10^3) = 24 \times 10^3 \text{ lb} \cdot \text{ft}$$

$$= 288 \times 10^3 \text{ lb} \cdot \text{in}$$

Bending: $S = \frac{1}{6}bh^2$ for rectangular section.

$$S = \frac{M_{\text{max}}}{\sigma_{\text{all}}} = \frac{288 \times 10^3}{1750} = 164.57 \text{ in}^3$$

$$h = \sqrt{\frac{6S}{b}} = \sqrt{\frac{(6)(164.57)}{5}} = 14.05 \text{ in.}$$

Shear: $I = \frac{1}{12}bh^3$ for rectangular section.

$$A = \frac{1}{2}bh$$

$$\bar{y} = \frac{1}{4}h$$

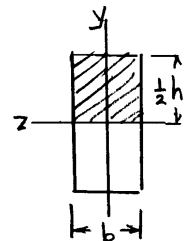
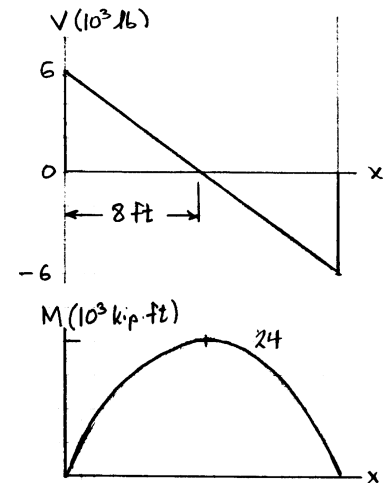
$$Q = A\bar{y} = (b)\left(\frac{1}{2}h\right)\left(\frac{1}{4}h\right) = \frac{1}{8}bh^2$$

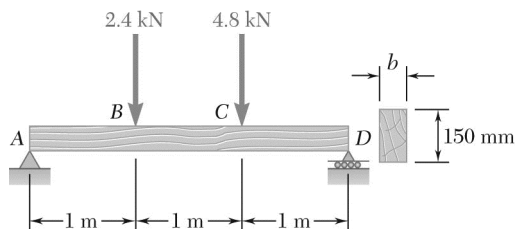
$$\tau_{\text{max}} = \frac{VQ}{Ib} = \frac{3V_{\text{max}}}{2bh}$$

$$h = \frac{3V_{\text{max}}}{2b\tau_{\text{max}}} = \frac{(3)(6 \times 10^3)}{(2)(5)(130)} = 13.85 \text{ in.}$$

The larger value of h is the minimum required depth.

$h = 14.05 \text{ in.} \blacktriangleleft$

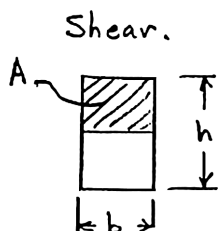
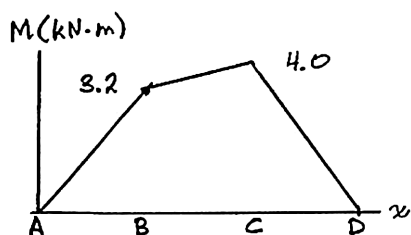
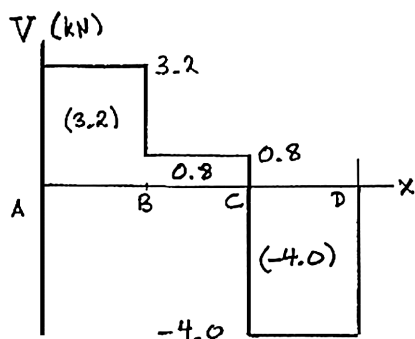




PROBLEM 6.16

For the beam and loading shown, determine the minimum required width b , knowing that for the grade of timber used, $\sigma_{\text{all}} = 12 \text{ MPa}$ and $\tau_{\text{all}} = 825 \text{ kPa}$.

SOLUTION



$$+\circlearrowleft M_D = 0: -3R_A + (2)(2.4) + (1)(4.8) = 0$$

$$R_A = 3.2 \text{ kN}$$

Draw shear and bending moment diagrams.

$$|V|_{\text{max}} = 4.0 \text{ kN} \quad |M|_{\text{max}} = 4.0 \text{ kN} \cdot \text{m}$$

$$\text{Bending: } S = \frac{|M|_{\text{max}}}{\sigma_{\text{all}}} = \frac{4.0 \times 10^3}{12 \times 10^6}$$

$$= 333.33 \times 10^{-6} \text{ m}^3 = 333.33 \times 10^3 \text{ mm}^3$$

For a rectangular cross section,

$$S = \frac{I}{c} = \frac{\frac{1}{12}bh^3}{\frac{1}{2}h} = \frac{1}{6}bh^2$$

$$b = \frac{6S}{h^2} = \frac{(6)(333.33 \times 10^3)}{150^2} = 88.9 \text{ mm}$$

$$A = \frac{1}{2}bh, \quad \bar{y} = \frac{1}{4}h$$

$$Q = A\bar{y} = \frac{1}{8}bh^2, \quad I = \frac{1}{12}bh^3$$

$$\tau = \frac{VQ}{It} = \frac{3V}{2bh}$$

$$bh = \frac{3}{2} \frac{V}{\tau} = \frac{3}{2} \frac{4.0 \times 10^3}{825 \times 10^3}$$

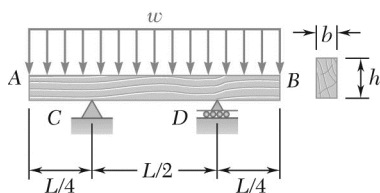
$$= 7.2727 \times 10^{-3} \text{ m}^2 = 7.2727 \times 10^3 \text{ mm}^2$$

$$b = \frac{bh}{h} = \frac{7.2727 \times 10^3}{150} = 48.5 \text{ mm}$$

The required value for b is the larger one.

$$b = 88.9 \text{ mm} \quad \blacktriangleleft$$

PROBLEM 6.17



A timber beam AB of length L and rectangular cross section carries a uniformly distributed load w and is supported as shown. (a) Show that the ratio τ_m/σ_m of the maximum values of the shearing and normal stresses in the beam is equal to $2h/L$, where h and L are, respectively, the depth and the length of the beam. (b) Determine the depth h and the width b of the beam, knowing that $L = 5$ m, $w = 8$ kN/m, $\tau_m = 1.08$ MPa, and $\sigma_m = 12$ MPa.

SOLUTION

$$R_A = R_B = \frac{wL}{2}$$

From shear diagram, $|V|_m = \frac{wL}{4}$ (1)

For rectangular section, $A = bh$ (2)

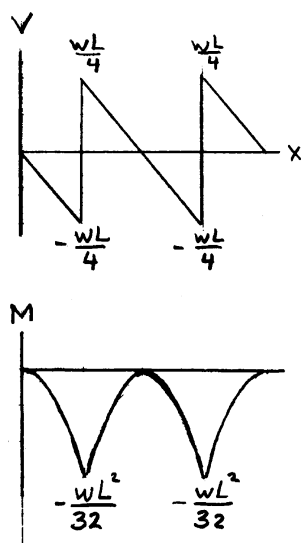
$$\tau_m = \frac{3}{2} \frac{V_m}{A} = \frac{3wL}{8bh} \quad (3)$$

From bending moment diagram, $|M|_m = \frac{wL^2}{32}$ (4)

For a rectangular cross section,

$$S = \frac{1}{6}bh^2 \quad (5)$$

$$\sigma_m = \frac{|M|_m}{S} = \frac{3wL^2}{16bh^2} \quad (6)$$



(a) Dividing Eq. (3) by Eq. (6), $\frac{\tau_m}{\sigma_m} = \frac{2h}{L} \blacktriangleleft$

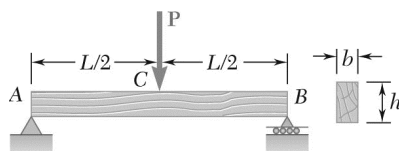
(b) Solving for h :

$$h = \frac{L\tau_m}{2\sigma_m} = \frac{(5)(1.08 \times 10^6)}{(2)(12 \times 10^6)} = 225 \times 10^{-3} \text{ m} \quad h = 225 \text{ mm} \blacktriangleleft$$

Solving Eq. (3) for b :

$$b = \frac{3wL}{8h\tau_m} = \frac{(3)(8 \times 10^3)(5)}{(8)(225 \times 10^{-3})(1.08 \times 10^6)} = 61.7 \times 10^{-3} \text{ m} \quad b = 61.7 \text{ mm} \blacktriangleleft$$

PROBLEM 6.18



A timber beam AB of length L and rectangular cross section carries a single concentrated load P at its midpoint C . (a) Show that the ratio τ_m/σ_m of the maximum values of the shearing and normal stresses in the beam is equal to $2h/L$, where h and L are, respectively, the depth and the length of the beam. (b) Determine the depth h and the width b of the beam, knowing that $L = 2$ m, $P = 40$ kN, $\tau_m = 960$ kPa, and $\sigma_m = 12$ MPa.

SOLUTION

Reactions:

$$R_A = R_B = P/2 \uparrow$$

$$(1) \quad V_{\max} = R_A = \frac{P}{2}$$

$$(2) \quad A = bh \text{ for rectangular section.}$$

$$(3) \quad \tau_m = \frac{3}{2} \frac{V_{\max}}{A} = \frac{3P}{4bh} \text{ for rectangular section.}$$

$$(4) \quad M_{\max} = \frac{PL}{4}$$

$$(5) \quad S = \frac{1}{6}bh^2 \text{ for rectangular section.}$$

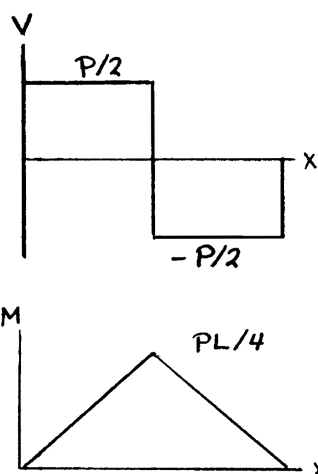
$$(6) \quad \sigma_m = \frac{M_{\max}}{S} = \frac{3PL}{2bh^2}$$

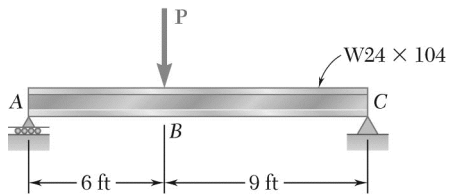
$$(a) \quad \frac{\tau_m}{\sigma_m} = \frac{h}{2L} \quad \blacktriangleleft$$

$$(b) \quad \text{Solving for } h, \quad h = \frac{2L\tau_m}{\sigma_m} = \frac{(2)(2)(960 \times 10^3)}{12 \times 10^6} = 320 \times 10^{-3} \text{ m} \quad h = 320 \text{ mm} \quad \blacktriangleleft$$

Solving Eq. (3) for b ,

$$b = \frac{3P}{4h\tau_m} = \frac{(3)(40 \times 10^3)}{(4)(320 \times 10^{-3})(960 \times 10^3)} = 97.7 \times 10^{-3} \text{ m} \quad b = 97.7 \text{ mm} \quad \blacktriangleleft$$

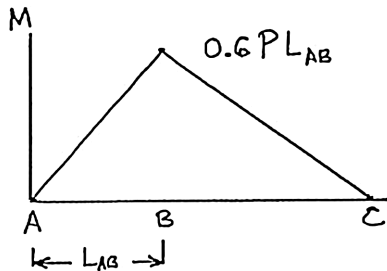
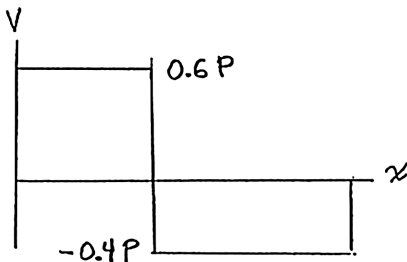




PROBLEM 6.19

For the wide-flange beam with the loading shown, determine the largest load **P** that can be applied, knowing that the maximum normal stress is 24 ksi and the largest shearing stress using the approximation $\tau_m = V/A_{\text{web}}$ is 14.5 ksi.

SOLUTION



$$+\circlearrowleft \sum M_C = 0: -15R_A + qP = 0$$

$$R_A = 0.6P$$

Draw shear and bending moment diagrams.

$$|V|_{\text{max}} = 0.6P \quad |M|_{\text{max}} = 0.6PL_{AB}$$

$$L_{AB} = 6 \text{ ft} = 72 \text{ in.}$$

Bending.

For W 24 x 104, $S = 258 \text{ in}^3$

$$S = \frac{|M|_{\text{max}}}{\sigma_{\text{all}}} = \frac{0.6PL_{AB}}{\sigma_{\text{all}}}$$

$$P = \frac{\sigma_{\text{all}} S}{0.6L_{AB}} = \frac{(24)(258)}{(0.6)(72)} = 143.3 \text{ kips}$$

Shear.

$$A_{\text{web}} = dt_w$$

$$= (24.1)(0.500)$$

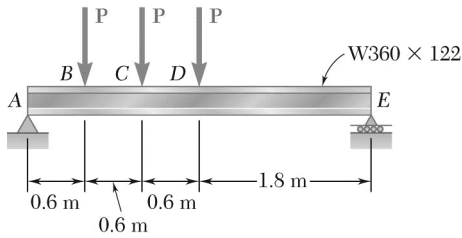
$$= 12.05 \text{ in}^2$$

$$\tau = \frac{|V|_{\text{max}}}{A_{\text{web}}} = \frac{0.6P}{A_{\text{web}}}$$

$$P = \frac{\tau A_{\text{web}}}{0.6} = \frac{(14.5)(12.05)}{0.6} = 291 \text{ kips}$$

The smaller value of **P** is the allowable value.

P = 143.3 kips ◀



PROBLEM 6.20

For the wide-flange beam with the loading shown, determine the largest load P that can be applied, knowing that the maximum normal stress is 160 MPa and the largest shearing stress using the approximation $\tau_m = V/A_{\text{web}}$ is 100 MPa.

SOLUTION

$$+\circlearrowleft \sum M_E = 0: -3.6R_A + 3.0P + 2.4P + 1.8P = 0$$

$$R_A = 2P \uparrow$$

Draw shear and bending moment diagrams.

$$M_B = 2PL_{AB}, \quad M_C = M_D = 3PL_{AB}$$

$$|V|_{\max} = 2P \quad |M|_{\max} = 3PL_{AB}$$

Bending. For W 360 $\times$ 122, $S = 2020 \times 10^3 \text{ mm}^3$
 $= 2020 \times 10^{-6} \text{ m}^3$

$$\frac{|M|_{\max}}{\sigma_{\text{all}}} = \frac{3PL_{AB}}{\sigma_{\text{all}}} = S$$

$$P = \frac{\sigma_{\text{all}} S}{3L_{AB}} = \frac{(160 \times 10^6)(2020 \times 10^3)}{(3)(0.6)} = 179.6 \times 10^3 \text{ N}$$

Shear. $A_{\text{web}} = dt_w = (363)(13.0)$
 $= 4.719 \times 10^3 \text{ mm}^2 = 4.719 \times 10^{-3} \text{ m}^2$

$$\tau = \frac{|V|_{\max}}{A_{\text{web}}} = \frac{2P}{A_{\text{web}}}$$

$$P = \frac{\tau A_{\text{web}}}{2} = \frac{(100 \times 10^6)(4.719 \times 10^{-3})}{2} = 236 \times 10^3 \text{ N}$$

The smaller value of P is the allowable one.

$$P = 179.6 \text{ kN} \quad \blacktriangleleft$$

PROBLEM 6.21

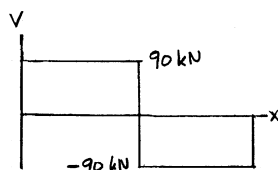
For the beam and loading shown, consider section $n-n$ and determine the shearing stress at (a) point a , (b) point b .

SOLUTION

Draw the shear diagram.

$$|V|_{\max} = 90 \text{ kN}$$

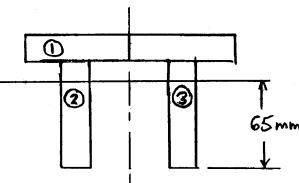
Part	$A(\text{mm}^2)$	$\bar{y}(\text{mm})$	$A\bar{y}(10^3 \text{ mm}^3)$	$d(\text{mm})$	$Ad^2(10^6 \text{ mm}^4)$	$\bar{I}(10^6 \text{ mm}^4)$
①	3200	90	288	25	2.000	0.1067
②	1600	40	64	-25	1.000	0.8533
③	1600	40	64	-25	1.000	0.8533
Σ	6400		416		4.000	1.8133



$$\bar{Y} = \frac{\Sigma A\bar{y}}{\Sigma A} = \frac{416 \times 10^3}{6400} = 65 \text{ mm}$$

$$I = \Sigma Ad^2 + \Sigma \bar{I} = (4.000 + 1.8133) \times 10^6 \text{ mm}^4$$

$$= 5.8133 \times 10^6 \text{ mm}^4 = 5.8133 \times 10^{-6} \text{ m}^4$$



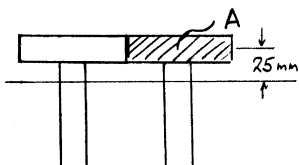
(a) $A = (80)(20) = 1600 \text{ mm}^2$

$$\bar{y} = 25 \text{ mm}$$

$$Q_a = A\bar{y} = 40 \times 10^3 \text{ mm}^3 = 40 \times 10^{-6} \text{ m}^3$$

$$\tau_a = \frac{VQ_a}{It} = \frac{(90 \times 10^3)(40 \times 10^{-6})}{(5.8133 \times 10^{-6})(20 \times 10^{-3})} = 31.0 \times 10^6 \text{ Pa}$$

$$\tau_a = 31.0 \text{ MPa} \quad \blacktriangleleft$$

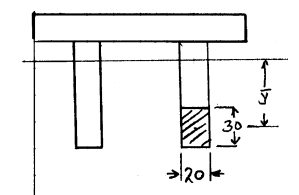


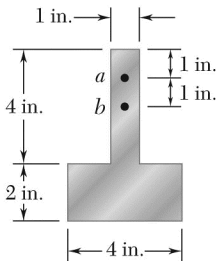
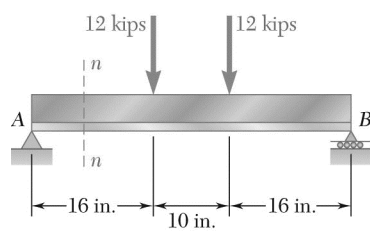
(b) $A = (30)(20) = 600 \text{ mm}^2$ $\bar{y} = 65 - 15 = 50 \text{ mm}$

$$Q_b = A\bar{y} = 30 \times 10^3 \text{ mm}^3 = 30 \times 10^{-6} \text{ m}^3$$

$$\tau_b = \frac{VQ_b}{It} = \frac{(90 \times 10^3)(30 \times 10^{-6})}{(5.8133 \times 10^{-6})(20 \times 10^{-3})} = 23.2 \times 10^6 \text{ Pa}$$

$$\tau_b = 23.2 \text{ MPa} \quad \blacktriangleleft$$





PROBLEM 6.22

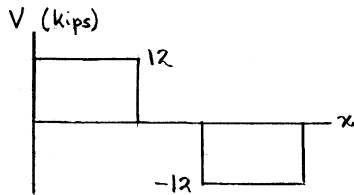
For the beam and loading shown, consider section $n-n$ and determine the shearing stress at (a) point a , (b) point b .

SOLUTION

$$R_A = R_B = 12 \text{ kips}$$

Draw shear diagram.

$$V = 12 \text{ kips}$$

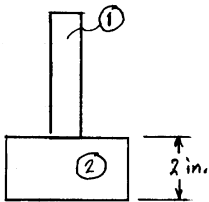


Determine section properties.

Part	$A(\text{in}^2)$	$\bar{y}(\text{in.})$	$A\bar{y}(\text{in}^3)$	$d(\text{in.})$	$Ad^2(\text{in}^4)$	$\bar{I}(\text{in}^4)$
①	4	4	16	2	16	5.333
②	8	1	8	-1	8	2.667
Σ	12		24		24	8.000

$$\bar{Y} = \frac{\Sigma A\bar{y}}{\Sigma A} = \frac{24}{12} = 2 \text{ in.}$$

$$I = \Sigma Ad^2 + \Sigma \bar{I} = 32 \text{ in}^4$$

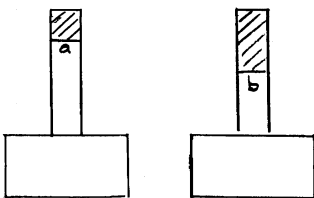


$$(a) \quad A = 1 \text{ in}^2 \quad \bar{y} = 3.5 \text{ in.} \quad Q_a = A\bar{y} = 3.5 \text{ in}^3$$

$$t = 1 \text{ in.}$$

$$\tau_a = \frac{VQ_a}{It} = \frac{(12)(3.5)}{(32)(1)}$$

$$\tau_a = 1.3125 \text{ ksi} \quad \blacktriangleleft$$



(a)

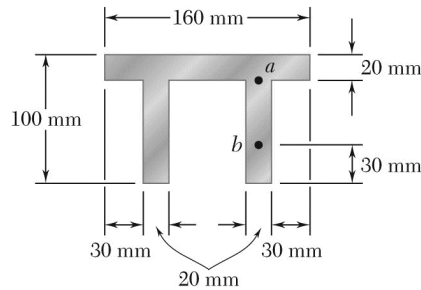
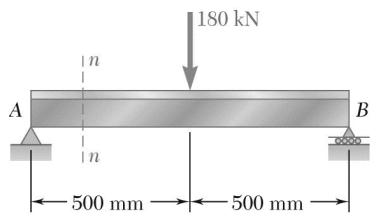
(b)

$$(b) \quad A = 2 \text{ in}^2 \quad \bar{y} = 3 \text{ in.} \quad Q_b = A\bar{y} = 6 \text{ in}^3$$

$$t = 1 \text{ in.}$$

$$\tau_b = \frac{VQ_b}{It} = \frac{(12)(6)}{(32)(1)}$$

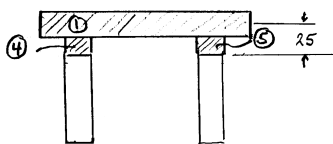
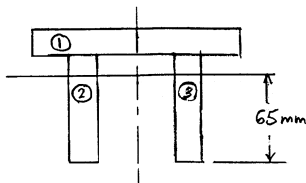
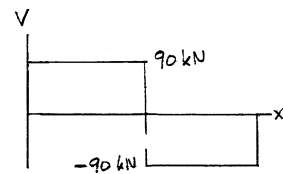
$$\tau_b = 2.25 \text{ ksi} \quad \blacktriangleleft$$



PROBLEM 6.23

For the beam and loading shown, determine the largest shearing stress in section $n-n$.

SOLUTION



Draw the shear diagram.

$$|V|_{\max} = 90 \text{ kN}$$

Part	$A(\text{mm}^2)$	x'	$A\bar{y}(\text{mm}^3)$	$d(\text{mm})$	$Ad^2(\text{mm}^4)$	$\bar{I}(\text{mm}^4)$
①	3200	90	288	25	2.000	0.1067
②	1600	40	64	-25	1.000	0.8533
③	1600	40	64	-25	1.000	0.8533
Σ	6400		416		4.000	1.8133

$$\bar{y} = \frac{\Sigma A\bar{y}}{\Sigma A} = \frac{416 \times 10^3}{6400} = 65 \text{ mm}$$

$$I = \Sigma Ad^2 + \Sigma \bar{I} = (4.000 + 1.8133) \times 10^6 \text{ mm}^4$$

$$= 5.8133 \times 10^6 \text{ mm}^4 = 5.8133 \times 10^{-6} \text{ m}^4$$

Part	$A(\text{mm}^2)$	$\bar{y}(\text{mm})$	$A\bar{y}(\text{mm}^3)$
①	3200	25	80
④	300	7.5	2.25
⑤	300	7.5	2.25
Σ			84.5

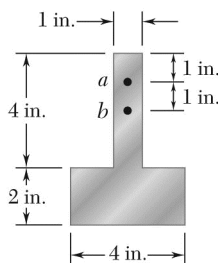
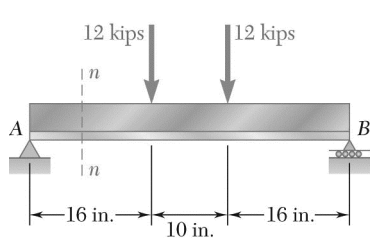
$$Q = \Sigma A\bar{y} = 84.5 \times 10^3 \text{ mm}^3 = 84.5 \times 10^{-6} \text{ m}^3$$

$$t = (2)(20) = 40 \text{ mm} = 40 \times 10^{-3} \text{ m}$$

$$\tau_{\max} = \frac{VQ}{It} = \frac{(90 \times 10^3)(84.5 \times 10^{-6})}{(5.8133 \times 10^{-6})(40 \times 10^{-3})}$$

$$= 32.7 \times 10^6 \text{ Pa}$$

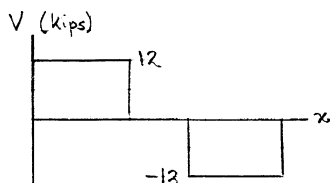
$$\tau_m = 32.7 \text{ MPa} \quad \blacktriangleleft$$



PROBLEM 6.24

For the beam and loading shown, determine the largest shearing stress in section $n-n$.

SOLUTION

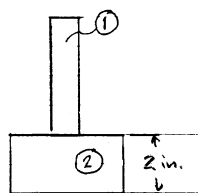


$$R_A = R_B = 12 \text{ kips}$$

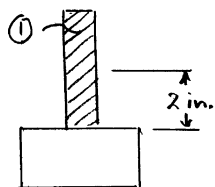
Draw shear diagram.

$$V = 12 \text{ kips}$$

Determine section properties.



Part	$A(\text{in}^2)$	$\bar{y}(\text{in.})$	$A\bar{y}(\text{in}^3)$	$d(\text{in.})$	$Ad^2(\text{in}^4)$	$\bar{I}(\text{in}^4)$
①	4	4	16	2	16	5.333
②	8	1	8	-1	8	2.667
Σ	12		24		24	8.000



$$\bar{Y} = \frac{\Sigma A\bar{y}}{\Sigma A} = \frac{24}{12} = 2 \text{ in.}$$

$$I = \Sigma Ad^2 + \Sigma \bar{I} = 32 \text{ in}^4$$

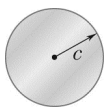
$$Q = A_1\bar{y}_1 = (4)(2) = 8 \text{ in}^3$$

$$t = 1 \text{ in.}$$

$$\tau_{\max} = \frac{VQ}{It} = \frac{(12)(8)}{(32)(1)}$$

$$\tau_{\max} = 3.00 \text{ ksi} \quad \blacktriangleleft$$

PROBLEM 6.25



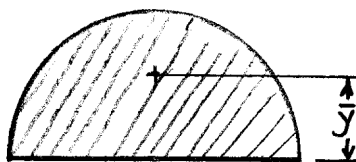
A beam having the cross section shown is subjected to a vertical shear V . Determine (a) the horizontal line along which the shearing stress is maximum, (b) the constant k in the following expression for the maximum shearing stress

$$\tau_{\max} = k \frac{V}{A}$$

where A is the cross-sectional area of the beam.

SOLUTION

$$I = \frac{\pi}{4} c^4 \quad \text{and} \quad A = \pi c^2$$



For semicircle,

$$A_s = \frac{\pi}{2} c^2 \quad \bar{y} = \frac{4c}{3\pi}$$

$$Q = A_s \bar{y} = \frac{\pi}{2} c^2 \cdot \frac{4c}{3\pi} = \frac{2}{3} c^3$$

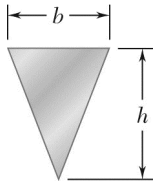
(a) $\tau_{\max}$ occurs at center where

$$t = 2c. \quad \blacktriangleleft$$

$$(b) \quad \tau_{\max} = \frac{VQ}{It} = \frac{V \cdot \frac{2}{3} c^3}{\frac{\pi}{4} c^4 \cdot 2c} = \frac{4V}{3\pi c^2} = \frac{4}{3} \frac{V}{A}$$

$$k = \frac{4}{3} = 1.333 \quad \blacktriangleleft$$

PROBLEM 6.26



A beam having the cross section shown is subjected to a vertical shear V . Determine (a) the horizontal line along which the shearing stress is maximum, (b) the constant k in the following expression for the maximum shearing stress

$$\tau_{\max} = k \frac{V}{A}$$

where A is the cross-sectional area of the beam.

SOLUTION

$$A = \frac{1}{2}bh \quad I = \frac{1}{36}bh^3$$

For a cut at location y ,

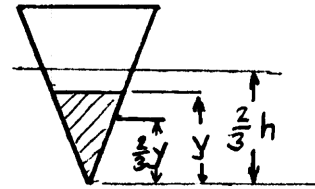
$$A(y) = \frac{1}{2} \left(\frac{by}{h} \right) y = \frac{by^2}{2h}$$

$$\bar{y}(y) = \frac{2}{3}h - \frac{2}{3}y$$

$$Q(y) = A\bar{y} = \frac{by^2}{3}(h - y)$$

$$t(y) = \frac{by}{h}$$

$$\tau(y) = \frac{VQ}{It} = \frac{V \frac{by^2}{3}(h - y)}{(\frac{1}{36}bh^3) \frac{by}{h}} = \frac{12Vy(h - y)}{bh^3} = \frac{12V}{bh^3}(hy - y^2)$$



(a) To find location of maximum of τ , set $\frac{d\tau}{dy} = 0$.

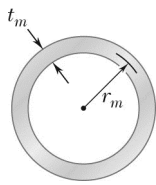
$$\frac{d\tau}{dy} = \frac{12V}{bh^3}(h - 2y_m) = 0$$

$$y_m = \frac{1}{2}h, \text{ i.e., at mid-height } \blacktriangleleft$$

$$(b) \quad \tau_m = \frac{12V}{bh^3}(hy_m - y_m^2) = \frac{12V}{bh^3} \left[\frac{1}{2}h^2 - \left(\frac{1}{2}h \right)^2 \right] = \frac{3V}{bh} = \frac{3}{2} \frac{V}{A}$$

$$k = \frac{3}{2} = 1.500 \quad \blacktriangleleft$$

PROBLEM 6.27



A beam having the cross section shown is subjected to a vertical shear V . Determine (a) the horizontal line along which the shearing stress is maximum, (b) the constant k in the following expression for the maximum shearing stress

$$\tau_{\max} = k \frac{V}{A}$$

where A is the cross-sectional area of the beam.

SOLUTION

For a thin-walled circular section,

$$A = 2\pi r_m t_m$$

$$J = Ar_m^2 = 2\pi r_m^3 t_m, \quad I = \frac{1}{2}J = \pi r_m^3 t_m$$



For a semicircular arc,

$$\bar{y} = \frac{2r_m}{\pi}$$

$$A_s = \pi r_m t_m$$

$$Q = A_s \bar{y} = \pi r_m t_m \frac{2r_m}{\pi} = 2r_m^2 t_m$$

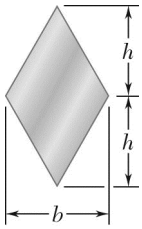
$$(a) \quad t = 2t_m$$

at neutral axis where maximum occurs. ◀

$$(b) \quad \tau_{\max} = \frac{VQ}{It} = \frac{V(2r_m^2 t_m)}{(\pi r_m^3 t_m)(2t_m)} = \frac{V}{\pi r_m t_m} = \frac{2V}{A}$$

$$k = 2.00 \quad \blacktriangleleft$$

PROBLEM 6.28



A beam having the cross section shown is subjected to a vertical shear V . Determine (a) the horizontal line along which the shearing stress is maximum, (b) the constant k in the following expression for the maximum shearing stress

$$\tau_{\max} = k \frac{V}{A}$$

where A is the cross-sectional area of the beam.

SOLUTION

$$A = 2 \left(\frac{1}{2} bh \right) = bh \quad I = 2 \left(\frac{1}{12} bh^3 \right) = \frac{1}{6} bh^3$$

For a cut at location y , where $y \leq h$,

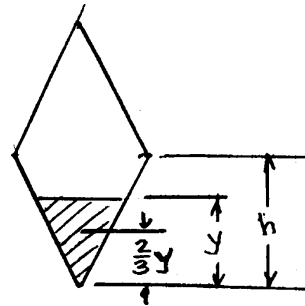
$$A(y) = \frac{1}{2} \left(\frac{by}{h} \right) y = \frac{by^2}{2h}$$

$$\bar{y}(y) = h - \frac{2}{3}y$$

$$Q(y) = A\bar{y} = \frac{by^2}{2} - \frac{by^3}{3h}$$

$$t(y) = \frac{by}{h}$$

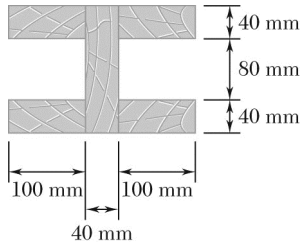
$$\tau(y) = \frac{VQ}{It} = V \frac{6}{bh^3} \cdot \frac{h}{by} \cdot \frac{by^2}{2} - \frac{by^3}{3h} = \frac{V}{bh} \left[3 \left(\frac{y}{h} \right) - 2 \left(\frac{y}{h} \right)^2 \right]$$



(a) To find location of maximum of τ , set $\frac{d\tau}{dy} = 0$.

$$\frac{d\tau}{dy} = \frac{V}{bh^2} [3 - 4 \frac{y_m}{h}] = 0 \quad \frac{y_m}{h} = \frac{3}{4}, \text{ i.e., } \pm \frac{1}{4}h \text{ from neutral axis.} \blacktriangleleft$$

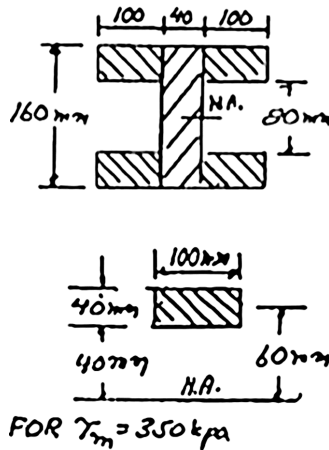
$$(b) \quad \tau(y_m) = \frac{V}{bh} \left[3 \left(\frac{3}{4} \right) - 2 \left(\frac{3}{4} \right)^2 \right] = \frac{9}{8} \frac{V}{bh} = 1.125 \frac{V}{A} \quad k = 1.125 \blacktriangleleft$$



PROBLEM 6.29

The built-up beam shown is made by gluing together five planks. Knowing that in the glued joints the average allowable shearing stress is 350 kPa, determine the largest permissible vertical shear in the beam.

SOLUTION



$$I = \frac{1}{12}(240 \text{ mm})(160 \text{ mm})^3 - \frac{1}{12}(200 \text{ mm})(80 \text{ mm})^3$$

$$= 73.4 \times 10^6 \text{ mm}^4$$

$$I = 73.4 \times 10^{-6} \text{ m}^4$$

$$t = 40 \text{ mm} = 0.04 \text{ m}$$

$$Q = (100 \text{ mm} \times 40 \text{ mm})(60 \text{ mm})$$

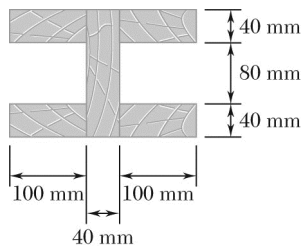
$$= 240 \times 10^3 \text{ mm}^3$$

$$Q = 240 \times 10^{-6} \text{ m}^3$$

For $\tau_m = 350 \text{ kPa}$,

$$\tau_m = \frac{VQ}{It}: \quad 350 \times 10^3 \text{ Pa} = \frac{V(240 \times 10^{-6} \text{ m}^3)}{(73.4 \times 10^{-6} \text{ m}^4)(0.04 \text{ m})}$$

$$V = 4.28 \text{ kN} \quad \blacktriangleleft$$

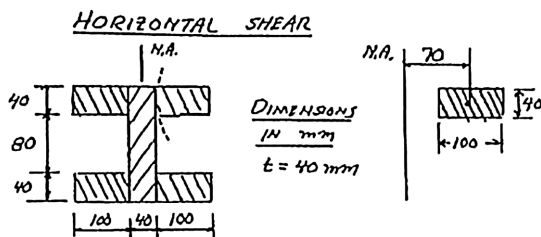


PROBLEM 6.30

For the beam of Prob. 6.29, determine the largest permissible horizontal shear.

PROBLEM 6.29 The built-up beam shown is made by gluing together five planks. Knowing that in the glued joints the average allowable shearing stress is 350 kPa, determine the largest permissible vertical shear in the beam.

SOLUTION



$$I = 2 \frac{1}{12} (40)(240)^3 + \frac{1}{12} (80)(40)^3$$

$$Q = (40 \times 100)70$$

$$= 280 \times 10^3 \text{ mm}^3$$

$$I = 92.6 \times 10^6 \text{ mm}^4 = 92.6 \times 10^{-6} \text{ m}^4$$

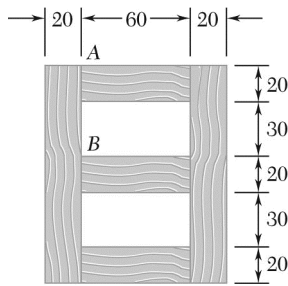
$$Q = (280 \times 10^{-6}) \text{ m}^3$$

For $\tau = 350 \text{ kPa}$,

$$\tau = \frac{VQ}{It}; \quad 350 \times 10^3 \text{ Pa} = \frac{V(280 \times 10^{-6} \text{ m}^3)}{(92.6 \times 10^{-6} \text{ m}^4)(0.04 \text{ m})}$$

$$V = 4630 \text{ N}$$

$$V = 4.63 \text{ kN} \quad \blacktriangleleft$$



Dimensions in mm

PROBLEM 6.31

Several wooden planks are glued together to form the box beam shown. Knowing that the beam is subjected to a vertical shear of 3 kN, determine the average shearing stress in the glued joint (a) at A, (b) at B.

SOLUTION

$$I_A = \frac{1}{12}bh^3 + Ad^2 = \frac{1}{12}(60)(20)^3 + (60)(20)(50)^2$$

$$= 3.04 \times 10^6 \text{ mm}^4$$

$$I_B = \frac{1}{12}bh^3 = \frac{1}{12}(60)(20)^3 = 0.04 \times 10^6 \text{ mm}^4$$

$$I_C = \frac{1}{12}bh^3 = \frac{1}{12}(20)(120)^3 = 2.88 \times 10^6 \text{ mm}^4$$

$$I = 2I_A + I_B + 2I_C = 11.88 \times 10^6 \text{ mm}^4 = 11.88 \times 10^{-6} \text{ m}^4$$

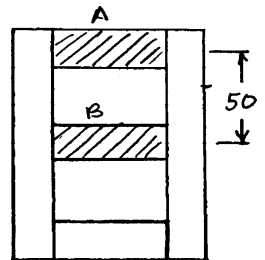
$$Q_A = A\bar{y} = (60)(20)(50) = 60 \times 10^3 \text{ mm}^3 = 60 \times 10^{-6} \text{ m}^3$$

$$t = 20 \text{ mm} + 20 \text{ mm} = 40 \text{ mm} = 40 \times 10^{-3} \text{ m}$$

$$(a) \quad \tau_A = \frac{VQ_A}{It} = \frac{(3 \times 10^3)(60 \times 10^{-6})}{(11.88 \times 10^{-6})(40 \times 10^{-3})} = 379 \times 10^3 \text{ Pa}$$

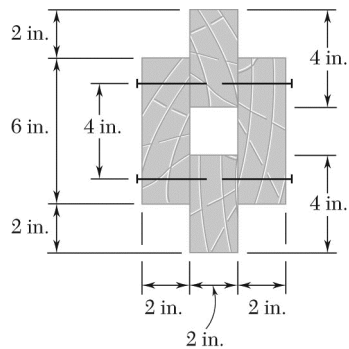
$$Q_B = 0$$

$$(b) \quad \tau_B = \frac{VQ_B}{It} = 0$$



$$\tau_A = 379 \text{ kPa} \quad \blacktriangleleft$$

$$\tau_B = 0 \quad \blacktriangleleft$$



PROBLEM 6.32

The built-up timber beam is subjected to a 1500-lb vertical shear. Knowing that the longitudinal spacing of the nails is $s = 2.5$ in. and that each nail is 3.5 in. long, determine the shearing force in each nail.

SOLUTION

$$I_1 = \frac{1}{12}(2)(4)^3 + (2)(4)(3)^2$$

$$= 82.6667 \text{ in}^4$$

$$I_2 = \frac{1}{12}(2)(6)^3 = 36 \text{ in}^4$$

$$I = 2I_1 + 2I_2$$

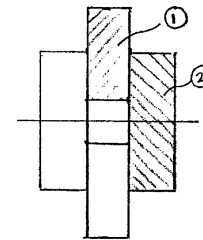
$$= 237.333 \text{ in}^4$$

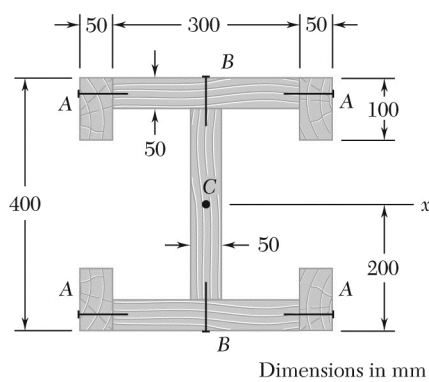
$$Q = A_1 \bar{y}_1 = (2)(4)(3) = 24 \text{ in}^3$$

$$q = \frac{VQ}{I} = \frac{(1500)(24)}{237.333} = 151.685 \text{ lb/in}$$

$$2F_{\text{nail}} = qs \quad F_{\text{nail}} = \frac{1}{2}qs = \left(\frac{1}{2}\right)(151.685)(2.5)$$

$$F_{\text{nail}} = 189.6 \text{ lb} \quad \blacktriangleleft$$





PROBLEM 6.33

The built-up wooden beam shown is subjected to a vertical shear of 8 kN. Knowing that the nails are spaced longitudinally every 60 mm at A and every 25 mm at B, determine the shearing force in the nails (a) at A, (b) at B. (Given: $I_x = 1.504 \times 10^9 \text{ mm}^4$.)

SOLUTION

$$I_x = 1.504 \times 10^9 \text{ mm}^4 = 1504 \times 10^{-6} \text{ m}^4$$

$$s_A = 60 \text{ mm} = 0.060 \text{ m}$$

$$s_B = 25 \text{ mm} = 0.025 \text{ m}$$

(a)

$$Q_A = Q_1 = A_1 \bar{y}_1 = (50)(100)(150) = 750 \times 10^3 \text{ mm}^3 \\ = 750 \times 10^{-6} \text{ m}^3$$

$$F_A = q_A s_A$$

$$= \frac{V Q_1 s_A}{I} = \frac{(8 \times 10^3)(750 \times 10^{-6})(0.060)}{1504 \times 10^{-6}}$$

$$F_A = 239 \text{ N} \quad \blacktriangleleft$$

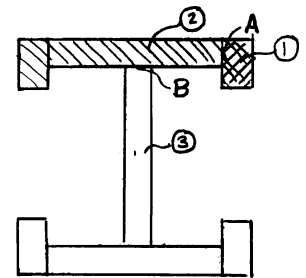
(b)

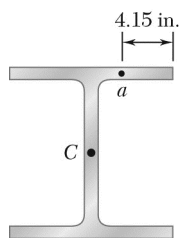
$$Q_2 = A_2 \bar{y}_2 = (300)(50)(175) = 2625 \times 10^3 \text{ mm}^3$$

$$Q_B = 2Q_1 + Q_2 = 4125 \times 10^3 \text{ mm}^3 \\ = 4125 \times 10^{-6} \text{ m}^3$$

$$F_B = q_B s_B = \frac{V Q_B s_B}{I} = \frac{(8 \times 10^3)(4125 \times 10^{-6})(0.025)}{1504 \times 10^{-6}}$$

$$F_B = 549 \text{ N} \quad \blacktriangleleft$$





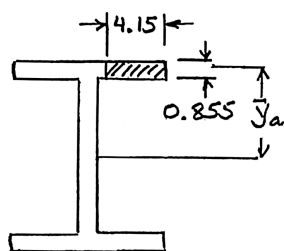
PROBLEM 6.34

Knowing that a vertical shear V of 50 kips is exerted on $W14 \times 82$ rolled-steel beam, determine the shearing stress (a) at point a , (b) at the centroid C .

SOLUTION

For $W14 \times 82$, $d = 14.3$ in., $b_f = 10.1$ in., $t_f = 0.855$ in., $t_w = 0.510$ in., $I = 881$ in⁴

(a)



$$A_a = (4.15)(0.855) = 3.5482 \text{ in}^2$$

$$\bar{y}_a = \frac{d}{2} - \frac{t_f}{2} = \frac{14.3}{2} - \frac{0.855}{2} = 6.7225 \text{ in.}$$

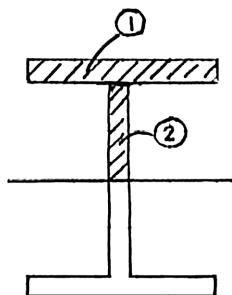
$$Q_a = A_a \bar{y}_a = 23.853 \text{ in}^3$$

$$t_a = t_f = 0.855 \text{ in.}$$

$$\tau_a = \frac{VQ_a}{It_a} = \frac{(50)(23.853)}{(881)(0.855)}$$

$$\tau_a = 1.583 \text{ ksi} \quad \blacktriangleleft$$

(b)



$$A_1 = b_f t_f (10.1)(0.855) = 8.6355 \text{ in}^2$$

$$\bar{y}_1 = \frac{d}{2} - \frac{t_f}{2} = 6.7225 \text{ in.}$$

$$A_2 = t_w \left(\frac{d}{2} - t_f \right) = (0.510)(6.295) = 3.2105 \text{ in}^2$$

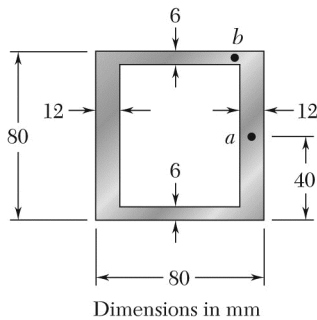
$$\bar{y}_2 = \frac{1}{2} \left(\frac{d}{2} - t_f \right) = \frac{1}{2} (6.295) = 3.1475 \text{ in.}$$

$$Q_C = A_1 \bar{y}_1 + A_2 \bar{y}_2 = (8.6355)(6.7225) + (3.2105)(3.1475) \\ = 68.157 \text{ in}^3$$

$$t_C = t_w = 0.510 \text{ in.}$$

$$\tau_C = \frac{VQ_C}{It_C} = \frac{(50)(68.157)}{(881)(0.510)}$$

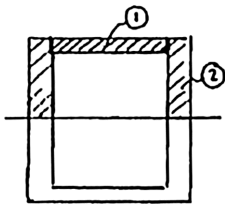
$$\tau_C = 7.59 \text{ ksi} \quad \blacktriangleleft$$



PROBLEM 6.35

An extruded aluminum beam has the cross section shown. Knowing that the vertical shear in the beam is 150 kN, determine the shearing stress at (a) point *a*, (b) point *b*.

SOLUTION



$$I = \frac{1}{12}(80)(80)^3 - \frac{1}{12}(56)(68)^3 = 1.9460 \times 10^6 \text{ mm}^4$$

$$= 1.946 \times 10^{-6} \text{ m}^4$$

$$(a) \quad Q_a = A_1 \bar{y}_1 + 2A_2 \bar{y}_2$$

$$= (56)(6)(37) + (2)(12)(40)(20) = 31.632 \times 10^3 \text{ mm}^3$$

$$= 31.632 \times 10^{-6} \text{ m}^3$$

$$t_a = (2)(12) = 24 \text{ mm} = 0.024 \text{ m}$$

$$\tau_a = \frac{VQ_a}{It_a} = \frac{(150 \times 10^3)(31.632 \times 10^{-6})}{(1.946 \times 10^{-6})(0.024)} = 101.6 \times 10^6 \text{ Pa}$$

$$\tau_a = 101.6 \text{ MPa} \quad \blacktriangleleft$$

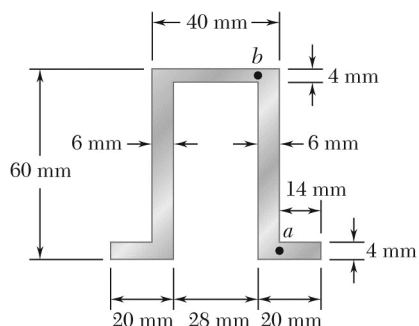
$$(b) \quad Q_b = A_1 \bar{y}_1 = (56)(6)(37) = 12.432 \times 10^3 \text{ mm}^3$$

$$= 12.432 \times 10^{-6} \text{ m}^3$$

$$t_b = (2)(6) = 12 \text{ mm} = 0.012 \text{ m}$$

$$\tau_b = \frac{VQ_b}{It_b} = \frac{(150 \times 10^3)(12.432 \times 10^{-6})}{(1.946 \times 10^{-6})(0.012)} = 79.9 \times 10^6 \text{ Pa}$$

$$\tau_b = 79.9 \text{ MPa} \quad \blacktriangleleft$$

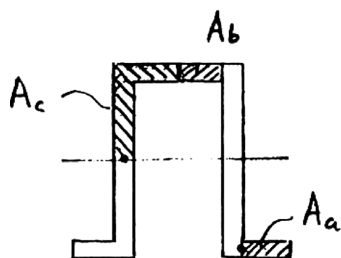


PROBLEM 6.36

Knowing that a given vertical shear V causes a maximum shearing stress of 75 MPa in the hat-shaped extrusion shown, determine the corresponding shearing stress (a) at point a , (b) at point b .

SOLUTION

Neutral axis lies 30 mm above bottom.



$$\tau_c = \frac{VQ_c}{It} \quad \tau_a = \frac{VQ_a}{It_a} \quad \tau_b = \frac{VQ_b}{It_b}$$

$$\frac{\tau_a}{\tau_c} = \frac{Q_a t_c}{Q_c t_a} \quad \frac{\tau_b}{\tau_c} = \frac{Q_b t_c}{Q_c t_b}$$

$$Q_c = (6)(30)(15) + (14)(4)(28) = 4260 \text{ mm}^3$$

$$t_c = 6 \text{ mm}$$

$$Q_a = (14)(4)(28) = 1568 \text{ mm}^3$$

$$t_a = 4 \text{ mm}$$

$$Q_b = (14)(4)(28) = 1568 \text{ mm}^3$$

$$t_b = 4 \text{ mm}$$

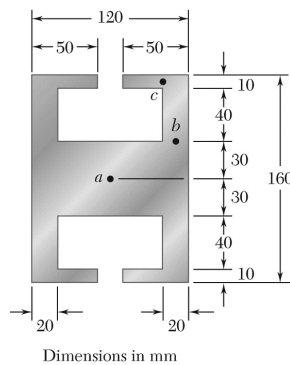
$$\tau_c = 75 \text{ MPa}$$

$$(a) \quad \tau_a = \frac{Q_a}{Q_c} \cdot \frac{t_c}{t_a} \tau_c = \frac{1568}{4260} \cdot \frac{6}{4} \cdot 75$$

$$\tau_a = 41.4 \text{ MPa} \quad \blacktriangleleft$$

$$(b) \quad \tau_b = \frac{Q_b}{Q_c} \cdot \frac{t_c}{t_b} \tau_c = \frac{1568}{4260} \cdot \frac{6}{4} \cdot 75$$

$$\tau_b = 41.4 \text{ MPa} \quad \blacktriangleleft$$



PROBLEM 6.37

Knowing that a given vertical shear V causes a maximum shearing stress of 75 MPa in an extruded beam having the cross section shown, determine the shearing stress at the three points indicated.

SOLUTION

$$\tau = \frac{VQ}{It} \quad \tau \text{ is proportional to } Q/t.$$

Point c :

$$\begin{aligned} Q_c &= (30)(10)(75) \\ &= 22.5 \times 10^3 \text{ mm}^3 \\ t_c &= 10 \text{ mm} \end{aligned}$$

$$Q_c/t_c = 2250 \text{ mm}^2$$

Point b :

$$\begin{aligned} Q_b &= Q_c + (20)(50)(55) \\ &= 77.5 \times 10^3 \text{ mm}^3 \\ t_b &= 20 \text{ mm} \end{aligned}$$

$$Q_b/t_b = 3875 \text{ mm}^2$$

Point a :

$$\begin{aligned} Q_a &= 2Q_b + (120)(30)(15) \\ &= 209 \times 10^3 \text{ mm}^3 \\ t_a &= 120 \text{ mm} \end{aligned}$$

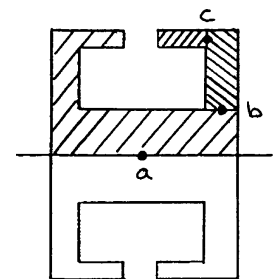
$$Q_a/t_a = 1741.67 \text{ mm}^2$$

$(Q/t)_m$ occurs at b .

$$\tau_m = \tau_b = 75 \text{ MPa}$$

$$\frac{\tau_a}{Q_a/t_a} = \frac{\tau_b}{Q_b/t_b} = \frac{\tau_c}{Q_c/t_c}$$

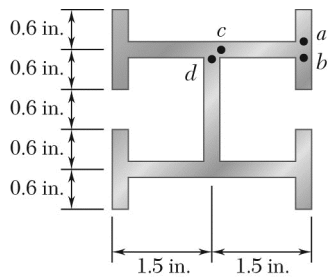
$$\frac{\tau_a}{1741.67 \text{ mm}^2} = \frac{75 \text{ MPa}}{3875 \text{ mm}^2} = \frac{\tau_a}{2250 \text{ mm}^2}$$



$$\tau_a = 33.7 \text{ MPa} \quad \blacktriangleleft$$

$$\tau_b = 75.0 \text{ MPa} \quad \blacktriangleleft$$

$$\tau_c = 43.5 \text{ MPa} \quad \blacktriangleleft$$



PROBLEM 6.38

An extruded beam has the cross section shown and a uniform wall thickness of 0.20 in. Knowing that a given vertical shear V causes a maximum shearing stress $\tau = 9$ ksi, determine the shearing stress at the four points indicated.

SOLUTION

$$Q_a = (0.2)(0.5)(0.5 - 0.25) = 0.125 \text{ in}^3$$

$$Q_b = (0.2)(0.5)(0.3 + 0.25) = 0.055 \text{ in}^3$$

$$Q_c = Q_a + Q_b + (1.4)(0.2)(0.9) = 0.432 \text{ in}^3$$

$$Q_d = 2Q_a + 2Q_b + (3.0)(0.2)(0.9) = 0.900 \text{ in}^3$$

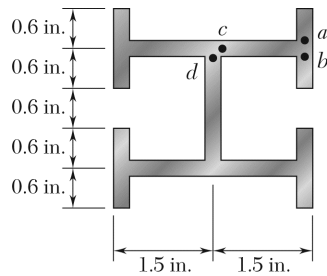
$$Q_m = Q_d + (0.2)(0.8)(0.4) = 0.964 \text{ in}^3$$

$$\tau = \frac{VQ}{It}$$

Since V , I , and t are constant, τ is proportional to Q .

$$\frac{\tau_a}{0.125} = \frac{\tau_b}{0.055} = \frac{\tau_c}{0.432} = \frac{\tau_d}{0.900} = \frac{\tau_m}{0.964} = \frac{9}{0.964}$$

$$\tau_a = 1.167 \text{ ksi}; \quad \tau_b = 0.513 \text{ ksi}; \quad \tau_c = 4.03 \text{ ksi}; \quad \tau_d = 8.40 \text{ ksi} \quad \blacktriangleleft$$



PROBLEM 6.39

Solve Prob. 6.38, assuming that the beam is subjected to a horizontal shear V .

PROBLEM 6.38 An extruded beam has the cross section shown and a uniform wall thickness of 0.20 in. Knowing that a given vertical shear V causes a maximum shearing stress $\tau = 9$ ksi, determine the shearing stress at the four points indicated.

SOLUTION

$$Q_a = (0.5)(0.2)(1.4) = 0.140 \text{ in}^3$$

$$Q_b = (0.5)(0.2)(1.4) = 0.140 \text{ in}^3$$

$$Q_c = Q_a + Q_b + (0.2)(1.4)(0.8) = 0.504 \text{ in}^3$$

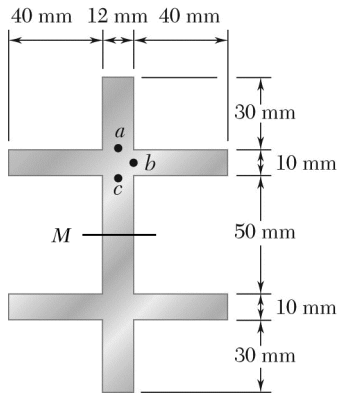
$$Q_d = 0$$

$$Q_m = Q_c = 0.504 \text{ in}^3$$

$$\tau = \frac{VQ}{It} \quad \text{Since } V, I, \text{ and } t \text{ are constant, } \tau \text{ is proportional to } Q.$$

$$\frac{\tau_a}{0.140} = \frac{\tau_b}{0.140} = \frac{\tau_c}{0.504} = \frac{\tau_d}{0} = \frac{\tau_m}{0.504} = \frac{9}{0.504}$$

$$\tau_a = 2.50 \text{ ksi}; \quad \tau_b = 2.50 \text{ ksi}, \quad \tau_c = 9.00 \text{ ksi}, \quad \tau_d = 0 \quad \blacktriangleleft$$



PROBLEM 6.40

Knowing that a given vertical shear V causes a maximum shearing stress of 50 MPa in a thin-walled member having the cross section shown, determine the corresponding shearing stress (a) at point a , (b) at point b , (c) at point c .

SOLUTION

$$Q_a = (12)(30)(25 + 10 + 15) = 18 \times 10^3 \text{ mm}^3$$

$$Q_b = (40)(10)(25 + 5) = 12 \times 10^3 \text{ mm}^3$$

$$Q_c = Q_a + 2Q_b + (12)(10)(25 + 5) = 45.6 \times 10^3 \text{ mm}^3$$

$$Q_m = Q_c + (12)(25)\left(\frac{25}{2}\right) = 49.35 \times 10^3 \text{ mm}^3$$

$$t_a = t_c = t_m = 12 \text{ mm}$$

$$t_b = 10 \text{ mm}$$

$$\tau_m = 50 \text{ MPa}$$

$$(a) \quad \frac{\tau_a}{\tau_m} = \frac{Q_a}{Q_m} \cdot \frac{t_m}{t_a} = \frac{18}{49.35} \cdot \frac{12}{12} = 0.3647$$

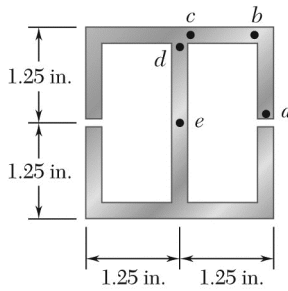
$$\tau_a = 18.23 \text{ MPa} \quad \blacktriangleleft$$

$$(b) \quad \frac{\tau_b}{\tau_m} = \frac{Q_b}{Q_m} \cdot \frac{t_m}{t_b} = \frac{12}{49.35} \cdot \frac{12}{10} = 0.2918$$

$$\tau_b = 14.59 \text{ MPa} \quad \blacktriangleleft$$

$$(c) \quad \frac{\tau_c}{\tau_m} = \frac{Q_c}{Q_m} \cdot \frac{t_m}{t_c} = \frac{45.6}{49.35} \cdot \frac{12}{12} = 0.9240$$

$$\tau_c = 46.2 \text{ MPa} \quad \blacktriangleleft$$



PROBLEM 6.41

The extruded aluminum beam has a uniform wall thickness of $\frac{1}{8}$ in. Knowing that the vertical shear in the beam is 2 kips, determine the corresponding shearing stress at each of the five points indicated.

SOLUTION

$$I = \frac{1}{12}(2.50)(2.50)^3 - \frac{1}{12}(2.125)(2.25)^3 = 1.2382 \text{ in}^4$$

$$t = 0.125 \text{ in. at all sections.}$$

$$V = 2 \text{ kips}$$

$$Q_a = 0 \quad \tau_a = \frac{VQ_a}{It} \quad \tau_a = 0 \quad \blacktriangleleft$$

$$Q_b = (0.125)(1.25)\left(\frac{1.25}{2}\right) = 0.09766 \text{ in}^3$$

$$\tau_b = \frac{VQ_b}{It} = \frac{(2)(0.09766)}{(1.2382)(0.125)} \quad \tau_b = 1.26 \text{ ksi} \quad \blacktriangleleft$$

$$Q_c = Q_b + (1.0625)(0.125)(1.1875) = 0.25537 \text{ in}^2$$

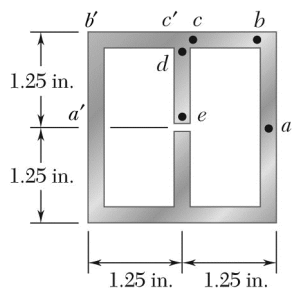
$$\tau_c = \frac{VQ_c}{It} = \frac{(2)(0.25537)}{(1.2382)(0.125)} \quad \tau_c = 3.30 \text{ ksi} \quad \blacktriangleleft$$

$$Q_d = 2Q_c + (0.125)^2(1.1875) = 0.52929$$

$$\tau_d = \frac{VQ_d}{It} = \frac{(2)(0.52929)}{(1.2382)(0.125)} \quad \tau_d = 6.84 \text{ ksi} \quad \blacktriangleleft$$

$$Q_e = Q_d + (0.125)(1.125)\left(\frac{1.125}{2}\right) = 0.60839$$

$$\tau_e = \frac{VQ_e}{It} = \frac{(2)(0.60839)}{(1.2382)(0.125)} \quad \tau_e = 7.86 \text{ ksi} \quad \blacktriangleleft$$



PROBLEM 6.42

The extruded aluminum beam has a uniform wall thickness of $\frac{1}{8}$ in. Knowing that the vertical shear in the beam is 2 kips, determine the corresponding shearing stress at each of the five points indicated.

SOLUTION

$$I = \frac{1}{12}(2.50)(2.50)^3 - \frac{1}{2}(2.125)(2.25)^3 = 1.2382 \text{ in}^4$$

Add symmetric points c' , b' , and a' .

$$Q_e = 0$$

$$Q_d = (0.125)(1.125)\left(\frac{1.125}{2}\right) = 0.07910 \text{ in}^3 \quad t_d = 0.125 \text{ in.}$$

$$Q_c = Q_e = (0.125)^2(1.1875) = 0.09765 \text{ in}^4 \quad t_c = 0.25 \text{ in.}$$

$$Q_b = Q_c + (2)(1.0625)(0.125)(1.1875) = 0.41308 \text{ in}^3 \quad t_b = 0.25 \text{ in.}$$

$$Q_a = Q_b + (2)(0.125)(1.25)\left(\frac{1.25}{2}\right) = 0.60839 \text{ in}^3 \quad t_a = 0.25 \text{ in.}$$

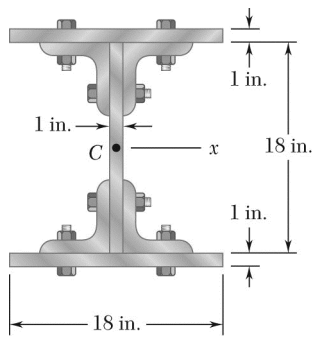
$$\tau_a = \frac{VQ_a}{It_a} = \frac{(2)(0.60839)}{(1.2382)(0.25)} \quad \tau_a = 3.93 \text{ ksi} \quad \blacktriangleleft$$

$$\tau_b = \frac{VQ_b}{It_b} = \frac{(2)(0.41308)}{(1.2382)(0.25)} \quad \tau_b = 2.67 \text{ ksi} \quad \blacktriangleleft$$

$$\tau_c = \frac{VQ_c}{It_c} = \frac{(2)(0.09765)}{(1.2382)(0.25)} \quad \tau_c = 0.63 \text{ ksi} \quad \blacktriangleleft$$

$$\tau_d = \frac{VQ_d}{It_d} = \frac{(2)(0.07910)}{(1.2382)(0.25)} \quad \tau_d = 1.02 \text{ ksi} \quad \blacktriangleleft$$

$$\tau_e = \frac{VQ_e}{It_e} \quad \tau_e = 0 \quad \blacktriangleleft$$



PROBLEM 6.43

Three 1×18-in. steel plates are bolted to four L6×6×1 angles to form a beam with the cross section shown. The bolts have a $\frac{7}{8}$ -in. diameter and are spaced longitudinally every 5 in. Knowing that the allowable average shearing stress in the bolts is 12 ksi, determine the largest permissible vertical shear in the beam. (Given: $I_x = 6123 \text{ in}^4$)

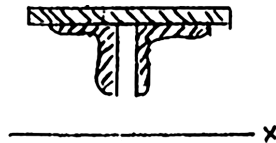
SOLUTION

Flange:
$$I_f = \frac{1}{12}(18)(1)^3 + (18)(1)(9.5)^2 = 1626 \text{ in}^4$$

Web:
$$I_w = \frac{1}{12}(1)(18)^3 = 486 \text{ in}^4$$

Angle:
$$\begin{aligned} \bar{I} &= 35.5 \text{ in}^4, & A &= 11.0 \text{ in}^2 \\ \bar{y} &= 1.86 \text{ in.} & d &= 9 - 1.86 = 7.14 \text{ in.} \\ I_a &= \bar{I} + Ad^2 = 596.18 \text{ in}^4 \end{aligned}$$

$$I = 2I_f + I_w + 4I_a = 6123 \text{ in}^4, \text{ which agrees with the given value.}$$



Flange:
$$Q_f = (18)(1)(9.5) = 171 \text{ in}^3$$

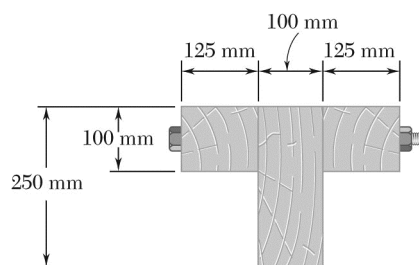
Angle:
$$\begin{aligned} Q_a &= Ad = (11.0)(7.14) = 78.54 \text{ in}^3 \\ Q &= Q_f + 2Q_a = 328.08 \text{ in}^3 \end{aligned}$$

$$A_{\text{bolt}} = \frac{\pi}{4} \left(\frac{7}{8} \right)^2 = 0.60132 \text{ in}^2$$

$$F_{\text{bolt}} = 2\tau_{\text{bolt}} A_{\text{bolt}} = (2)(12)(0.60132) = 14.4317 \text{ kips}$$

$$q_{\text{all}} = \frac{F_{\text{bolt}}}{s} = \frac{14.4317}{5} = 2.8863 \text{ kip/s}$$

$$q = \frac{VQ}{I} \quad V_{\text{all}} = \frac{q_{\text{all}} I}{Q} = \frac{(2.8863)(6123)}{328.08} \quad V_{\text{all}} = 53.9 \text{ kips} \quad \blacktriangleleft$$



PROBLEM 6.44

Three planks are connected as shown by bolts of 14-mm diameter spaced every 150 mm along the longitudinal axis of the beam. For a vertical shear of 10 kN, determine the average shearing stress in the bolts.

SOLUTION

Locate neutral axis and compute moment of inertia.

Part	$A(\text{mm}^2)$	$\bar{y}(\text{mm})$	$A\bar{y}(\text{mm}^3)$	$d(\text{mm})$	$Ad^2(\text{mm}^4)$	$\bar{I}(\text{mm}^4)$
①	12500	200	2.5×10^6	37.5	17.5781×10^6	10.4167×10^6
②	25000	125	3.125×10^6	37.5	35.156×10^6	130.208×10^6
③	12500	200	2.5×10^6	37.5	17.5781×10^6	10.4167×10^6
Σ	50000		8.125×10^6		70.312×10^6	151.04×10^6

$$\bar{Y} = \frac{\Sigma A\bar{y}}{\Sigma A} = \frac{8.125 \times 10^6}{50 \times 10^3} = 162.5 \text{ mm}$$

$$I = \Sigma Ad^2 + \Sigma \bar{I} = 221.35 \times 10^6 \text{ mm}^4$$

$$= 221.35 \times 10^{-6} \text{ m}^4$$

$$Q = A_1 \bar{y}_1 = (12500)(37.5) = 468.75 \times 10^3 \text{ mm}^3$$

$$= 468.75 \times 10^{-6} \text{ m}^3$$

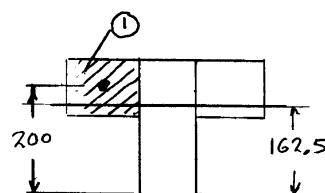
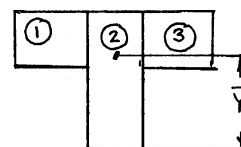
$$q = \frac{VQ}{I} = \frac{(10 \times 10^3)(468.75 \times 10^{-6})}{221.35 \times 10^{-6}}$$

$$= 21.177 \times 10^3 \text{ N/m}$$

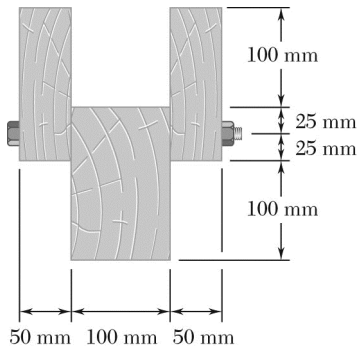
$$F_{\text{bolt}} = qs = (21.177 \times 10^3)(150 \times 10^{-3}) = 3.1765 \times 10^3 \text{ N}$$

$$A_{\text{bolt}} = \frac{\pi}{4}(14)^2 = 153.938 \text{ mm}^2 = 153.938 \times 10^{-6} \text{ m}^2$$

$$\tau_{\text{bolt}} = \frac{F_{\text{bolt}}}{A_{\text{bolt}}} = \frac{3.1765 \times 10^3}{153.938 \times 10^{-6}} = 20.6 \times 10^6 \text{ Pa}$$



$$\tau_{\text{bolt}} = 20.6 \text{ MPa} \quad \blacktriangleleft$$

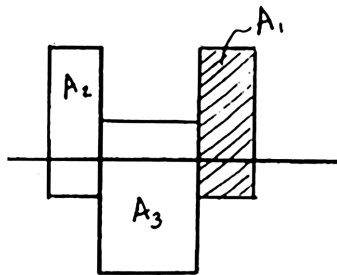


PROBLEM 6.45

A beam consists of three planks connected as shown by steel bolts with a longitudinal spacing of 225 mm. Knowing that the shear in the beam is vertical and equal to 6 kN and that the allowable average shearing stress in each bolt is 60 MPa, determine the smallest permissible bolt diameter that can be used.

SOLUTION

Part	$A(\text{mm}^2)$	$\bar{y}(\text{mm})$	$A\bar{y}^2(10^6\text{mm}^4)$	$\bar{I}(10^6\text{mm}^4)$
①	7500	50	18.75	14.06
②	7500	50	18.75	14.06
③	15,000	-50	37.50	28.12
Σ			75.00	56.25



$$I = \Sigma A\bar{y}^2 + \Sigma \bar{I} = 131.25 \times 10^6 \text{ mm}^4 = 131.25 \times 10^{-6} \text{ m}^4$$

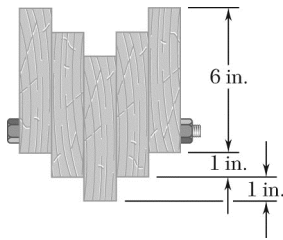
$$Q = A_1 \bar{y}_1 = (7500)(50) = 375 \times 10^3 \text{ mm}^3 \\ = 375 \times 10^{-6} \text{ m}^3$$

$$F_{\text{bolt}} = \tau_{\text{bolt}} A_{\text{bolt}} = qs = \frac{VQs}{I}$$

$$A_{\text{bolt}} = \frac{VQs}{\tau_{\text{bolt}} I} = \frac{(6 \times 10^3)(375 \times 10^{-6})(0.225)}{(6 \times 10^6)(131.25 \times 10^6)} = 64.286 \times 10^{-6} \text{ m}^2 \\ = 64.286 \text{ mm}^2$$

$$d_{\text{bolt}} = \sqrt{\frac{4A_{\text{bolt}}}{\pi}} = \sqrt{\frac{(4)(64.286)}{\pi}}$$

$$d_{\text{bolt}} = 9.05 \text{ mm} \quad \blacktriangleleft$$



PROBLEM 6.46

A beam consists of five planks of 1.5×6 -in. cross section connected by steel bolts with a longitudinal spacing of 9 in. Knowing that the shear in the beam is vertical and equal to 2000 lb and that the allowable average shearing stress in each bolt is 7500 psi, determine the smallest permissible bolt diameter that can be used.

SOLUTION

Part	$A(\text{in}^2)$	$\bar{y}_0(\text{in.})$	$A\bar{y}_0(\text{in}^3)$	$\bar{y}(\text{in.})$	$A\bar{y}^2(\text{in}^4)$	$\bar{I}(\text{in}^4)$
①	9	5	45	0.8	5.76	27
②	9	4	36	-0.2	0.36	27
③	9	3	27	-1.2	12.96	27
④	9	4	36	-0.2	0.36	27
⑤	9	5	45	0.8	5.76	27
Σ	45		189		25.20	135

$$\bar{Y}_0 = \frac{\Sigma A\bar{y}}{\Sigma A} = \frac{189}{45} = 4.2 \text{ in.}$$

$$I = \Sigma A\bar{d}^2 + \Sigma \bar{I} = 160.2 \text{ in}^4$$

Between ① and ②:

$$Q_{12} = Q_1 = A\bar{y}_1 = (9)(0.8) = 7.2 \text{ in}^3$$

Between ② and ③:

$$Q_{23} = Q_1 + A\bar{y}_2 = 7.2 + (9)(-0.2) = 5.4 \text{ in}^3$$

$$q = \frac{VQ}{I}$$

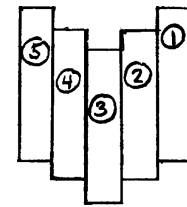
Maximum q is based on $Q_{12} = 7.2 \text{ in}^3$.

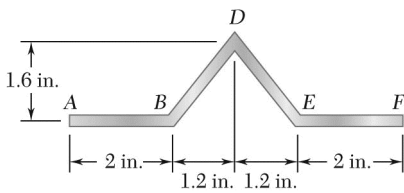
$$q = \frac{(2000)(7.2)}{160.2} = 89.888 \text{ lb/in}$$

$$F_{\text{bolt}} = qs = (89.888)(9) = 809 \text{ lb}$$

$$\tau_{\text{bolt}} = \frac{F_{\text{bolt}}}{A_{\text{bolt}}} \quad A_{\text{bolt}} = \frac{F_{\text{bolt}}}{\tau_{\text{bolt}}} = \frac{809}{7500} = 0.1079 \text{ in}^2$$

$$A_{\text{bolt}} = \frac{\pi}{4} d_{\text{bolt}}^2 \quad d_{\text{bolt}} = \sqrt{\frac{4A_{\text{bolt}}}{\pi}} = \sqrt{\frac{(4)(0.1079)}{\pi}} \quad d_{\text{bolt}} = 0.371 \text{ in.} \blacktriangleleft$$





PROBLEM 6.47

A plate of $\frac{1}{4}$ -in. thickness is corrugated as shown and then used as a beam. For a vertical shear of 1.2 kips, determine (a) the maximum shearing stress in the section, (b) the shearing stress at point B. Also, sketch the shear flow in the cross section.

SOLUTION

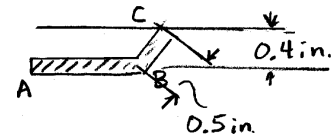
$$L_{BD} = \sqrt{(1.2)^2 + (1.6)^2} = 2.0 \text{ in.} \quad A_{BD} = (0.25)(2.0) = 0.5 \text{ in}^2$$

Locate neutral axis and compute moment of inertia.

Part	$A(\text{in}^2)$	$\bar{y}(\text{in.})$	$A\bar{y} (\text{in}^3)$	$d(\text{in.})$	$Ad^2(\text{in}^4)$	$\bar{I}(\text{in}^4)$
AB	0.5	0	0	0.4	0.080	neglect
BD	0.5	0.8	0.4	0.4	0.080	*0.1067
DE	0.5	0.8	0.4	0.4	0.080	*0.1067
EF	0.5	0	0	0.4	0.080	neglect
Σ	2.0		0.8		0.320	0.2133

$$^* \frac{1}{12} A_{BD} h^2 = \frac{1}{12} (0.5)(1.6)^2 = 0.1067 \text{ in}^4 \quad \bar{Y} = \frac{\Sigma A\bar{y}}{\Sigma A} = \frac{0.8}{2.0} = 0.4 \text{ in.}$$

$$I = \Sigma Ad^2 + \Sigma \bar{I} = 0.5333 \text{ in}^4$$



(a)

$$Q_m = Q_{AB} + Q_{BC}$$

$$Q_{AB} = (2)(0.25)(0.4) = 0.2 \text{ in}^3$$

$$Q_{BC} = (0.5)(0.25)(0.2) = 0.025 \text{ in}^3$$

$$Q_m = 0.225 \text{ in}^3$$

$$\tau_m = \frac{VQ_m}{It} = \frac{(1.2)(0.225)}{(0.5333)(0.25)}$$

$$\tau_m = 2.03 \text{ ksi} \quad \blacktriangleleft$$

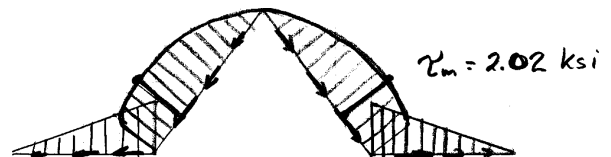
(b)

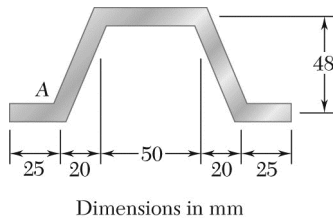
$$Q_B = Q_{AB} = 0.2 \text{ in}^3$$

$$\tau_B = \frac{VQ_B}{It} = \frac{(1.2)(0.2)}{(0.5333)(0.25)}$$

$$\tau_B = 1.800 \text{ ksi} \quad \blacktriangleleft$$

$$\tau_D = 0$$





PROBLEM 6.48

A plate of 4-mm thickness is bent as shown and then used as a beam. For a vertical shear of 12 kN, determine (a) the shearing stress at point *A*, (b) the maximum shearing stress in the beam. Also, sketch the shear flow in the cross section.

SOLUTION

$$\tan \alpha = \frac{20}{48} \quad \alpha = 22.62^\circ$$

Slanted side: $A_s = (4 \sec \alpha)(48) = 208 \text{ mm}^2$

$$\bar{I}_s = \frac{1}{12}(4 \sec \alpha)(48)^3 = 39.936 \times 10^3 \text{ mm}^4$$

Top: $I_T = \frac{1}{12}(50)(4)^3 + (50)(4)(24)^2 = 115.46 \times 10^3 \text{ mm}^4$

Bottom: $I_B = I_T = 115.46 \times 10^3 \text{ mm}^4$

$$I = 2\bar{I}_s + I_T + I_B = 310.8 \times 10^3 \text{ mm}^4 = 310.8 \times 10^{-9} \text{ m}^4$$

(a) $Q_A = (25)(4)(24) = 2.4 \times 10^3 \text{ mm}^3 = 2.4 \times 10^{-6} \text{ m}^3$

$$t = 4 \text{ mm} = 4 \times 10^{-3} \text{ m}$$

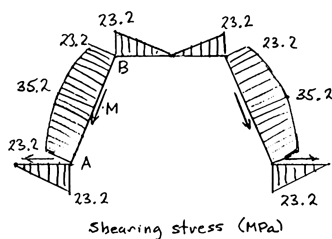
$$\tau_A = \frac{VQ_A}{It} = \frac{(12 \times 10^3)(2.4 \times 10^{-6})}{(310.8 \times 10^{-9})(4 \times 10^{-3})} = 23.2 \times 10^6 \text{ Pa} \quad \tau_A = 23.2 \text{ MPa} \quad \blacktriangleleft$$

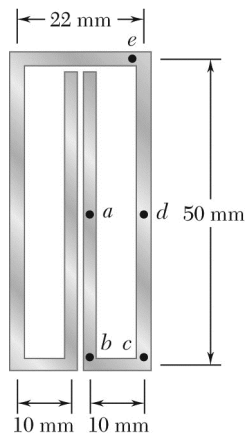
(b) Maximum shearing occurs at point *M*, 24 mm above the bottom

$$Q_M = Q_A + (4 \sec \alpha)(24)(12) = 2.4 \times 10^3 + 1.248 \times 10^3 = 3.648 \times 10^3 \text{ mm}^3 \\ = 3.648 \times 10^{-6} \text{ m}^3$$

$$\tau_M = \frac{VQ_M}{It} = \frac{(12 \times 10^3)(3.648 \times 10^{-6})}{(310.8 \times 10^{-9})(4 \times 10^{-3})} = 35.2 \times 10^6 \text{ Pa} \quad \tau_M = 35.2 \text{ MPa} \quad \blacktriangleleft$$

$$Q_B = Q_A \quad \tau_B = \tau_A = 23.2 \text{ MPa}$$





PROBLEM 6.49

A plate of 2-mm thickness is bent as shown and then used as a beam. For a vertical shear of 5 kN, determine the shearing stress at the five points indicated and sketch the shear flow in the cross section.

SOLUTION

$$I = 2 \left[\frac{1}{12} (2)(48)^3 + \frac{1}{12} (2)(52)^3 + \frac{1}{12} (20)(2)^3 + (20)(2)(25)^2 \right]$$

$$= 133.75 \times 10^3 \text{ mm}^4 = 133.75 \times 10^{-9} \text{ m}^4$$

$$Q_a = (2)(24)(12) = 576 \text{ mm}^3 = 576 \times 10^{-9} \text{ m}^3$$

$$Q_b = 0$$

$$Q_c = Q_b - (12)(2)(25) = -600 \text{ mm}^3 = -600 \times 10^{-9} \text{ m}^3$$

$$Q_d = Q_c - (2)(24)(12) = -1.176 \times 10^3 \text{ mm}^3 = -1.176 \times 10^{-6} \text{ m}^3$$

$$Q_e = Q_d + (2)(26)(13) = -600 \text{ mm}^3 = -500 \times 10^{-9} \text{ m}^3$$

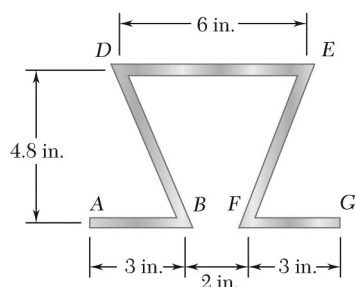
$$\tau_a = \frac{VQ_a}{It} = \frac{(5 \times 10^3)(576 \times 10^{-9})}{(133.75 \times 10^{-9})(2 \times 10^{-3})} = 10.77 \times 10^6 \text{ Pa} \quad \tau_a = 10.76 \text{ MPa} \quad \blacktriangleleft$$

$$\tau_b = \frac{VQ_b}{It} \quad \tau_b = 0 \quad \blacktriangleleft$$

$$\tau_c = \frac{VQ_c}{It} = \frac{(5 \times 10^3)(600 \times 10^{-9})}{(133.75 \times 10^{-9})(2 \times 10^{-3})} = 11.21 \times 10^6 \text{ Pa} \quad \tau_c = 11.21 \text{ MPa} \quad \blacktriangleleft$$

$$\tau_d = \frac{VQ_d}{It} = \frac{(5 \times 10^3)(1.176 \times 10^{-6})}{(133.75 \times 10^{-9})(2 \times 10^{-3})} = 22.0 \times 10^6 \text{ Pa} \quad \tau_d = 22.0 \text{ MPa} \quad \blacktriangleleft$$

$$\tau_e = \frac{VQ_e}{It} = \frac{(5 \times 10^3)(500 \times 10^{-9})}{(133.75 \times 10^{-9})(2 \times 10^{-3})} = 9.35 \times 10^6 \text{ Pa} \quad \tau_e = 9.35 \text{ MPa} \quad \blacktriangleleft$$



PROBLEM 6.50

A plate of thickness t is bent as shown and then used as a beam. For a vertical shear of 600 lb, determine (a) the thickness t for which the maximum shearing stress is 300 psi, (b) the corresponding shearing stress at point E. Also, sketch the shear flow in the cross section.

SOLUTION

$$L_{BD} = L_{EF} = \sqrt{4.8^2 + 2^2} = 5.2 \text{ in.}$$

Neutral axis lies at 2.4 in. above AB.

Calculate I .

$$I_{AB} = (3t)(2.4)^2 = 17.28t$$

$$I_{BD} = \frac{1}{12}(5.2t)(4.8)^2 = 9.984t$$

$$I_{DE} = (6t)(2.4)^2 = 34.56t$$

$$I_{EF} = I_{DB} = 9.984t$$

$$I_{FG} = I_{AB} = 17.28t$$

$$I = \Sigma I = 89.09t$$

(a) At point C, $Q_C = Q_{AB} + Q_{BC} = (3t)(2.4) + (2.6t)(1.2) = 10.32t$

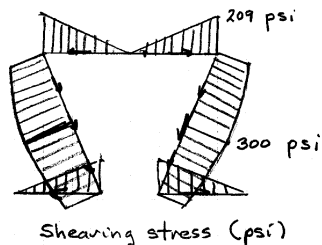
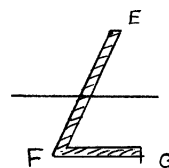
$$\tau = \frac{VQ_C}{It} \quad \therefore \quad t = \frac{VQ}{\tau I} = \frac{(600)(10.32t)}{(300)(89.09t)} = 0.23168 \text{ in.} \quad t = 0.232 \text{ in.} \quad \blacktriangleleft$$

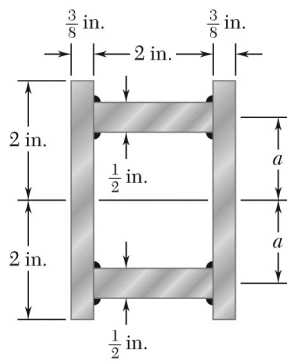
(b) $I = (89.09)(0.23168) = 20.64 \text{ in}^4$

$$Q_E = Q_{EF} + Q_{FG}$$

$$= 0 + (3)(0.23168)(2.4) = 1.668 \text{ in}^3$$

$$\tau_E = \frac{VQ_E}{It} = \frac{(600)(1.668)}{(20.64)(0.23168)} \quad \tau_E = 209 \text{ psi} \quad \blacktriangleleft$$





PROBLEM 6.51

The design of a beam calls for connecting two vertical rectangular $\frac{3}{8} \times 4$ -in. plates by welding them to horizontal $\frac{1}{2} \times 2$ -in. plates as shown. For a vertical shear V , determine the dimension a for which the shear flow through the welded surface is maximum.

SOLUTION

$$I = (2) \left(\frac{1}{12} \right) \left(\frac{3}{8} \right) (4)^3 + (2) \left(\frac{1}{12} \right) (2) \left(\frac{1}{2} \right)^3 + (2)(2) \left(\frac{1}{2} \right) a^2$$

$$= 4.041667 + 2a^2 \text{ in}^4$$

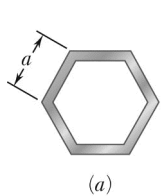
$$Q = (2) \left(\frac{1}{2} \right) a = a \text{ in}^3$$

$$q = \frac{VQ}{I} = \frac{Va}{4.041667 + 2a^2} \quad \text{Set } \frac{dq}{da} = 0.$$

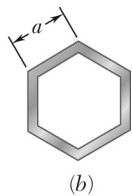
$$\frac{dq}{da} = \left[\frac{(4.041667 + 2a^2) - (a)(4a)}{(4.041667 + 2a^2)^2} \right] V = 0$$

$$2a^2 = 4.041667$$

$$a = 1.422 \text{ in.} \quad \blacktriangleleft$$



(a)

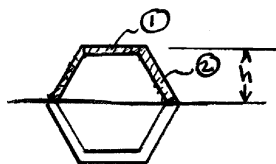


(b)

PROBLEM 6.52

An extruded beam has a uniform wall thickness t . Denoting by V the vertical shear and by A the cross-sectional area of the beam, express the maximum shearing stress as $\tau_{\max} = k(V/A)$ and determine the constant k for each of the two orientations shown.

SOLUTION



$$(a) \quad h = \frac{\sqrt{3}}{2}a$$

$$A_1 = A_2 = at$$

$$I_1 = A_1 h^2 = at h^2 = \frac{3}{4}a^3 t$$

$$I_2 = \frac{1}{3}A_2 h^2 = \frac{1}{3}at \frac{3}{4}a^2 = \frac{1}{4}a^3 t$$

$$I = 2I_1 + 4I_2 = \frac{5}{2}a^3 t$$

$$Q_1 = A_1 h = \frac{\sqrt{3}}{2}a^2 t$$

$$Q_2 = A_2 \frac{h}{2} = \frac{\sqrt{3}}{4}a^2 t$$

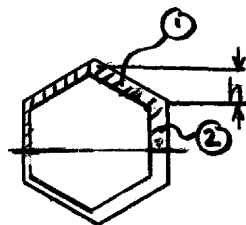
$$Q_m = Q_1 + 2Q_2 = \sqrt{3}a^2 t$$

$$\tau_m = \frac{VQ}{I(2t)} = \frac{V\sqrt{3}a^2 t}{\left(\frac{5}{2}a^3 t\right)(2t)} = \frac{\sqrt{3}}{5} \frac{V}{at}$$

$$= \frac{6\sqrt{3}}{5} \frac{V}{6at} = \frac{6\sqrt{3}}{5} \frac{V}{A} = k \frac{V}{A}$$

$$k = \frac{6\sqrt{3}}{5}$$

$$k = 2.08 \blacktriangleleft$$



$$(b) \quad h = \frac{a}{2}$$

$$A_1 = at$$

$$A_2 = \frac{1}{2}at$$

$$I_1 = \bar{I}_1 + A_1 d^2$$

$$= \frac{1}{12}ath^2 + at\left(\frac{a}{2} + \frac{h}{2}\right)^2$$

$$= \frac{1}{48}a^3 t + \frac{9}{16}a^3 t = \frac{7}{12}a^3 t$$

$$I_2 = \frac{1}{3}t\left(\frac{a}{2}\right)^3 = \frac{1}{24}a^3 t$$

$$I = 4I_1 + 4I_2 = \frac{5}{2}a^3 t$$

$$Q_1 = at\left(\frac{a}{2} + \frac{h}{2}\right) = \frac{3}{4}a^2 t$$

$$Q_2 = \left(\frac{1}{2}at\right)\left(\frac{a}{4}\right) = \frac{1}{8}a^2 t$$

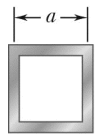
$$Q = 2Q_1 + 2Q_2 = \frac{7}{4}a^3 t$$

$$\tau_m = \frac{VQ}{I(2t)} = \frac{V \cdot \frac{7}{4}a^3 t}{\left(\frac{5}{2}a^3 t\right)(2t)}$$

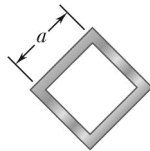
$$= \frac{7}{20} \frac{V}{at} = \frac{42}{20} \frac{V}{6at} = \frac{21}{10} \frac{V}{A}$$

$$= k \frac{V}{A}$$

$$k = \frac{21}{10} = 2.10 \blacktriangleleft$$



(a)

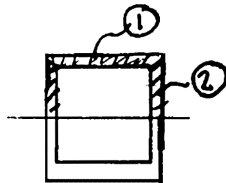


(b)

PROBLEM 6.53

An extruded beam has a uniform wall thickness t . Denoting by V the vertical shear and by A the cross-sectional area of the beam, express the maximum shearing stress as $\tau_{\max} = k(V/A)$ and determine the constant k for each of the two orientations shown.

SOLUTION



$$(a) \quad I_1 = (at) \left(\frac{a}{2} \right)^2 = \frac{1}{4} a^3 t$$

$$I_2 = \frac{1}{3} t \left(\frac{a}{2} \right)^3 = \frac{1}{24} a^3 t$$

$$I = 2I_1 + 4I_2 = \frac{2}{3} a^3 t$$

$$Q_1 = (at) \left(\frac{a}{2} \right) = \frac{1}{2} a^2 t$$

$$Q_2 = \left(\frac{1}{2} at \right) \left(\frac{a}{4} \right) = \frac{1}{8} a^2 t$$

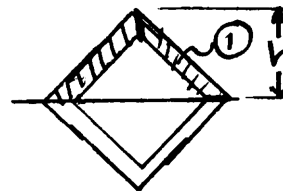
$$Q = Q_1 + 2Q_2 = \frac{3}{4} a^2 t$$

$$\tau_{\max} = \frac{VQ}{I(2t)} = \frac{V \left(\frac{3}{4} a^2 t \right)}{\left(\frac{2}{3} a^3 t \right) (2t)}$$

$$= \frac{9}{16} \frac{V}{at} = \frac{9}{4} \frac{V}{4at} = \frac{9}{4} \frac{V}{A}$$

$$= k \frac{V}{A}$$

$$k = \frac{9}{4} = 2.25 \quad \blacktriangleleft$$



$$(b) \quad h = \frac{1}{2} \sqrt{2} a$$

$$I_1 = \frac{1}{3} A_1 h^2 = \left(\frac{1}{3} at \right) \left(\frac{\sqrt{2}}{2} a \right)^2$$

$$= \frac{1}{6} a^3 t$$

$$I = 4I_1 = \frac{2}{3} a^3 t$$

$$Q_1 = at \left(\frac{h}{2} \right) = \frac{1}{4} \sqrt{2} a^2 t$$

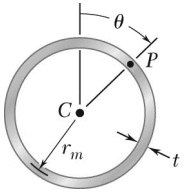
$$Q = 2Q_1 = \frac{1}{2} \sqrt{2} a^2 t$$

$$\tau_{\max} = \frac{VQ}{I(2t)} = \frac{V \left(\frac{1}{2} \sqrt{2} a^2 t \right)}{\left(\frac{2}{3} a^3 t \right) (2t)}$$

$$= \frac{3\sqrt{2}}{8} \frac{V}{at} = \frac{3\sqrt{2}}{2} \frac{V}{4at}$$

$$= \frac{3\sqrt{2}}{2} \frac{V}{A} = k \frac{V}{A}$$

$$k = \frac{3\sqrt{2}}{2} = 2.12 \quad \blacktriangleleft$$



PROBLEM 6.54

(a) Determine the shearing stress at point P of a thin-walled pipe of the cross section shown caused by a vertical shear V . (b) Show that the maximum shearing stress occurs for $\theta = 90^\circ$ and is equal to $2V/A$, where A is the cross-sectional area of the pipe.

SOLUTION

$$A = 2\pi r_m t \quad J = Ar_m^2 = 2\pi r_m^3 t \quad I = \frac{1}{2}J = \pi r_m^3 t$$

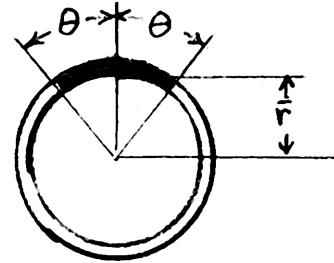
$$\bar{r} = \frac{\sin \theta}{\theta} \text{ for a circular arc.}$$

$$A_p = 2r\theta t$$

$$Q_p = A_p \bar{r} = 2rt \sin \theta$$

$$(a) \quad \tau_p = \frac{VQ_p}{I(2t)} = \frac{(V)(2rt \sin \theta)}{(\pi r_m^3 t)(2t)}$$

$$(b) \quad \tau_m = \frac{2V \sin \frac{\pi}{2}}{2\pi r_m t}$$



$$\tau_p = \frac{V \sin \theta}{\pi r_m t} \quad \blacktriangleleft$$

$$\tau_m = \frac{2V}{A} \quad \blacktriangleleft$$

PROBLEM 6.55

For a beam made of two or more materials with different moduli of elasticity, show that Eq. (6.6)

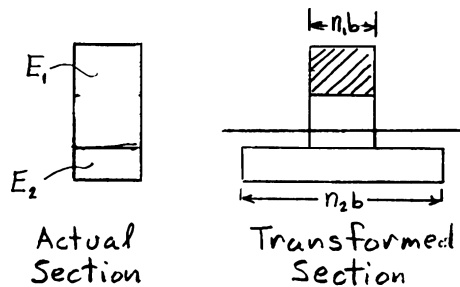
$$\tau_{ave} = \frac{VQ}{It}$$

remains valid provided that both Q and I are computed by using the transformed section of the beam (see Sec. 4.6), and provided further that t is the actual width of the beam where τ_{ave} is computed.

SOLUTION

Let E_{ref} be a reference modulus of elasticity.

$$n_1 = \frac{E_1}{E_{ref}}, \quad n_2 = \frac{E_2}{E_{ref}}, \quad \text{etc.}$$



Widths b of actual section are multiplied by n 's to obtain the transformed section. The bending stress distribution in the cross section is given by

$$\sigma_x = -\frac{nMy}{I}$$

where I is the moment of inertia of the transformed cross section and y is measured from the centroid of the transformed section.

The horizontal shearing force over length Δx is

$$\Delta H = -\int (\Delta \sigma_x) dA = \int \frac{n(\Delta M)y}{I} dA = \frac{(\Delta M)}{I} \int ny dA = \frac{Q(\Delta M)}{I}$$

$$Q = \int ny dA = \text{first moment of transformed section.}$$

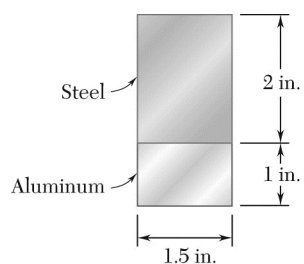
Shear flow:

$$q = \frac{\Delta H}{\Delta x} = \frac{\Delta M}{\Delta x} \frac{Q}{I} = \frac{VQ}{I}$$

q is distributed over actual width t , thus

$$\tau = \frac{q}{t}.$$

$$\tau = \frac{VQ}{It}$$



PROBLEM 6.56

A steel bar and an aluminum bar are bonded together as shown to form a composite beam. Knowing that the vertical shear in the beam is 4 kips and that the modulus of elasticity is 29×10^6 psi for the steel and 10.6×10^6 psi for the aluminum, determine (a) the average stress at the bonded surface, (b) the maximum shearing stress in the beam. (*Hint:* Use the method indicated in Prob. 6.55.)

SOLUTION

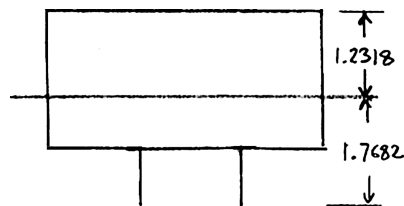
$n = 1$ in aluminum.

$$n = \frac{29 \times 10^6 \text{ psi}}{10.6 \times 10^6 \text{ psi}} = 2.7358 \text{ in steel.}$$

Part	$nA(\text{in}^2)$	$\bar{y}(\text{in.})$	$nA\bar{y}(\text{in}^3)$	$d(\text{in.})$	$nAd^2(\text{in}^2)$	$n\bar{I}(\text{in}^4)$
Steel	8.2074	2.0	16.4148	0.2318	0.4410	2.7358
Alum.	1.5	0.5	0.75	1.2682	2.4125	0.1250
Σ	9.7074		17.1648		2.8535	2.8608

$$\bar{Y} = \frac{\Sigma nA\bar{y}}{\Sigma A} = \frac{17.1648}{9.7074} = 1.7682 \text{ in.}$$

$$I = \Sigma nAd^2 + \Sigma n\bar{I} = 5.7143 \text{ in}^4$$

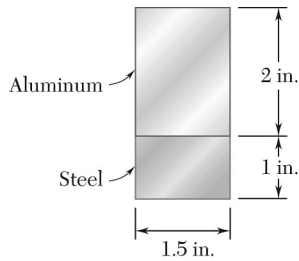


(a) At the bonded surface, $Q = (1.5)(1.2682) = 1.9023 \text{ in}^3$

$$\tau = \frac{VQ}{It} = \frac{(4)(1.9023)}{(5.7143)(1.5)} \quad \tau = 0.888 \text{ ksi} \quad \blacktriangleleft$$

(b) At the neutral axis, $Q = (2.7358)(1.5)(1.2318) \left(\frac{1.2318}{2} \right) = 3.1133 \text{ in}^3$

$$\tau_{\max} = \frac{VQ}{It} = \frac{(4)(3.1133)}{(5.7143)(1.5)} \quad \tau_{\max} = 1.453 \text{ ksi} \quad \blacktriangleleft$$



PROBLEM 6.57

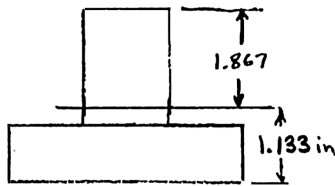
A steel bar and an aluminum bar are bonded together as shown to form a composite beam. Knowing that the vertical shear in the beam is 4 kips and that the modulus of elasticity is 29×10^6 psi for the steel and 10.6×10^6 psi for the aluminum, determine (a) the average stress at the bonded surface, (b) the maximum shearing stress in the beam. (*Hint:* Use the method indicated in Prob. 6.55.)

SOLUTION

$n = 1$ in aluminum.

$$n = \frac{29 \times 10^6 \text{ psi}}{10.6 \times 10^6 \text{ psi}} = 2.7358 \text{ in steel.}$$

Part	$nA(\text{in}^2)$	$\bar{y}(\text{in.})$	$nA\bar{y}(\text{in}^3)$	$d(\text{in.})$	$nAd^2(\text{in}^2)$	$n\bar{I}(\text{in}^4)$
Alum.	3.0	2.0	6.0	0.8665	2.2525	1.0
Steel	4.1038	0.5	2.0519	0.6335	1.6469	0.3420
Σ	7.1038		8.0519		3.8994	1.3420



$$\bar{Y} = \frac{\Sigma nA\bar{y}}{\Sigma nA} = \frac{8.0519}{7.1038} = 1.1335 \text{ in.}$$

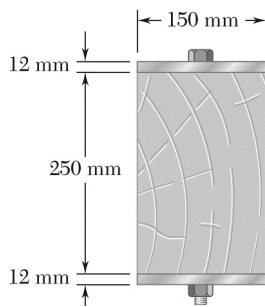
$$I = \Sigma nAd^2 + \Sigma n\bar{I} = 5.2414 \text{ in}^4$$

(a) At the bonded surface, $Q = (1.5)(2)(0.8665) = 2.5995 \text{ in}^3$

$$\tau = \frac{VQ}{It} = \frac{(4)(2.5995)}{(5.2414)(1.5)} \quad \tau = 1.323 \text{ ksi} \quad \blacktriangleleft$$

(b) At the neutral axis, $Q = (1.5)(1.8665)\left(\frac{1.8665}{2}\right) = 2.6129 \text{ in}^3$

$$\tau_{\max} = \frac{VQ}{It} = \frac{(4)(2.6129)}{(5.2814)(1.5)} \quad \tau_{\max} = 1.329 \text{ ksi} \quad \blacktriangleleft$$



PROBLEM 6.58

A composite beam is made by attaching the timber and steel portions shown with bolts of 12-mm diameter spaced longitudinally every 200 mm. The modulus of elasticity is 10 GPa for the wood and 200 GPa for the steel. For a vertical shear of 4 kN, determine (a) the average shearing stress in the bolts, (b) the shearing stress at the center of the cross section. (*Hint:* Use the method indicated in Prob. 6.55.)

SOLUTION

Let

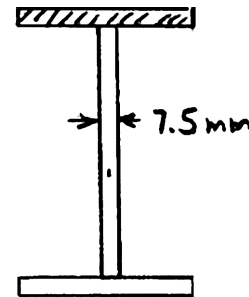
$$E_{\text{ref}} = E_s = 200 \text{ GPa}$$

$$n_s = 1 \quad n_w = \frac{E_w}{E_s} = \frac{10 \text{ GPa}}{200 \text{ GPa}} = \frac{1}{20}$$

Widths of transformed section:

$$b_s = 150 \text{ mm} \quad b_w = \left(\frac{1}{20}\right)(150) = 7.5 \text{ mm}$$

$$\begin{aligned} I &= 2 \left[\frac{1}{12} (150)(12)^3 + (150)(12)(125 + 6)^2 \right] + \frac{1}{12} (7.5)(250)^3 \\ &= 2[0.0216 \times 10^6 + 30.890 \times 10^6] + 9.766 \times 10^6 \\ &= 71.589 \times 10^6 \text{ mm}^4 = 71.589 \times 10^{-6} \text{ m}^4 \end{aligned}$$



$$(a) \quad Q_1 = (150)(12)(125 + 6) = 235.8 \times 10^3 \text{ mm}^3 = 235.8 \times 10^{-6} \text{ m}^3$$

$$q = \frac{VQ_1}{I} = \frac{(4 \times 10^3)(235.8 \times 10^{-6})}{71.589 \times 10^{-6}} = 13.175 \times 10^3 \text{ N/m}$$

$$F_{\text{bolt}} = qs = (23.187 \times 10^3)(200 \times 10^{-3}) = 2.635 \times 10^3 \text{ N}$$

$$A_{\text{bolt}} = \frac{\pi}{4} d_{\text{bolt}}^2 = \left(\frac{\pi}{4}\right)(12)^2 = 113.1 \text{ mm}^2 = 113.1 \times 10^{-6} \text{ m}^2$$

$$\tau_{\text{bolt}} = \frac{F_{\text{bolt}}}{A_{\text{bolt}}} = \frac{2.635 \times 10^3}{113.1 \times 10^{-6}} = 23.3 \times 10^6 \text{ Pa}$$

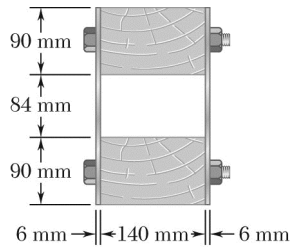
$$\tau_{\text{bolt}} = 23.3 \text{ MPa} \quad \blacktriangleleft$$

$$(b) \quad Q_2 = Q_1 + (7.5)(125)(62.5) = 235.8 \times 10^3 + 58.594 \times 10^3 = 294.4 \times 10^3 \text{ mm}^3 = 294.4 \times 10^{-6} \text{ m}^3$$

$$t = 150 \text{ mm} = 150 \times 10^{-3} \text{ m}$$

$$\tau_c = \frac{VQ_2}{It} = \frac{(4 \times 10^3)(294.4 \times 10^{-6})}{(71.589 \times 10^{-6})(150 \times 10^{-3})} = 109.7 \times 10^3 \text{ Pa}$$

$$\tau_c = 109.7 \text{ kPa} \quad \blacktriangleleft$$



PROBLEM 6.59

A composite beam is made by attaching the timber and steel portions shown with bolts of 12-mm diameter spaced longitudinally every 200 mm. The modulus of elasticity is 10 GPa for the wood and 200 GPa for the steel. For a vertical shear of 4 kN, determine (a) the average shearing stress in the bolts, (b) the shearing stress at the center of the cross section. (Hint: Use the method indicated in Prob. 6.55.)

SOLUTION

Let steel be the reference material.

$$n_s = 1.0 \quad n_w = \frac{E_w}{E_s} = \frac{10 \text{ GPa}}{200 \text{ GPa}} = 0.05$$

Depth of section: $d = 90 + 84 + 90 = 264 \text{ mm}$

For steel portion: $I_s = 2 \frac{1}{12} b d^3 = (2) \left(\frac{1}{12} \right) (6)(264)^3 = 18.400 \times 10^6 \text{ mm}^4$

For the wooden portion: $I_w = \frac{1}{12} b (d_1^3 - d_2^3) = \frac{1}{12} (140)(264^3 - 84^3) = 207.75 \times 10^6 \text{ mm}^4$

For the transformed section: $I = n_s I_s + n_w I_w$

$$I = (1.0)(18.400 \times 10^6) + (0.05)(207.75 \times 10^6) = 28.787 \times 10^6 \text{ mm}^4 = 28.787 \times 10^{-6} \text{ m}^4$$

(a) Shearing stress in the bolts.

For the upper wooden portion $Q_w = (90)(140)(42 + 45) = 1.0962 \times 10^6 \text{ mm}^3$

For the transformed wooden portion

$$Q = n_w Q_w = (0.05)(1.0962 \times 10^6) = 54.81 \times 10^3 \text{ mm}^3 = 54.81 \times 10^{-6} \text{ m}^3$$

Shear flow on upper wooden portion.

$$q = \frac{VQ}{I} = \frac{(4000)(54.81 \times 10^{-6})}{28.787 \times 10^{-6}} = 7616 \text{ N/m}$$

$$F_{\text{bolt}} = qs = (7616)(0.200) = 1523.2 \text{ N}$$

$$A_{\text{bolt}} = \frac{\pi}{4} d_{\text{bolt}}^2 = \frac{\pi}{4} (12)^2 = 113.1 \text{ mm}^2 = 113.1 \times 10^{-6} \text{ m}^2$$

Double shear: $\tau_{\text{bolt}} = \frac{F_{\text{bolt}}}{2A_{\text{bolt}}} = \frac{1523.2}{(2)(113.1 \times 10^{-6})}$

$$= 6.73 \times 10^6 \text{ Pa}$$

$$\tau_{\text{bolt}} = 6.73 \text{ MPa} \quad \blacktriangleleft$$

PROBLEM 6.59 (Continued)

(b) Shearing stress at the center of the cross section.

For two steel plates $Q_s = (2)(6)(90 + 42)(90 - 42) = 76.032 \times 10^3 \text{ mm}^3 = 76.032 \times 10^{-6} \text{ m}^3$

For the neutral axis $Q = 54.81 \times 10^{-6} + 76.032 \times 10^{-6} = 130.842 \times 10^{-6} \text{ m}^3$

Shear flow across the neutral axis

$$q = \frac{VQ}{I} = \frac{(4000)(130.842 \times 10^{-6})}{28.787 \times 10^{-6}} = 18.181 \times 10^3 \text{ N/m}$$

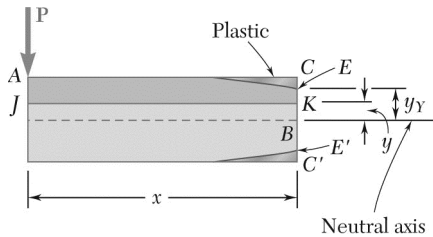
Double thickness $2t = 12 \text{ mm} = 0.012 \text{ m}$

Shearing stress $\tau = \frac{q}{2t} = \frac{18.181 \times 10^3}{0.012} = 1.515 \times 10^6 \text{ Pa}$

$$\tau = 1.515 \text{ MPa} \quad \blacktriangleleft$$

PROBLEM 6.60

Consider the cantilever beam AB discussed in Sec. 6.8 and the portion $ACKJ$ of the beam that is located to the left of the transverse section CC' and above the horizontal plane JK , where K is a point at a distance $y < y_Y$ above the neutral axis. (See Figure). (a) Recalling that $\sigma_x = \sigma_Y$ between C and E and $\sigma_x = (\sigma_Y/y_Y)y$ between E and K , show that the magnitude of the horizontal shearing force H exerted on the lower face of the portion of beam $ACKJ$ is



$$H = \frac{1}{2} b \sigma_Y \left(2c - y_Y - \frac{y^2}{y_Y} \right)$$

(b) Observing that the shearing stress at K is

$$\tau_{xy} = \lim_{\Delta A \rightarrow 0} \frac{\Delta H}{\Delta A} = \lim_{\Delta x \rightarrow 0} \frac{1}{b} \frac{\Delta H}{\Delta x} = \frac{1}{b} \frac{\partial H}{\partial x}$$

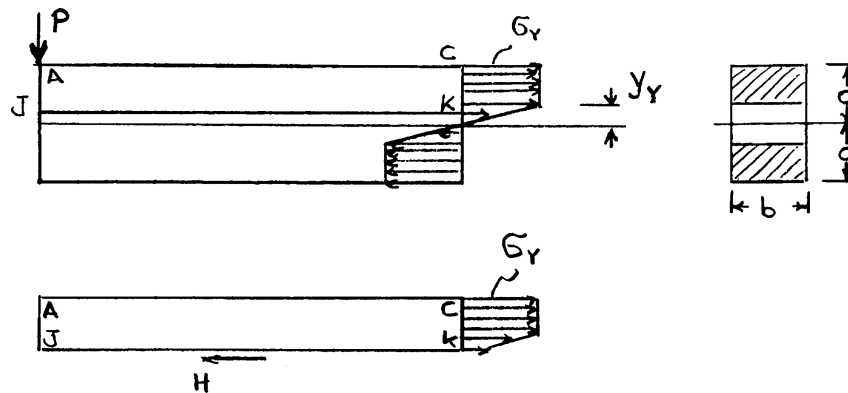
and recalling that y_Y is a function of x defined by Eq. (6.14), derive Eq. (6.15).

SOLUTION

Point K is located a distance y above the neutral axis.

The stress distribution is given by

$$\sigma = \sigma_Y \frac{y}{y_Y} \quad \text{for } 0 \leq y < y_Y \quad \text{and} \quad \sigma = \sigma_Y \quad \text{for } y_Y \leq y \leq c.$$



PROBLEM 6.60 (Continued)

For equilibrium of horizontal forces acting on $ACKJ$,

$$\begin{aligned}
 H &= \int \sigma dA = \int_y^{y_Y} \frac{\sigma_Y y b}{y_Y} dy + \int_{y_Y}^c \sigma_Y b dy \\
 &= \frac{\sigma_Y b}{y_Y} \left(\frac{y_Y^2 - y^2}{2} \right) + \sigma_Y b (c - y_Y) \\
 H &= \frac{1}{2} b \sigma_Y \left(2c - y_Y - \frac{y^2}{y_Y} \right) \quad \blacktriangleleft (a)
 \end{aligned}$$

Note that y_Y is a function of x .

$$\begin{aligned}
 \tau_{xy} &= \frac{1}{b} \frac{\partial H}{\partial x} = \frac{1}{2} \sigma_Y \left(-\frac{\partial y_Y}{\partial x} + \frac{y^2}{y_Y^2} \frac{dy_Y}{dx} \right) \\
 &= -\frac{1}{2} \sigma_Y \left(1 - \frac{y^2}{y_Y^2} \right) \frac{dy_Y}{dx}
 \end{aligned}$$

But

$$M = Px = \frac{3}{2} M_y \left(1 - \frac{1}{3} \frac{y_Y^2}{c^2} \right)$$

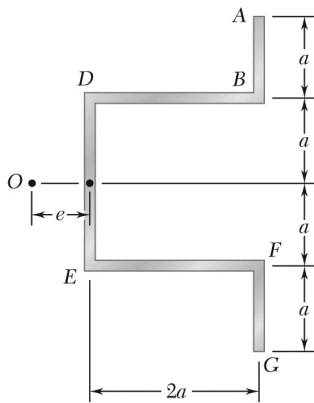
Differentiating,

$$\frac{dM}{dx} = P = \frac{3}{2} M_y \left(-\frac{2}{3} \frac{y_Y}{c^2} \frac{dy_Y}{dx} \right)$$

$$\frac{dy_Y}{dx} = -\frac{P c^2}{y_Y M_y} = -\frac{P c^2}{y_Y \frac{2}{3} \sigma_Y b c^2} = -\frac{3}{2} \frac{P}{\sigma_Y b y_Y}$$

Then

$$\tau_{xy} = \frac{1}{2} \sigma_Y \left(1 - \frac{y^2}{y_Y^2} \right) \frac{3}{2} \frac{P}{\sigma_Y b \sigma_Y} = \frac{3P}{4b y_Y} \left(1 - \frac{y^2}{y_Y^2} \right) \quad \blacktriangleleft (b)$$



PROBLEM 6.61

Determine the location of the shear center O of a thin-walled beam of uniform thickness having the cross section shown.

SOLUTION

$$I_{AB} = I_{FG} = \frac{1}{12}ta^3 + (ta)\left(\frac{3a}{2}\right)^2 = \frac{7}{3}ta^3$$

$$I_{DB} = I_{EF} = (2at)a^2 + \frac{1}{12}(2a)t^3 \approx 2a^3t$$

$$I_{DE} = \frac{1}{12}t(2a)^3 = \frac{2}{3}ta^3 \quad I = \Sigma I = \frac{28}{3}ta^3$$

Part AB:

$$A = t(2a - y); \quad \bar{y} = \frac{2a + y}{3}$$

$$Q = A\bar{y} = \frac{1}{2}t(2a - y)(2a + y)$$

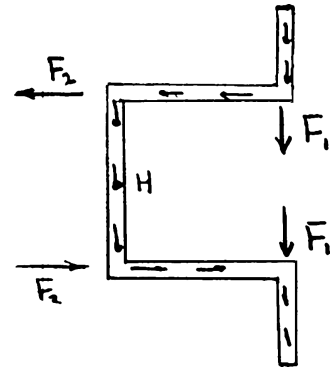
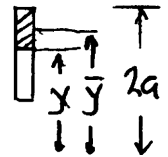
$$= \frac{1}{2}t(4a^2 - y^2)$$

$$\tau = \frac{VQ}{It} = \frac{V}{2I}(4a^2 - y^2)$$

$$F_1 = \int \tau dA = \int_a^{2a} \frac{V}{2I}(4a^2 - y^2)tdy$$

$$= \frac{Vt}{2I} \left(4a^2y - \frac{y^3}{3} \right) \Big|_a^{2a} = \frac{Vta^3}{2I} \left[(4)(2) - \frac{(2)^3}{3} - (4)(1) + \left(\frac{1}{3} \right) \right]$$

$$= \frac{5}{6} \frac{Vta^3}{I} = \frac{5}{56}V$$



PROBLEM 6.61 (Continued)

Part DB:

$$Q = (ta) \frac{3a}{2} + txa$$

$$= ta \left(\frac{3a}{2} + x \right)$$

$$\tau = \frac{VQ}{It} = \frac{Va}{I} \left(\frac{3a}{2} + x \right)$$

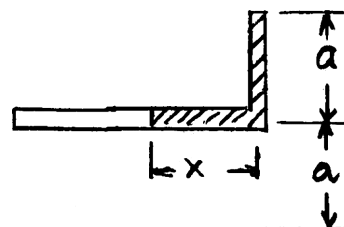
$$F_2 = \int \tau dA = \int_0^{2a} \frac{Va}{I} \left(\frac{3a}{2} + x \right) t dx = \frac{Vta}{I} \int_0^{2a} \left(\frac{3a}{2} + x \right) dx$$

$$= \frac{Vta}{I} \left(\frac{3ax}{2} + \frac{x^2}{2} \right) \Big|_0^{2a} = \frac{Vta^3}{I} \left[\frac{(3)(2)}{2} + \frac{(2)^2}{2} \right]$$

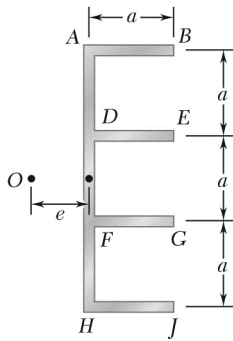
$$= 5 \frac{Vta^3}{I} = \frac{15}{28} V$$

$$+\circlearrowleft \Sigma M_H = +\circlearrowleft \Sigma M_H: \quad Ve = F_2(2a) - 2F_1(2a)$$

$$= \frac{30}{28} Va - \frac{20}{56} Va = \frac{5}{7} Va$$



$$e = \frac{5}{7} a = 0.714a \quad \blacktriangleleft$$



PROBLEM 6.62

Determine the location of the shear center O of a thin-walled beam of uniform thickness having the cross section shown.

SOLUTION

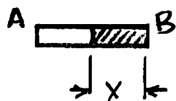
$$I_{AB} = I_{HJ} = at \left(\frac{3a}{2} \right)^2 + \frac{1}{12} at^3 \approx \frac{9}{4} ta^3$$

$$I_{DE} = I_{FG} = at \left(\frac{a}{2} \right)^2 + \frac{1}{12} at^3 \approx \frac{1}{4} ta^3$$

$$I_{AH} = \frac{1}{12} t(3a)^3 = \frac{9}{4} ta^3$$

$$I = \Sigma I = \frac{29}{4} ta^3$$

Part AB:

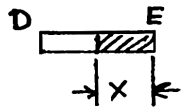


$$A = tx \quad \bar{y} = \frac{3a}{2} \quad Q = \frac{3}{2} atx$$

$$\tau = \frac{VQ}{It} = \frac{V \cdot \frac{3}{2} atx}{\frac{29}{4} ta^3 t} = \frac{6Vx}{29a^2 t}$$

$$F_1 = \int \tau dA = \int_0^a \frac{6Vx}{29a^2 t} t dx = \frac{6V}{29a^2} \int_0^a x dx = \frac{3}{29} V$$

Part DE:



$$A = tx \quad \bar{y} = \frac{a}{2} \quad Q = \frac{1}{2} atx$$

$$\tau = \frac{VQ}{It} = \frac{V \cdot \frac{1}{2} atx}{\frac{29}{4} ta^3 t} = \frac{2Vx}{29a^2 t}$$

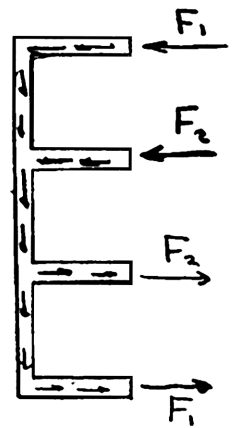
$$F_2 = \int \tau dA = \int_0^a \frac{2Vx}{29a^2 t} t dx = \frac{2V}{29a^2} \int_0^a x dx = \frac{1}{29} V$$

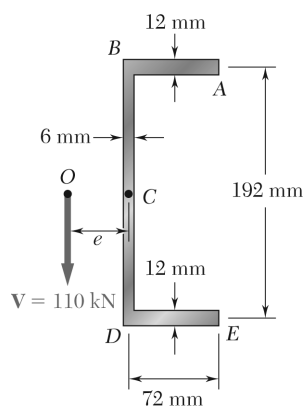
$$+\circlearrowleft \Sigma M_K = +\circlearrowleft \Sigma M_K: \quad Ve = F_1(3a) + F_2(a)$$

$$= \frac{9}{29} Va + \frac{1}{29} Va = \frac{10}{29} Va$$

$$e = \frac{10}{29} a$$

$$e = 0.345a \quad \blacktriangleleft$$





PROBLEM 6.63

An extruded beam has the cross section shown. Determine (a) the location of the shear center O , (b) the distribution of the shearing stresses caused by the vertical shearing force V shown applied at O .

SOLUTION

$$I = 2 \left[\left(\frac{1}{12} \right) (72)(12)^3 + (72)(12) \left(\frac{192}{2} \right)^2 \right] + \frac{1}{12} (6)(192)^3$$

$$= 19.4849 \times 10^6 \text{ mm}^4 = 19.4849 \times 10^{-6} \text{ m}^4$$

Part AB : $A = 12x$ $Q = A\bar{y} = (12x) \left(\frac{192}{2} \right) = 1152x$

$$q = \frac{VQ}{I} = \frac{1152Vx}{I}$$

$x = 0$ at point A . $x = l_{AB} = 72 \text{ mm}$ at point B .

$$F_1 = \int_{x_A}^{x_B} q dx = \int_0^{72} \frac{1152Vx}{I} dx = \frac{1152V}{I} \frac{(72)^2}{2}$$

$$= \frac{(576)(72)^2}{19.4849 \times 10^6} V = 0.153246V$$

$$+\circlearrowleft \Sigma M_C = +\circlearrowleft M_C: Ve = (0.153246V)(192)$$

(a) $e = 29.423 \text{ mm}$

(b) Point A : $x = 0$ $Q = 0$, $q = 0$

Point B in part AB : $x = 72 \text{ mm}$

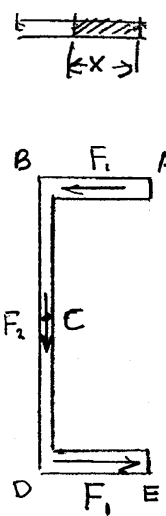
$$Q = (1152)(72) = 82.944 \times 10^3 \text{ mm}^3 = 82.944 \times 10^{-6} \text{ m}^3$$

$$t = 12 \text{ mm} = 0.012 \text{ m}$$

$$\tau_B = \frac{VQ}{It} = \frac{(110 \times 10^3)(82.944 \times 10^{-6})}{(19.4849 \times 10^6)(0.012)}$$

$$= 39.0 \times 10^6 \text{ Pa}$$

$$\tau_B = 39.0 \text{ MPa in } AB \blacktriangleleft$$



$$e = 29.4 \text{ mm} \blacktriangleleft$$

$$\tau_A = 0 \blacktriangleleft$$

PROBLEM 6.63 (Continued)

Part *BD*:

Point *B*:

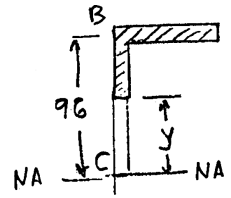
$$y = 96 \text{ mm} \quad Q = 82.944 \times 10^3 \text{ mm}^3 = 82.944 \times 10^{-6} \text{ m}^3$$

$$t = 6 \text{ mm} = 0.006 \text{ m}$$

$$\tau_B = \frac{VQ}{It} = \frac{(110 \times 10^3)(82.944 \times 10^{-6})}{(19.4849 \times 10^{-6})(0.006)}$$

$$= 78.0 \times 10^6 \text{ Pa}$$

$$\tau_B = 78.0 \text{ MPa in } BD \quad \blacktriangleleft$$



Point *C*:

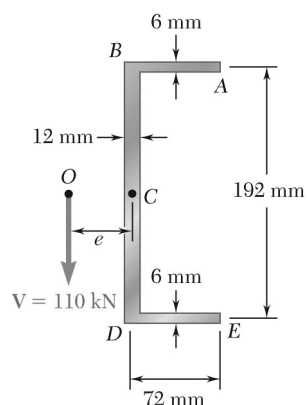
$$y = 0, \quad t = 6 \text{ mm} = 0.006 \text{ m}$$

$$Q = 82.944 \times 10^3 + (6)(96)\left(\frac{96}{2}\right) = 110.592 \times 10^3 \text{ mm}^3 = 110.592 \times 10^{-6} \text{ m}^3$$

$$\tau = \frac{VQ}{It} = \frac{(110.10^3)(110.592 \times 10^{-6})}{(19.4849 \times 10^{-6})(0.006)} = 104.1 \times 10^6 \text{ Pa}$$

$$\tau_C = 104.1 \text{ MPa} \quad \blacktriangleleft$$

PROBLEM 6.64



An extruded beam has the cross section shown. Determine (a) the location of the shear center O , (b) the distribution of the shearing stresses caused by the vertical shearing force V shown applied at O .

SOLUTION

$$I = 2 \left[\left(\frac{1}{12} \right) (72)(6)^3 + (72)(6) \left(\frac{192}{2} \right)^2 \right] + \frac{1}{12} (12)(192)^3$$

$$= 15.0431 \times 10^6 \text{ mm}^4 = 15.0431 \times 10^{-6} \text{ m}^4$$

Part AB: $A = 6x$ $Q = A\bar{y} = (6x) \left(\frac{192}{2} \right) = 576x$

$$q = \frac{VQ}{I} = \frac{576Vx}{I}$$

$x = 0$ at point A. $x = l_{AB} = 72 \text{ mm}$ at point B.

$$F_1 = \int_{x_A}^{x_B} q dx = \int_0^{72} \frac{576Vx}{I} dx = \frac{576V}{I} \frac{(72)^2}{2}$$

$$= \frac{(288)(72)^2}{15.0431 \times 10^6} V = 0.099247V$$

$$+\curvearrowright M_C = +\curvearrowright M_C: Ve = (0.099247)V(192)$$

(a) $e = 19.0555 \text{ mm}$

(b) Point A: $x = 0$ $Q = 0$, $q = 0$

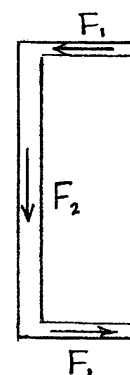
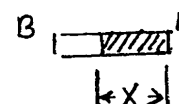
Point B in part AB: $x = 72 \text{ mm}$

$$Q = (576)(72) = 41.472 \times 10^3 \text{ mm}^3 = 41.472 \times 10^{-6} \text{ m}^3$$

$$t = 6 \text{ mm} = 0.006 \text{ m}$$

$$\tau_B = \frac{VQ}{It} = \frac{(110 \times 10^3)(41.472 \times 10^{-6})}{(15.0431 \times 10^{-6})(0.006)}$$

$$= 50.5 \times 10^6 \text{ Pa}$$



$e = 19.06 \text{ mm} \quad \blacktriangleleft$

$\tau_A = 0 \quad \blacktriangleleft$

$\tau_B = 50.5 \text{ MPa} \quad \blacktriangleleft$

PROBLEM 6.64 (Continued)

Part *BD*:

Point *B*: $y = 96 \text{ mm}$ $Q = 41.472 \times 10^3 \text{ mm}^3 = 41.472 \times 10^{-6} \text{ m}^3$

$$t = 12 \text{ mm} = 0.012 \text{ m}$$

$$\tau_B = \frac{VQ}{It} = \frac{(110 \times 10^3)(41.472 \times 10^{-6})}{(15.0431 \times 10^{-6})(0.012)}$$

$$= 25.271 \times 10^6 \text{ Pa}$$

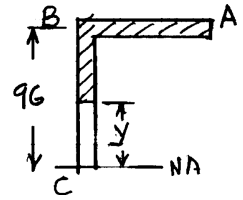
$$\tau_B = 25.3 \text{ MPa} \quad \blacktriangleleft$$

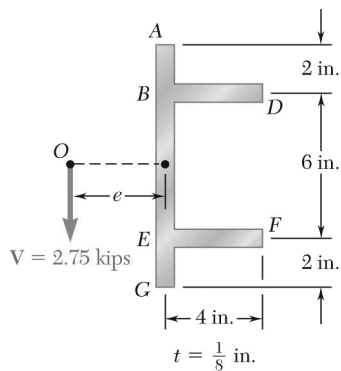
Point *C*: $y = 0$, $t = 0.012 \text{ m}$

$$Q = 41.472 \times 10^3 + (12)(96)\left(\frac{96}{2}\right) = 96.768 \times 10^3 \text{ mm}^3 = 96.768 \times 10^{-6} \text{ m}^3$$

$$\tau_C = \frac{VQ}{It} = \frac{(110 \times 10^3)(96.768 \times 10^{-6})}{(15.0431 \times 10^{-6})(0.012)} = 58.967 \times 10^6 \text{ Pa}$$

$$\tau_C = 59.0 \text{ MPa} \quad \blacktriangleleft$$





PROBLEM 6.65

An extruded beam has the cross section shown. Determine (a) the location of the shear center O , (b) the distribution of the shearing stresses caused by the vertical shearing force V shown applied at O .

SOLUTION

Part	A (in ²)	d (in.)	Ad^2 (in ⁴)	$\bar{I}$ (in ⁴)
BD	0.50	3	4.50	≈ 0
$ABEG$	1.25	0	0	10.417
EF	0.50	3	4.50	≈ 0
Σ	2.25		9.00	10.417

$$I = \Sigma Ad^2 + \Sigma \bar{I} = 19.417 \text{ in}^4$$

(a) Part BD :

$$Q(x) = 3tx$$

$$q(x) = \frac{VQ(x)}{I} = \frac{V}{I}(3tx)$$

$$F_{BD} = \frac{3Vt}{I} \int_0^4 x dx = \frac{3Vt}{I}(8) = \frac{24Vt}{I}$$

Its moment about H :

$$(M_{BD})_H = 3F_{BD} = \frac{72Vt}{I}$$

Part EF : By same method,

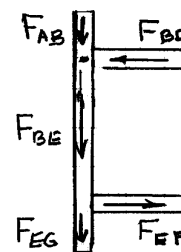
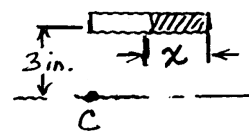
$$F_{EF} = \frac{24Vt}{I} \quad (M_{EF})_H = \frac{72Vt}{I}$$

Moments of F_{AB} , F_{BE} , and F_{EG} about H are zero.

$$Ve = \Sigma M_H = \frac{72Vt}{I} + \frac{72Vt}{I} = \frac{144Vt}{I}$$

$$e = \frac{144t}{I} = \frac{(144)(0.125)}{19.417}$$

$$e = 0.927 \text{ in.} \quad \blacktriangleleft$$



PROBLEM 6.65 (Continued)

(b) At A , D , F , and G : $Q = 0$

$$\tau_A = \tau_D = \tau_F = \tau_G = 0 \quad \blacktriangleleft$$

Just above B :

$$Q_1 = Q_{AB} = (2t)(4) = 8t$$

$$\tau_1 = \frac{VQ_1}{It} = \frac{(2.75)(8t)}{(19.417)t} \quad \tau_1 = 1.133 \text{ ksi} \quad \blacktriangleleft$$

Just to the right of B :

$$Q_2 = Q_{BD} = (3)t(4) = 12t$$

$$\tau_2 = \frac{VQ_2}{It} = \frac{(2.75)(12t)}{(19.417)t} \quad \tau_2 = 1.700 \text{ ksi} \quad \blacktriangleleft$$

Just below B :

$$Q_3 = Q_1 + Q_2 = 20t$$

$$\tau_3 = \frac{VQ_3}{It} = \frac{(2.75)(20t)}{(19.417)t} \quad \tau_3 = 2.83 \text{ ksi} \quad \blacktriangleleft$$

At H (neutral axis):

$$Q_H = Q_3 + Q_{BH} = 20t + t(3)(1.5) = 24.5t$$

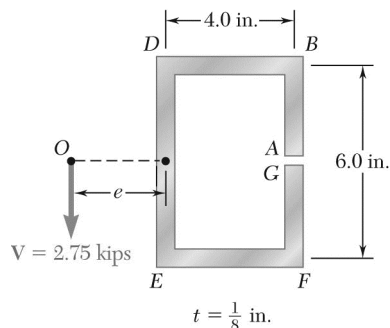
$$\tau_H = \frac{VQ_H}{It} = \frac{(2.75)(24.5t)}{(19.417)t} \quad \tau_H = 3.47 \text{ ksi} \quad \blacktriangleleft$$

By symmetry:

$$\tau_4 = \tau_3 = 2.83 \text{ ksi} \quad \blacktriangleleft$$

$$\tau_5 = \tau_2 = 1.700 \text{ ksi} \quad \blacktriangleleft$$

$$\tau_6 = \tau_1 = 1.133 \text{ ksi} \quad \blacktriangleleft$$



PROBLEM 6.66

An extruded beam has the cross section shown. Determine (a) the location of the shear center O , (b) the distribution of the shearing stresses caused by the vertical shearing force V shown applied at O .

SOLUTION

$$I_{AB} = \frac{1}{3}(0.125)(3)^3 = 1.125 \text{ in}^4$$

$$I_{BD} = \frac{1}{12}(4)(0.125)^3 + (4)(0.125)(3)^2 = 4.50065 \text{ in}^4$$

$$I_{DE} = \frac{1}{12}(0.125)(6)^3 = 2.25 \text{ in}^4$$

$$I_{EF} = I_{BD} = 4.50065 \text{ in}^4$$

$$I_{FG} = I_{AB} = 1.125 \text{ in}^4$$

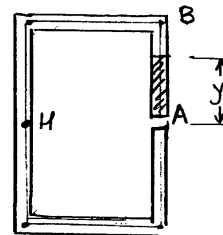
$$I = \Sigma I = 13.50 \text{ in}^4$$

(a) Part AB: $Q(y) = ty \frac{y}{2} = 0.5ty^2$

$$q(y) = \frac{VQ(y)}{I} = \frac{0.5Vt}{I} y^2$$

$$F_{AB} = \int_0^3 q(y) dy = \frac{0.5Vt}{I} \int_0^3 y^2 dy = 4.5 \frac{Vt}{I} \uparrow$$

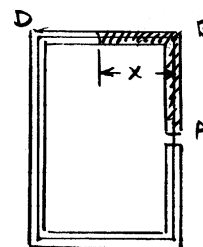
Its moment about H is $4F_{AB} = 18 \frac{Vt}{I} \curvearrowright$



Part BD: $Q(x) = Q_B + xt(3) = (4.5 + 3x)t$

$$q(x) = \frac{Vq(x)}{I} = \frac{Vt}{I} (4.5 + 3x)$$

$$F_{BD} = \int_0^4 q(x) dx = \frac{Vt}{I} \int_0^4 (4.5 + 3x) dx = 42 \frac{Vt}{I} \leftarrow$$



Its moment about H : $3F_{BD} = 126 \frac{Vt}{I} \curvearrowright$

$$Q_D = [4.5 + (3)(4)]t = 16.5t$$

PROBLEM 6.66 (Continued)

Part EF : By symmetry with part BD , $F_{EF} = 42 \frac{Vt}{I} \rightarrow$

Its moment about H is $3 F_{EF} = 126 \frac{Vt}{I} \curvearrowright$

Part FG : By symmetry with part AB , $F_{FG} = 4.5 \frac{VT}{I} \uparrow$

Its moment about H is 4. $F_{FG} = 18 \frac{Vt}{I} \curvearrowright$

Moment about H of force in part DE is zero.

$$Ve = \Sigma M_H = \frac{Vt}{I}(18 + 126 + 0 + 126 + 18) = \frac{144Vt}{I}$$

$$e = \frac{144t}{I} = \frac{(2.88)(0.125)}{13.50}$$

$$e = 2.67 \text{ in.} \quad \blacktriangleleft$$

$$(b) \quad Q_A = Q_G = 0$$

$$\tau_A = \tau_G = 0 \quad \blacktriangleleft$$

$$Q_B = Q_F = 4.5t$$

$$\tau_B = \tau_F = \frac{VQ_B}{It} = \frac{(2.75)(4.5t)}{13.50t}$$

$$\tau_B = \tau_F = 0.917 \text{ ksi} \quad \blacktriangleleft$$

$$Q_D = Q_E = 16.5t$$

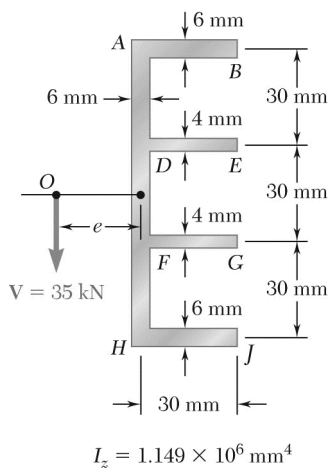
$$\tau_D = \tau_E = \frac{VQ_D}{It} = \frac{(2.75)(16.5t)}{13.50t}$$

$$\tau_D = \tau_E = 3.36 \text{ ksi} \quad \blacktriangleleft$$

At H (neutral axis): $Q_H = Q_D + t(3)(1.5) = 21t$

$$\tau_H = \frac{VQ_H}{It} = \frac{(2.75)(21t)}{13.50t}$$

$$\tau_H = 4.28 \text{ ksi} \quad \blacktriangleleft$$



PROBLEM 6.67

An extruded beam has the cross section shown. Determine (a) the location of the shear center O , (b) the distribution of the shearing stresses caused by the vertical shearing force V shown applied at O .

SOLUTION

$$I_{AB} = I_{HJ} = \frac{1}{12}(30)(6)^3 + (30)(6)(45)^2 = 0.365 \times 10^6 \text{ mm}^4$$

$$I_{DE} = I_{FG} = \frac{1}{12}(30)(4)^3 + (30)(4)(15)^2 = 0.02716 \times 10^6 \text{ mm}^4$$

$$I_{AH} = \frac{1}{12}(6)(90)^3 = 0.3645 \times 10^6 \text{ mm}^4$$

$$I = \Sigma I = 1.14882 \times 10^6 \text{ mm}^4$$

(a) For a typical flange,

$$A(s) = ts$$

$$Q(s) = yts$$

$$q(s) = \frac{VQ(s)}{I} = \frac{V y t s}{I}$$

$$F = \int_0^b q(s) ds = \frac{V y t b^2}{2I}$$

Flange AB :

$$F_{AB} = \frac{V(45)(6)(30)^2}{(2)(1.14882 \times 10^6)} = 0.10576V \leftarrow$$

Flange DE :

$$F_{DE} = \frac{V(15)(4)(30)^2}{(2)(1.14882 \times 10^6)} = 0.023502V \leftarrow$$

Flange FG :

$$F_{FG} = 0.023502V \rightarrow$$

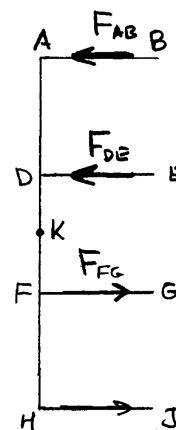
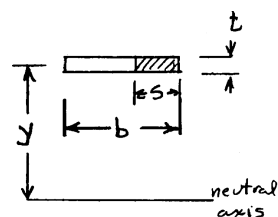
Flange HJ :

$$F_{HJ} = 0.10576V \rightarrow$$

$$+\curvearrowright \Sigma M_K = + \curvearrowright \Sigma M_K: Ve = 45 F_{AB} + 15 F_{DE} + 15 F_{FG} + 45 F_{HJ} = 10.223V$$

Dividing by V ,

$$e = 10.22 \text{ mm} \quad \blacktriangleleft$$



PROBLEM 6.67 (Continued)

(b) Calculation of shearing stresses.

$$V = 35 \times 10^3 \text{ N} \quad I = 1.14882 \times 10^{-6} \text{ m}^4$$

At B , E , G , and J ,

$$\tau = 0 \quad \blacktriangleleft$$

At A and H ,

$$Q = (30)(6)(45) = 8.1 \times 10^3 \text{ mm}^3 = 8.1 \times 10^{-6} \text{ m}^3$$

$$t = 6 \times 10^{-3} \text{ m}$$

$$\tau = \frac{VQ}{It} = \frac{(35 \times 10^3)(8.1 \times 10^{-6})}{(1.14882 \times 10^{-6})(6 \times 10^{-3})} = 41.1 \times 10^6 \text{ Pa} \quad \tau = 41.1 \text{ MPa} \quad \blacktriangleleft$$

Just above D and just below F :

$$Q = 8.1 \times 10^3 + (6)(30)(30) = 13.5 \times 10^3 \text{ mm}^3 = 13.5 \times 10^{-6} \text{ m}^3$$

$$t = 6 \times 10^{-3} \text{ m}$$

$$\tau = \frac{VQ}{It} = \frac{(35 \times 10^3)(13.5 \times 10^{-6})}{(1.14882 \times 10^{-6})(6 \times 10^{-3})} = 68.5 \times 10^6 \text{ Pa} \quad \tau = 68.5 \text{ MPa} \quad \blacktriangleleft$$

Just to right of D and just to the right of F :

$$Q = (30)(4)(15) = 1.8 \times 10^3 \text{ mm}^3 = 1.8 \times 10^{-6} \text{ m}^3 \quad t = 4 \times 10^{-3} \text{ m}$$

$$\tau = \frac{VQ}{It} = \frac{(35 \times 10^3)(1.8 \times 10^{-6})}{(1.14882 \times 10^{-6})(4 \times 10^{-3})} = 13.71 \times 10^6 \text{ Pa} \quad \tau = 13.71 \text{ MPa} \quad \blacktriangleleft$$

Just below D and just above F :

$$Q = 13.5 \times 10^3 + 1.8 \times 10^3 = 15.3 \times 10^3 \text{ mm}^3 = 15.3 \times 10^{-6} \text{ m}^3$$

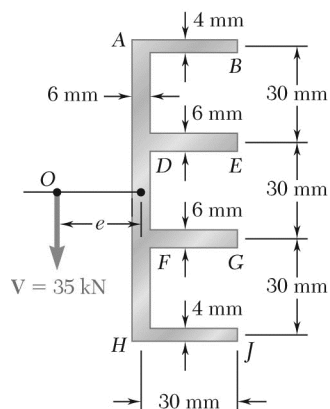
$$t = 6 \times 10^{-3} \text{ m}$$

$$\tau = \frac{VQ}{It} = \frac{(35 \times 10^3)(15.3 \times 10^{-6})}{(1.14882 \times 10^{-6})(6 \times 10^{-3})} = 77.7 \times 10^6 \text{ Pa} \quad \tau = 77.7 \text{ MPa} \quad \blacktriangleleft$$

At K ,

$$Q = 15.3 \times 10^3 + (6)(15)(7.5) = 15.975 \times 10^3 \text{ mm}^3 = 15.975 \times 10^{-6} \text{ m}^3$$

$$\tau = \frac{VQ}{It} = \frac{(35 \times 10^3)(15.975 \times 10^{-6})}{(1.14882 \times 10^{-6})(6 \times 10^{-3})} = 81.1 \times 10^6 \text{ Pa} \quad \tau = 81.1 \text{ MPa} \quad \blacktriangleleft$$



PROBLEM 6.68

An extruded beam has the cross section shown. Determine (a) the location of the shear center O , (b) the distribution of the shearing stresses caused by the vertical shearing force V shown applied at O .

$$I_z = 0.933 \times 10^6 \text{ mm}^4$$

SOLUTION

$$I_{AB} = I_{HJ} = \frac{1}{12}(30)(4)^3 + (30)(4)(45)^2 = 0.24316 \times 10^6 \text{ mm}^4$$

$$I_{DE} = I_{FG} = \frac{1}{12}(30)(6)^3 + (30)(6)(15)^2 = 0.04104 \times 10^6 \text{ mm}^4$$

$$I_{AH} = \frac{1}{12}(6)(90)^3 = 0.3645 \times 10^6 \text{ mm}^4$$

$$I = \Sigma I = 0.9329 \times 10^6 \text{ mm}^4$$

(a) For a typical flange,

$$A(s) = ts$$

$$Q(s) = yts$$

$$q(s) = \frac{VQ(s)}{I} = \frac{V y t s}{I}$$

$$F = \int_0^b q(s) ds = \frac{V y t b^2}{2I}$$

Flange AB :

$$F_{AB} = \frac{V(45)(4)(30)^2}{(2)(0.9329 \times 10^6)} = 0.086826 V \leftarrow$$

Flange DE :

$$F_{DE} = \frac{V(15)(6)(30)^2}{(2)(0.9329 \times 10^6)} = 0.043413 V \leftarrow$$

Flange FG :

$$F_{FG} = 0.043413 V \rightarrow$$

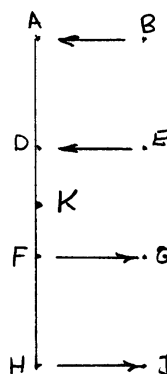
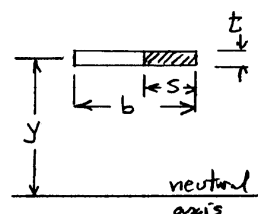
Flange HJ :

$$F_{HJ} = 0.086826 V \rightarrow$$

$$\begin{aligned} \sum M_K = 0: \quad V e &= 45 F_{AB} + 15 F_{DE} + 15 F_{FG} \\ &\quad + 45 F_{HJ} = 9.1167 V \end{aligned}$$

Dividing by V ,

$$e = 9.12 \text{ mm} \blacktriangleleft$$



PROBLEM 6.68 (Continued)

(b) Calculation of shearing stresses.

$$V = 35 \times 10^3 \text{ N} \quad I = 0.9329 \times 10^{-6} \text{ m}^4$$

At *B*, *E*, *G*, and *J*,

$$\tau = 0 \quad \blacktriangleleft$$

At *A* and *H*,

$$Q = (30)(4)(45) = 5.4 \times 10^3 \text{ mm}^3 = 5.4 \times 10^{-6} \text{ m}^3$$

$$q = \frac{VQ}{I} = \frac{(35 \times 10^3)(5.4 \times 10^{-6})}{0.9329 \times 10^{-6}} = 202.59 \times 10^3 \text{ N/m}$$

Just to the right of *A* and *H*: $t = 4 \times 10^{-3} \text{ m}$

$$\tau = \frac{q}{t} = \frac{202.59 \times 10^3}{4 \times 10^{-3}} = 50.6 \times 10^6 \text{ Pa} \quad \tau = 50.6 \text{ MPa} \quad \blacktriangleleft$$

Just below *A* and just above *H*: $t = 6 \times 10^{-3} \text{ m}$

$$\tau = \frac{q}{t} = \frac{202.59 \times 10^3}{6 \times 10^{-3}} = 33.8 \times 10^6 \text{ Pa} \quad \tau = 33.8 \text{ MPa} \quad \blacktriangleleft$$

Just above *D* and just below *F*: $t = 6 \times 10^{-3} \text{ m}$

$$Q = 5.4 \times 10^3 + (6)(30)(30) = 10.8 \times 10^3 \text{ mm}^3 = 10.8 \times 10^{-6} \text{ m}^3$$

$$\tau = \frac{VQ}{It} = \frac{(35 \times 10^3)(10.8 \times 10^{-6})}{(0.9329 \times 10^{-6})(6 \times 10^{-3})} = 67.5 \times 10^6 \text{ Pa} \quad \tau = 67.5 \text{ MPa} \quad \blacktriangleleft$$

Just to the right of *D* and *E*: $t = 6 \times 10^{-3} \text{ m}$

$$Q = (30)(6)(15) = 2.7 \times 10^3 \text{ mm}^2 = 2.7 \times 10^{-6} \text{ m}^3$$

$$\tau = \frac{VQ}{It} = \frac{(35 \times 10^3)(2.7 \times 10^{-6})}{(0.9329 \times 10^{-6})(6 \times 10^{-3})} = 16.88 \times 10^6 \text{ Pa} \quad \tau = 16.88 \text{ MPa} \quad \blacktriangleleft$$

Just below *D* and just above *F*: $t = 6 \times 10^{-3}$

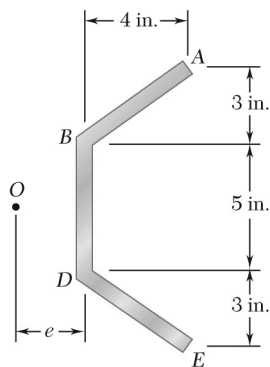
$$Q = 10.8 \times 10^3 + 2.7 \times 10^3 = 13.5 \times 10^3 \text{ mm}^3 = 13.5 \times 10^{-6} \text{ m}^3$$

$$\tau = \frac{VQ}{It} = \frac{(35 \times 10^3)(13.5 \times 10^{-6})}{(0.9329 \times 10^{-6})(6 \times 10^{-3})} = 84.4 \times 10^6 \text{ Pa} \quad \tau = 84.4 \text{ MPa} \quad \blacktriangleleft$$

At *K*,

$$Q = 13.5 \times 10^3 + (6)(15)(7.5) = 14.175 \times 10^3 \text{ mm}^3 = 14.175 \times 10^{-6} \text{ m}^3$$

$$\tau = \frac{VQ}{It} = \frac{(35 \times 10^3)(14.175 \times 10^{-6})}{(0.9329 \times 10^{-6})(6 \times 10^{-3})} = 88.6 \times 10^6 \text{ Pa} \quad \tau = 88.6 \text{ MPa} \quad \blacktriangleleft$$



PROBLEM 6.69

Determine the location of the shear center O of a thin-walled beam of uniform thickness having the cross section shown.

SOLUTION

$$L_{AB} = \sqrt{4^2 + 3^2} = 5 \text{ in.} \quad A_{AB} = 5t$$

$$I_{AB} = \frac{1}{12} A_{AB} h^2 + A_{AB} d^2 = \frac{1}{12} (5t)(3)^2 + (5t)(4)^2 = 83.75 t \text{ in}^4$$

$$I_{BD} = \frac{1}{12} (t)(5)^3 = 10.417 t \text{ in}^4$$

$$I = 2I_{AB} + I_{BD} = 177.917 t \text{ in}^4$$

In part BD , $Q = Q_{AB} + Q_{BY}$

$$Q = (5t)(4) + (2.5 - y)t \left(\frac{1}{2} \right) (2.5 + y)$$

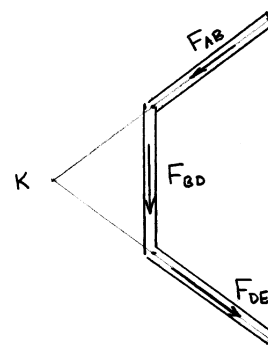
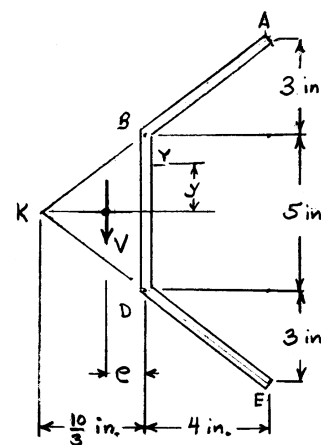
$$= 20t + 3.125t - \frac{1}{2}ty^2 = \left(23.125 - \frac{1}{2}y^2 \right) t$$

$$\begin{aligned} \tau &= \frac{VQ}{It} \quad F_{BD} = \int \tau dA = \int_{-2.5}^{2.5} \frac{V(23.125 - \frac{1}{2}y^2)t}{It} \cdot t dy \\ &= \frac{Vt}{I} \int_{-2.5}^{2.5} \left(23.125 - \frac{1}{2}y^2 \right) dy = \frac{Vt}{I} \left[23.125y - \frac{1}{6}y^3 \right]_{-2.5}^{2.5} \\ &= \frac{Vt}{I} \cdot 2 \left[(23.125)(2.5) - \frac{(2.5)^3}{6} \right] = \frac{Vt(110.417)}{177.917t} = 0.62061V \end{aligned}$$

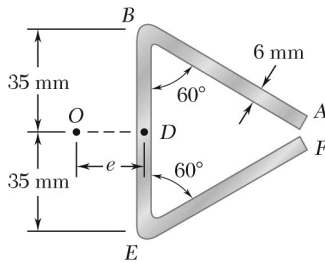
$$+\circlearrowleft \Sigma M_K = +\circlearrowleft \Sigma M_K: -V \left(\frac{10}{3} - e \right) = -\frac{10}{3} (0.62061V)$$

$$e = \frac{10}{3} [1 - 0.62061]$$

$$e = 1.265 \text{ in.} \quad \blacktriangleleft$$



Note that the lines of action of F_{AB} and F_{DE} pass through point K . Thus, these forces have zero moment about point K .



PROBLEM 6.70

Determine the location of the shear center O of a thin-walled beam of uniform thickness having the cross section shown.

SOLUTION

$$I_{DB} = \frac{1}{3}(6)(35)^3 = 85.75 \times 10^3 \text{ mm}^4$$

$$L_{AB} = 70 \text{ mm} \quad A_{AB} = (70)(6) = 420 \text{ mm}^2$$

$$I_{AB} = \frac{1}{3} A_{AB} h^2 = \left(\frac{1}{3} \right) (420)(35)^2 = 171.5 \times 10^3 \text{ mm}^4$$

$$I = (2)(85.75 \times 10^3) + (2)(171.5 \times 10^3) = 514.5 \times 10^3 \text{ mm}^4$$

Part AB:

$$A = ts = 6s$$

$$\bar{y} = \frac{1}{2}s \sin 30^\circ = \frac{1}{4}s$$

$$Q = A\bar{y} = \frac{3}{2}s^2$$

$$\tau = \frac{VQ}{It} = \frac{3Vs^2}{It}$$

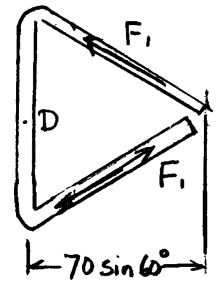
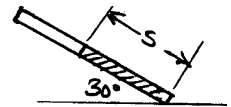
$$F_1 = \int \tau dA = \int_0^{70} \frac{3Vs^2}{2It} t ds = \frac{3V}{I} \int_0^{70} s^2 ds$$

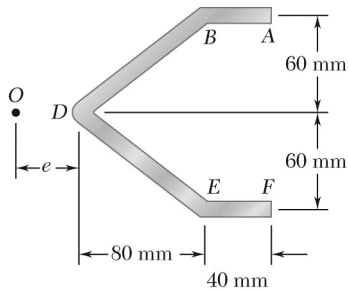
$$= \frac{(3)(70)^3}{(2)(3)I} V = \frac{1}{3} V$$

$$+\circlearrowleft \Sigma M_D = +\circlearrowleft \Sigma M_D: \quad Ve = 2[(F_1 \cos 60^\circ)(70 \sin 60^\circ)] \\ = 20.2V$$

Dividing by V ,

$$e = 20.2 \text{ mm} \quad \blacktriangleleft$$





PROBLEM 6.71

Determine the location of the shear center O of a thin-walled beam of uniform thickness having the cross section shown.

SOLUTION

$$I_{AB} = (40t)(60)^2 = 144 \times 10^3 t$$

$$L_{DB} = \sqrt{80^2 + 60^2} = 100 \text{ mm} \quad A_{DB} = 100t$$

$$I_{DB} = \frac{1}{3} A_{DB} h^2 = \frac{1}{3} (100t)(60)^2 = 120 \times 10^3 t$$

$$I = 2I_{AB} + 2I_{DB} = 528 \times 10^3 t$$

Part AB:

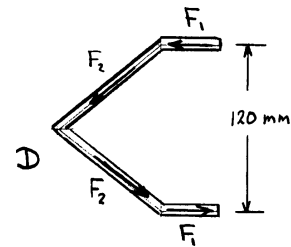
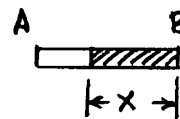
$$A = tx \quad \bar{y} = 60 \text{ mm}$$

$$Q = A\bar{y} = 60tx \text{ mm}^3$$

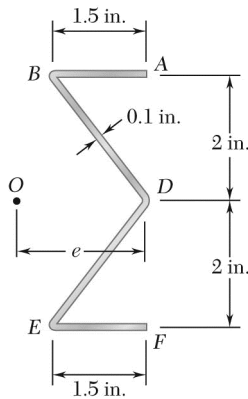
$$\tau = \frac{VQ}{It} = \frac{V(60tx)}{It} = \frac{60Vx}{I}$$

$$\begin{aligned} F_1 &= \int \tau dA = \int_0^{40} \frac{60Vx}{I} t dx = \frac{60Vt}{I} \int_0^{40} x dx \\ &= \frac{60Vt}{I} \frac{x^2}{2} \bigg|_0^{40} = \frac{(60)(30)^2 Vt}{(2)(528 \times 10^3)t} = 0.051136V \end{aligned}$$

$$+\circlearrowleft \Sigma M_D = +\circlearrowleft \Sigma M_D: \quad Ve = (0.051136V)(120)$$



$$e = 6.14 \text{ mm} \quad \blacktriangleleft$$



PROBLEM 6.72

Determine the location of the shear center O of a thin-walled beam of uniform thickness having the cross section shown.

SOLUTION

$$I_{AB} = \frac{1}{12}(1.5)(0.1)^3 + (1.5)(0.1)(2)^2 = 0.600125 \text{ in}^4$$

$$L_{BD} = \sqrt{1.5^2 + 2^2} = 2.5 \text{ in.} \quad A_{BD} = (2.5)(0.1) = 0.25 \text{ in}^2$$

$$I_{BD} = \frac{1}{3} A_{BD} h^2 = \frac{1}{3}(0.25)(2)^2 = 0.33333 \text{ in}^4$$

$$I = 2I_{AB} + 2I_{BD} = 1.86692 \text{ in}^4$$

Part AB:

$$A(x) = tx = 0.1x, \quad \bar{y} = 2 \text{ in.}$$

$$Q(x) = A(x)\bar{y} = 0.2x \text{ in}^3$$

$$q(x) = \frac{VQ(x)}{I} = \frac{0.2Vx}{I}$$

$$F_1 = \int_0^{1.5} q(x) dx = \frac{0.2V}{I} \int_0^{1.5} x dx$$

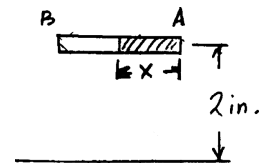
$$= \frac{(0.2)(1.5)^2}{2} \frac{V}{I} = 0.225 \frac{V}{I}$$

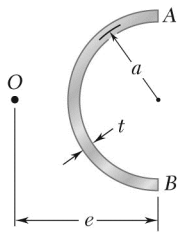
Likewise, by symmetry in part EF: $F_1 = 0.225 \frac{V}{I}$

$$+\circlearrowleft \Sigma M_D = +\circlearrowleft \Sigma M_D: \quad Ve = 4F_1 = 0.9 \frac{V}{I} = 0.482V$$

Dividing by V ,

$$e = 0.482 \text{ in.} \quad \blacktriangleleft$$





PROBLEM 6.73

Determine the location of the shear center O of a thin-walled beam of uniform thickness having the cross section shown.

SOLUTION

For a thin-walled hollow circular cross section, $A = 2\pi at$

$$J = a^2 A = 2\pi a^3 t \quad I = \frac{1}{2} J = \pi a^3 t$$

For the half-pipe section, $I = \frac{\pi}{2} a^3 t$

Use polar coordinate θ for partial cross section.

$$A = st = a \theta t \quad s = \text{arc length}$$

$$\bar{r} = a \frac{\sin \alpha}{\alpha} \quad \text{where} \quad \alpha = \frac{\theta}{2}$$

$$\bar{y} = \bar{r} \cos \alpha = a \frac{\sin \alpha \cos \alpha}{\alpha}$$

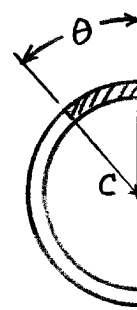
$$Q = A\bar{y} = a\theta t a \frac{\sin \alpha \cos \alpha}{\alpha} = a^2 t (2 \sin \alpha \cos \alpha) \\ = a^2 t \sin 2\alpha = a^2 t \sin \theta$$

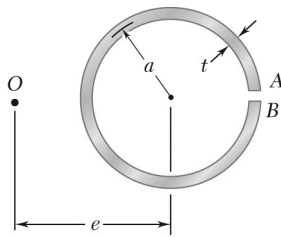
$$\tau = \frac{VQ}{It} = \frac{Va^2}{I} \sin \theta$$

$$M_H = \int a \tau dA = \int_0^\pi a \frac{Va^2}{I} \sin \theta t a d\theta = \frac{Va^4 t}{I} - \cos \theta \Big|_0^\pi \\ = 2 \frac{Va^4 t}{I} = \frac{4}{\pi} Va$$

But $M_H = Ve$, hence

$$e = \frac{4}{\pi} a = 1.273a \quad \blacktriangleleft$$





PROBLEM 6.74

Determine the location of the shear center O of a thin-walled beam of uniform thickness having the cross section shown.

SOLUTION

For whole cross section,

$$A = 2\pi at$$

$$J = Aa^2 = 2\pi a^3 t \quad I = \frac{1}{2} J = \pi a^3 t$$

Use polar coordinate θ for partial cross section.

$$A = st = a\theta t$$

s = arc length

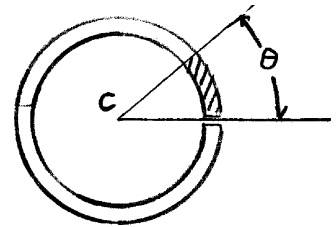
$$\bar{r} = a \frac{\sin \alpha}{\alpha} \quad \text{where} \quad \alpha = \frac{1}{2} \theta$$

$$\bar{y} = \bar{r} \sin \alpha = a \frac{\sin^2 \alpha}{\alpha}$$

$$\begin{aligned} Q = A\bar{y} &= a\theta t a \frac{\sin^2 \alpha}{\alpha} = a^2 t \cdot 2 \sin^2 \alpha \\ &= a^2 t \cdot 2 \sin^2 \frac{\theta}{2} = a^2 t (1 - \cos \theta) \end{aligned}$$

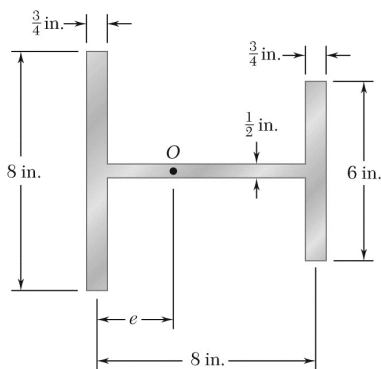
$$\tau = \frac{VQ}{It} = \frac{Va^2}{I} (1 - \cos \theta)$$

$$\begin{aligned} M_C &= \int a \tau dA = \int_0^{2\pi} \frac{Va^3}{I} (1 - \cos \theta) t a d\theta = \frac{Va^4 t}{I} (\theta - \sin \theta) \Big|_0^{2\pi} \\ &= \frac{2\pi Va^4 t}{\pi a^3 t} = 2aV \end{aligned}$$



But $M_C = Ve$, hence

$$e = 2a \quad \blacktriangleleft$$



PROBLEM 6.75

A thin-walled beam has the cross section shown. Determine the location of the shear center O of the cross section.

SOLUTION

Right flange:

$$I = \frac{1}{12} t_1 h_1^3 + \frac{1}{12} t_2 h_2^3$$

$$A = \left(\frac{1}{2} h_2 - y \right) t_2$$

$$\bar{y} = \frac{1}{2} \left(\frac{1}{2} h_2 + y \right) t_2$$

$$Q = A \bar{y}$$

$$= \frac{1}{2} \left(\frac{1}{2} h_2 - y \right) \left(\frac{1}{2} h_2 + y \right) t_2$$

$$= \frac{1}{2} \left(\frac{1}{4} h_2^2 - y^2 \right) t_2$$

$$\tau = \frac{VQ}{It_2} = \frac{V}{2It_2} \left(\frac{1}{4} h_2^2 - y^2 \right) t_2$$

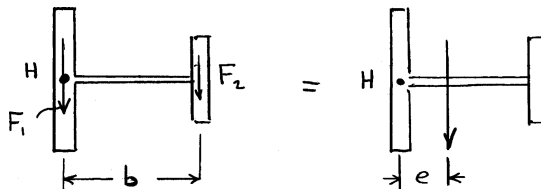
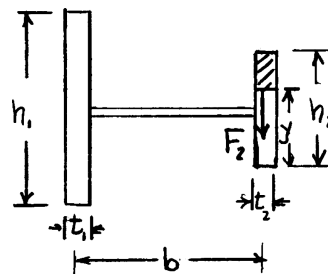
$$F_2 = \int \tau dA = \int_{-h_2/2}^{h_2/2} \frac{Vt_2}{2It_2} \left(\frac{1}{4} h_2^2 - y^2 \right) dy = \frac{Vt_2}{2I} \left(\frac{1}{4} h_2^2 y - \frac{y^3}{3} \right) \bigg|_{-h_2/2}^{h_2/2}$$

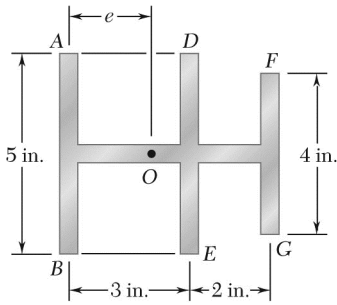
$$= \frac{Vt_2}{2I} \left\{ \frac{1}{4} h_2^2 \frac{h_2}{2} - \frac{1}{3} \left(\frac{h_2}{2} \right)^3 + \frac{1}{4} h_2^2 \frac{h_2}{2} - \frac{1}{3} \left(\frac{h_2}{2} \right)^3 \right\} = \frac{Vt_2 h_2^3}{12I} = \frac{Vt_2 h_2^3}{t_1 h_1^3 + t_2 h_2^3}$$

$$+\circlearrowleft M_H = +\circlearrowleft M_H: -Ve = -F_2 b = -V \frac{t_2 h_2^3 b}{t_1 h_1^3 + t_2 h_2^3}$$

$$e = \frac{t_2 h_2^3 b}{t_1 h_1^3 + t_2 h_2^3} = \frac{(0.75)(6)^3(8)}{(0.75)(8)^3 + (0.75)(6)^3}$$

$$e = 2.37 \text{ in.} \quad \blacktriangleleft$$





PROBLEM 6.76

A thin-walled beam has the cross section shown. Determine the location of the shear center O of the cross section.

SOLUTION

Let $h_1 = \overline{AB}$, $h_2 = \overline{DE}$, and $h_3 = \overline{FG}$. $I = \frac{1}{12}t(h_1^3 + h_2^3 + h_3^3)$

Part AB : $A = \left(\frac{1}{2}h_1 - y\right)t$ $\bar{y} = \frac{1}{2}\left(\frac{1}{2}h_1 + y\right)$

$$Q = A\bar{y} = \frac{1}{2}t\left(\frac{1}{2}h_1 - y\right)\left(\frac{1}{2}h_1 + y\right) = \frac{1}{2}t\left(\frac{1}{4}h_1^2 - y^2\right)$$

$$\tau = \frac{VQ}{It} = \frac{V}{2I}\left(\frac{1}{4}h_1^2 - y^2\right)$$

$$F_1 = \int \tau dA = \int_{-\frac{1}{2}h_1}^{\frac{1}{2}h_1} \frac{V}{2I}\left(\frac{1}{4}h_1^2 - y^2\right)tdy = \frac{Vt}{2I}\left(\frac{1}{4}h_1^2y - \frac{y^3}{3}\right)\bigg|_{-\frac{1}{2}h_1}^{\frac{1}{2}h_1}$$

$$= \frac{Vt}{I}\left(\frac{1}{4}h_1^2\frac{1}{2}h_1 - \frac{1}{3}\left(\frac{h_1}{2}\right)^3\right) = \frac{Vth_1^3}{12I} = \frac{h_1^3V}{h_1^3 + h_2^3 + h_3^3}$$

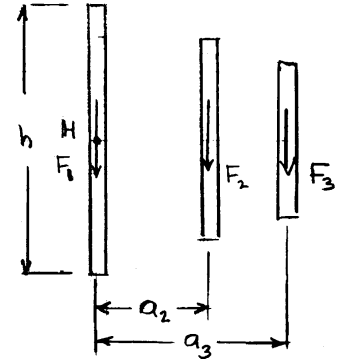
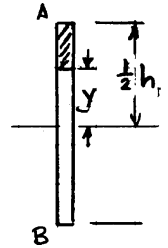
Likewise, for part DE , $F_2 = \frac{h_2^3V}{h_1^3 + h_2^3 + h_3^3}$

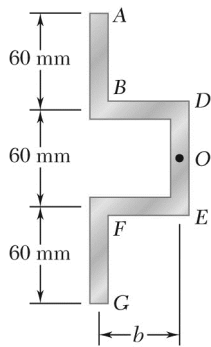
and for part FG , $F_3 = \frac{h_3^3V}{h_1^3 + h_2^3 + h_3^3}$

$$\curvearrowright \Sigma M_H = \curvearrowright \Sigma M_H: Ve = a_2F_2 + a_3F_3 = \frac{a_2h_2^3 + a_3h_3^3}{h_1^3 + h_2^3 + h_3^3}V$$

Dividing by V , $e = \frac{a_2h_2^3 + a_3h_3^3}{h_1^3 + h_2^3 + h_3^3} = \frac{(3)(5)^3 + (5)(4)^3}{5^3 + 5^3 + 4^3}$

$e = 2.21 \text{ in.} \quad \blacktriangleleft$





PROBLEM 6.77

A thin-walled beam of uniform thickness has the cross section shown. Determine the dimension b for which the shear center O of the cross section is located at the point indicated.

SOLUTION

Part AB :

$$\begin{aligned}
 A(s) &= ts & \bar{y}(s) &= y_A - \frac{1}{2}s \\
 Q(s) &= A(s)\bar{y}(s) = ty_A s - \frac{1}{2}ts^2 \\
 q(s) &= \frac{VQ(s)}{I} = \frac{Vt}{I} \left(y_A s - \frac{1}{2}s^2 \right) \\
 F_{AB} &= \int_0^{l_{AB}} q(s) ds \\
 &= \frac{Vt}{I} \left(\frac{y_A l_{AB}^2}{2} - \frac{l_{AB}^3}{6} \right) \downarrow
 \end{aligned}$$

At B :

$$Q_B = ty_A l_{AB} - \frac{1}{2}t l_{AB}^2$$

By symmetry,

$$F_{FG} = F_{AB}$$

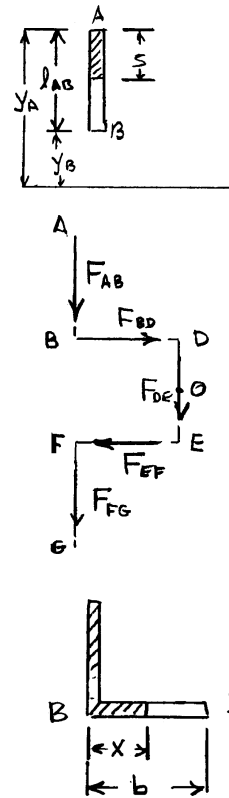
Part BD :

$$\begin{aligned}
 A(x) &= tx \\
 Q(x) &= Q_B + y_B A(x) = Q_B + t y_B x \\
 q(x) &= \frac{VQ(x)}{I} = \frac{V}{I} (Q_B + t y_B x) \\
 F_{BD} &= \int_0^b q(x) dx = \frac{V}{I} \left(Q_B b + \frac{1}{2} t y_B b^2 \right) \rightarrow
 \end{aligned}$$

By symmetry,

$$F_{EF} = F_{BD}$$

F_{DE} is not required, since its moment about O is zero.



PROBLEM 6.77 (Continued)

$$\curvearrowright \Sigma M_O = 0: \quad b(F_{AB} + F_{FG}) - y_B F_{BD} + y_F F_{EF} = 0$$

$$2b F_{AB} - 2 y_B F_{BD} = 0$$

$$2b \cdot \frac{Vt}{I} \left(\frac{y_A l_{AB}^2}{2} - \frac{l_{AB}^3}{6} \right) - 2y_B \frac{V}{I} \left(Q_B b + \frac{1}{2} t y_B b^2 \right)$$

$$\frac{2Vt}{I} \left\{ \frac{1}{2} y_A l_{AB}^2 - \frac{1}{6} l_{AB}^3 \right\} b - \frac{2Vt}{I} \left\{ \left(y_A l_{AB} - \frac{1}{2} l_{AB}^2 \right) y_B b - \frac{1}{2} y_B^2 b^2 \right\} = 0$$

Dividing by $\frac{2Vt}{I}$ and substituting numerical data,

$$\left\{ \frac{1}{2} (90)(60)^2 - \frac{1}{6} (60)^3 \right\} b - \left\{ (90)(60) - \frac{1}{2} (60)^2 \right\} (30)b + \frac{1}{2} (30)^2 b^2 = 0$$

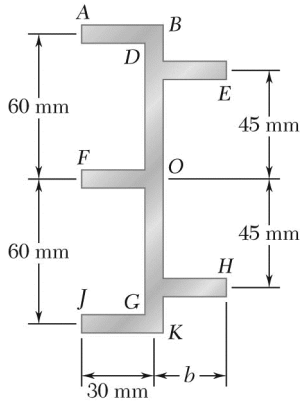
$$126 \times 10^3 b - 108 \times 10^3 b + 450 b^2 = 0$$

$$18 \times 10^3 b - 450 b^2 = 0$$

$$b = 0 \quad \text{and} \quad b = 40.0 \text{ mm} \quad \blacktriangleleft$$

PROBLEM 6.78

A thin-walled beam of uniform thickness has the cross section shown. Determine the dimension b for which the shear center O of the cross section is located at the point indicated.



SOLUTION

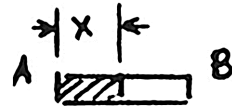
Part AB:

$$A = tx \quad \bar{y} = 60 \text{ mm}$$

$$Q = A\bar{y} = 60tx \text{ mm}^3$$

$$\tau = \frac{VQ}{It} = \frac{60 Vx}{I}$$

$$\begin{aligned} F_1 &= \int \tau dA = \int_0^{30} \frac{60 Vx}{I} t dx = \frac{60Vt}{I} \int_0^{30} x dx \\ &= \frac{60 Vt}{I} \frac{x^2}{2} \bigg|_0^{30} = \frac{(60)(30)^2}{2} \frac{Vt}{I} = 27 \times 10^3 \frac{Vt}{I} \end{aligned}$$



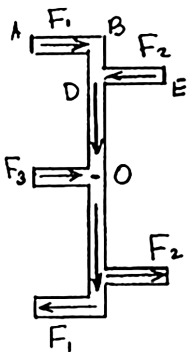
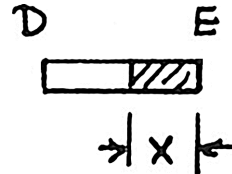
Part DE:

$$A = tx \quad \bar{y} = 45 \text{ mm}$$

$$Q = A\bar{y} = 45tx$$

$$\tau = \frac{VQ}{It} = \frac{45 Vx}{I}$$

$$F_2 = \int \tau dA = \int_0^b \frac{45 Vx}{I} t dx = \frac{45Vt}{I} \int_0^b x dx = \frac{45b^2 Vt}{2I}$$



$$+\circlearrowleft \sum M_O = +\circlearrowleft \sum M_O: \quad 0 = (2)(45)F_2 - (2)(60)F_1$$

$$\left[(45)^2 b^2 - (2)(60)(27 \times 10^3) \right] \frac{Vt}{I} = 0$$

$$b^2 = \frac{(2)(60)(27 \times 10^3)}{45^2} = 1600 \text{ mm}^2$$

$$b = 40 \text{ mm} \quad \blacktriangleleft$$

Note that the pair of F_1 forces form a couple. Likewise, the pair of F_2 forces form a couple. The lines of action of the forces in $BDOGK$ pass through point O .

PROBLEM 6.79

For the angle shape and loading of Sample Prob. 6.6, check that $\int qdz = 0$ along the horizontal leg of the angle and $\int qdy = P$ along its vertical leg.

SOLUTION

Refer to Sample Prob. 6.6.

Along horizontal leg: $\tau_f = \frac{3P(a-z)(a-3z)}{4ta^3} = \frac{3P}{4ta^3}(a^2 - 4az + 3z^2)$

$$\begin{aligned}\int qdz &= \int_0^a \tau_f t dz = \frac{3P}{4a^3} \int_0^a (a^2 - 4az + 3z^2) dz \\ &= \frac{3P}{4a^3} \left(a^2 z - 4a \frac{z^2}{2} + 3 \frac{z^3}{3} \right) \bigg|_0^a \\ &= \frac{3P}{4a^3} (a^3 - 2a^3 + a^3) = 0\end{aligned}$$

Along vertical leg: $\tau_e = \frac{3P(a-y)(a+5y)}{4ta^3} = \frac{3P}{4ta^3}(a^2 + 4ay - 5y^2)$

$$\begin{aligned}\int qdy &= \int_0^a \tau_e t dy = \frac{3P}{4a^3} \int_0^a (a^2 + 4ay - 5y^2) dy \\ &= \frac{3P}{4a^3} \left(a^2 y + 4a \frac{y^2}{2} - 5 \frac{y^3}{3} \right) \bigg|_0^a \\ &= \frac{3P}{4a^3} \left(a^3 + 2a^3 - \frac{5}{3}a^3 \right) = \frac{3P}{4a^3} \cdot \frac{4}{3}a^3 = P\end{aligned}$$

PROBLEM 6.80

For the angle shape and loading of Sample Prob. 6.6, (a) determine the points where the shearing stress is maximum and the corresponding values of the stress, (b) verify that the points obtained are located on the neutral axis corresponding to the given loading.

SOLUTION

Refer to Sample Prob. 6.6.

(a) Along vertical leg:
$$\tau_e = \frac{3P(a-y)(a+5y)}{4ta^3} = \frac{3P}{4ta^3}(a^2 + 4ay - 5y^2)$$

$$\frac{d\tau_e}{dy} = \frac{3P}{4ta^3}(4a - 10y) = 0 \quad y = \frac{2}{5}a \quad \blacktriangleleft$$

$$\tau_m = \frac{3P}{4ta^3} \left[a^2 + (4a) \left(\frac{2}{5}a \right) - (5) \left(\frac{2}{5}a \right)^2 \right] = \frac{3P}{4ta^3} \left(\frac{9}{5}a^2 \right) \quad \tau_m = \frac{27}{20} \frac{P}{ta} \quad \blacktriangleleft$$

Along horizontal leg:
$$\tau_f = \frac{3P(a-z)(a-3z)}{4ta^3} = \frac{3P}{4ta^3}(a^2 - 4az + 3z^2)$$

$$\frac{d\tau_f}{dz} = \frac{3P}{4ta^3}(-4a + 6z) = 0 \quad z = \frac{2}{3}a \quad \blacktriangleleft$$

$$\tau_m = \frac{3P}{4ta^3} \left[a^2 - (4a) \left(\frac{2}{3}a \right) + (3) \left(\frac{2}{3}a \right)^2 \right] = \frac{3P}{4ta^3} \left(-\frac{5}{3}a^2 \right) \quad \tau_m = -\frac{1}{4} \frac{P}{ta} \quad \blacktriangleleft$$

At the corner:
$$y = 0, \quad z = 0, \quad \tau = \frac{3}{4} \frac{P}{ta} \quad \blacktriangleleft$$

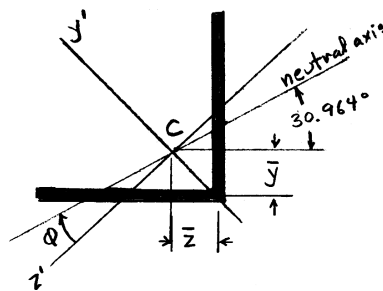
(b)
$$I_{y'} = \frac{1}{3}ta^3 \quad I_{z'} = \frac{1}{12}ta^3 \quad \theta = 45^\circ$$

$$\tan \phi = \frac{I_{z'}}{I_{y'}} \tan \theta = \frac{1}{4} \quad \phi = 14.036^\circ$$

$$\theta - \phi = 45 - 14.036 = 30.964^\circ$$

$$\bar{y} = \frac{\Sigma A \bar{y}}{\Sigma A} = \frac{at(a/2)}{2at} = \frac{1}{4}a$$

$$\bar{z} = \frac{\Sigma A \bar{z}}{\Sigma A} = \frac{at(a/2)}{2at} = \frac{1}{4}a$$



PROBLEM 6.80 (Continued)

Neutral axis intersects vertical leg at

$$y = \bar{y} + \bar{z} \tan 30.964^\circ$$

$$= \left(\frac{1}{4} + \frac{1}{4} \tan 30.964^\circ \right) a = 0.400a$$

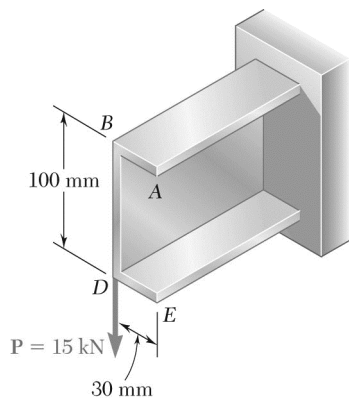
$$y = \frac{2}{5}a \quad \blacktriangleleft$$

Neutral axis intersects horizontal leg at

$$z = \bar{z} + \bar{y} \tan (45^\circ + \varphi)$$

$$= \left(\frac{1}{4} + \frac{1}{4} \tan 59.036^\circ \right) a = 0.667a$$

$$z = \frac{2}{3}a \quad \blacktriangleleft$$



PROBLEM 6.81*

A steel plate, 160 mm wide and 8 mm thick, is bent to form the channel shown. Knowing that the vertical load **P** acts at a point in the midplane of the web of the channel, determine (a) the torque **T** that would cause the channel to twist in the same way that it does under the load **P**, (b) the maximum shearing stress in the channel caused by the load **P**.

SOLUTION

Use results of Example 6.06 with $b = 30$ mm, $h = 100$ mm, and $t = 8$ mm.

$$e = \frac{b}{2 + \frac{h}{3b}} = \frac{30}{2 + \frac{100}{(3)(30)}} = 9.6429 \text{ mm} = 9.6429 \times 10^{-3} \text{ m}$$

$$I = \frac{1}{12} t h^2 (6b + h) = \frac{1}{12} (8)(100)^2 [(16)(30) + 100] = 1.86667 \times 10^6 \text{ mm}^4 = 1.86667 \times 10^{-6} \text{ m}^4$$

$$V = 15 \times 10^3 \text{ N}$$

(a) $T = Ve = (15 \times 10^3)(9.6429 \times 10^{-3})$

Stress at neutral axis due to V :

$$\begin{aligned} Q &= bt \frac{h}{2} + t \left(\frac{h}{2} \right) \left(\frac{h}{4} \right) = \frac{1}{8} th(h + 4b) \\ &= \frac{1}{8} (8)(100)[100 + (4)(30)] = 22 \times 10^3 \text{ mm}^3 = 22 \times 10^{-6} \text{ m}^3 \\ t &= 8 \times 10^{-3} \text{ m} \end{aligned}$$

$$\tau_v = \frac{VQ}{It} = \frac{(15 \times 10^3)(22 \times 10^{-6})}{(1.86667 \times 10^{-6})(8 \times 10^{-3})} = 22.10 \times 10^6 \text{ Pa} = 22.10 \text{ MPa}$$

Stress due to T :

$$a = 2b + h = 160 \text{ mm} = 0.160 \text{ m}$$

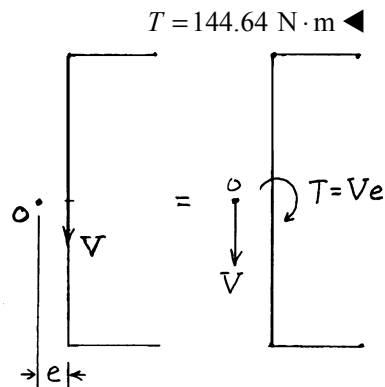
$$c_1 = \frac{1}{3} \left(1 - 0.630 \frac{t}{a} \right) = \frac{1}{3} \left[1 - (0.630) \frac{8}{160} \right] = 0.3228$$

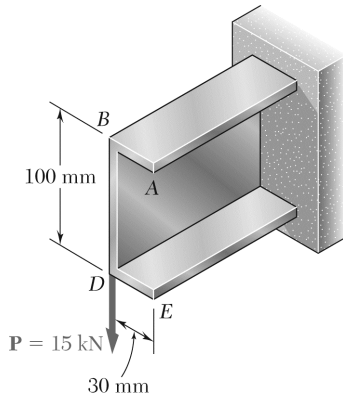
$$\tau_v = \frac{T}{c_1 a t^2} = \frac{144.64}{(0.3228)(0.160)(8 \times 10^{-3})^2} = 43.76 \times 10^6 \text{ Pa} = 43.76 \text{ MPa}$$

(b) By superposition,

$$\tau_{\max} = \tau_v + \tau_T$$

$$\tau_{\max} = 65.9 \text{ MPa} \quad \blacktriangleleft$$





PROBLEM 6.82*

Solve Prob. 6.81, assuming that a 6-mm-thick plate is bent to form the channel shown.

PROBLEM 6.81* A steel plate, 160 mm wide and 8 mm thick, is bent to form the channel shown. Knowing that the vertical load $\mathbf{P}$ acts at a point in the midplane of the web of the channel, determine (a) the torque $\mathbf{T}$ that would cause the channel to twist in the same way that it does under the load $\mathbf{P}$, (b) the maximum shearing stress in the channel caused by the load $\mathbf{P}$.

SOLUTION

Use results of Example 6.06 with $b = 30$ mm, $h = 100$ mm, and $t = 6$ mm.

$$e = \frac{b}{2 + \frac{h}{3b}} = \frac{30}{2 + \frac{100}{(3)(30)}} = 9.6429 \text{ mm} = 9.6429 \times 10^{-3} \text{ m}$$

$$I = \frac{1}{12} t h^2 (6b + h) = \frac{1}{12} (6)(100)^2 [(6)(30) + 100] = 1.400 \times 10^6 \text{ mm}^4 = 1.400 \times 10^{-6} \text{ m}^4$$

$$V = 15 \times 10^3 \text{ N}$$

$$(a) \quad T = Ve = (15 \times 10^3)(9.6429 \times 10^{-3})$$

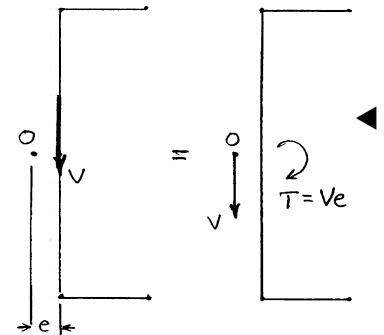
Stress at neutral axis due to V :

$$Q = bt \frac{h}{2} + t \left(\frac{h}{2} \right) \left(\frac{h}{4} \right) = \frac{1}{8} t h (h + 4b)$$

$$= \frac{1}{8} (6)(100) [100 + (4)(30)] = 16.5 \times 10^3 \text{ mm}^3 = 16.5 \times 10^{-6} \text{ m}^3$$

$$t = 6 \times 10^{-3} \text{ m}$$

$$\tau_V = \frac{VQ}{It} = \frac{(15 \times 10^3)(16.5 \times 10^{-6})}{(1.400 \times 10^{-6})(6 \times 10^{-3})} = 29.46 \times 10^6 \text{ Pa} = 29.46 \text{ MPa}$$



Stress due to T :

$$a = 2b + h = 160 \text{ mm} = 0.160 \text{ m}$$

$$c_1 = \frac{1}{3} \left(1 - 0.630 \frac{t}{a} \right) = \frac{1}{3} \left[1 - (0.630) \left(\frac{6}{160} \right) \right] = 0.32546$$

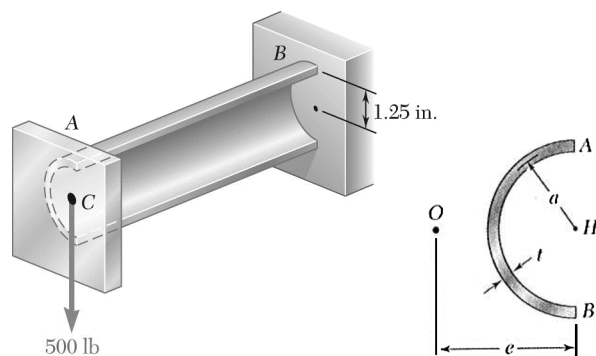
$$\tau_V = \frac{T}{c_1 a t^2} = \frac{144.64}{(0.32546)(0.160)(6 \times 10^{-3})^2} = 77.16 \times 10^6 \text{ Pa} = 77.16 \text{ MPa}$$

(b) By superposition,

$$\tau_{\max} = \tau_V + \tau_T$$

$$\tau_{\max} = 106.6 \text{ MPa} \quad \blacktriangleleft$$

PROBLEM 6.83*



The cantilever beam AB , consisting of half of a thin-walled pipe of 1.25-in. mean radius and $\frac{3}{8}$ -in. wall thickness, is subjected to a 500-lb vertical load. Knowing that the line of action of the load passes through the centroid C of the cross section of the beam, determine (a) the equivalent force-couple system at the shear center of the cross section, (b) the maximum shearing stress in the beam. (*Hint:* The shear center O of this cross section was shown in Prob. 6.73 to be located twice as far from its vertical diameter as its centroid C .)

SOLUTION

From the solution to Prob. 6.73,

$$I = \frac{\pi}{2} a^3 t \quad Q = a^2 t \sin \theta$$

$$e = \frac{4}{\pi} a \quad Q_{\max} = a^2 t$$

For a half-pipe section, the distance from the center of the semi-circle to the centroid is

$$\bar{x} = \frac{2}{\pi} a$$

At each section of the beam, the shearing force V is equal to P . Its line of action passes through the centroid C . The moment arm of its moment about the shear center O is

$$d = e - \bar{x} = \frac{4}{\pi} a - \frac{2}{\pi} a = \frac{2}{\pi} a$$

(a) Equivalent force-couple system at O .

$$V = P \quad M_0 = Vd = \frac{2}{\pi} Pa$$

$$\text{Data: } P = 500 \text{ lb} \quad a = 1.25 \text{ in.}$$

$$V = 500 \text{ lb} \quad \blacktriangleleft$$

$$M_0 = \left(\frac{2}{\pi} \right) (500)(1.25)$$

$$M_0 = 398 \text{ lb} \cdot \text{in} \quad \blacktriangleleft$$

PROBLEM 6.83* (Continued)

(b) Shearing stresses.

(1) Due to V : $\tau_V = \frac{VQ_{\max}}{It}$

$$\tau_V = \frac{(P)(a^2t)}{\left(\frac{\pi}{2}a^3t\right)(t)} = \frac{2P}{\pi at} = \frac{(2)(500)}{\pi(1.25)(0.375)} = 679 \text{ psi}$$

(2) Due to the torque. M_0 ,

For a long rectangular section of length l and width t , the shearing stress due to torque M_0 is

$$\tau_M = \frac{M_0}{c_1 l t^2} \quad \text{where} \quad c_1 = \frac{1}{3} \left(1 - 0.630 \frac{t}{l} \right)$$

Data: $l = \pi a = \pi(1.25) = 3.927 \text{ in.} \quad t = 0.375 \text{ in.} \quad c_1 = 0.31328$

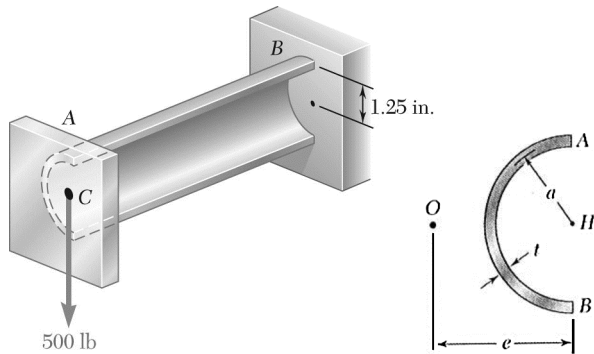
$$\tau_M = \frac{397.9}{(0.31328)(3.927)(0.375)^2} = 2300 \text{ psi}$$

By superposition, $\tau = \tau_V + \tau_M = 679 \text{ psi} + 2300 \text{ psi}$

$\tau = 2980 \text{ psi} \quad \blacktriangleleft$

PROBLEM 6.84*

Solve Prob. 6.83, assuming that the thickness of the beam is reduced to $\frac{1}{4}$ in.



PROBLEM 6.83* The cantilever beam AB , consisting of half of a thin-walled pipe of 1.25-in. mean radius and $\frac{3}{8}$ -in. wall thickness, is subjected to a 500-lb vertical load. Knowing that the line of action of the load passes through the centroid C of the cross section of the beam, determine (a) the equivalent force-couple system at the shear center of the cross section, (b) the maximum shearing stress in the beam. (*Hint:* The shear center O of this cross section was shown in Prob. 6.73 to be located twice as far from its vertical diameter as its centroid C .)

SOLUTION

From the solution to Prob. 6.73,

$$I = \pi a^3 t \quad Q = a^2 t \sin \theta$$

$$e = \frac{4}{\pi} a \quad Q_{\max} = a^2 t$$

For a half-pipe section, the distance from the center of the semi-circle to the centroid is

$$\bar{x} = \frac{2}{\pi} a$$

At each section of the beam, the shearing force V is equal to P . Its line of action passes through the centroid C . The moment arm of its moment about the shear center O is

$$d = e - \bar{x} = \frac{4}{\pi} a - \frac{2}{\pi} a = \frac{2}{\pi} a$$

(a) Equivalent force-couple system at O .

$$V = P \quad M_0 = Vd = \frac{2}{\pi} Pa$$

Data: $P = 500 \text{ lb} \quad a = 1.25 \text{ in.}$

$$V = 500 \text{ lb} \quad \blacktriangleleft$$

$$M_0 = 398 \text{ lb} \cdot \text{in} \quad \blacktriangleleft$$

PROBLEM 6.84* (Continued)

(b) Shearing stresses.

(1) Due to V , $\tau_V = \frac{VQ_{\max}}{It}$

$$\tau_V = \frac{(P)(a^2t)}{\left(\frac{\pi}{2}a^3t(t)\right)} = \frac{2P}{\pi at} = \frac{(2)(500)}{\pi(1.25)(0.250)} = 1019 \text{ psi}$$

(2) Due to the torque. M_0 ,

For a long rectangular section of length l and width t , the shearing stress due to torque M_0 is

$$\tau_M = \frac{M_0}{c_1 l t^2} \quad \text{where} \quad c_1 = \frac{1}{3} \left(1 - 0.630 \frac{t}{l} \right)$$

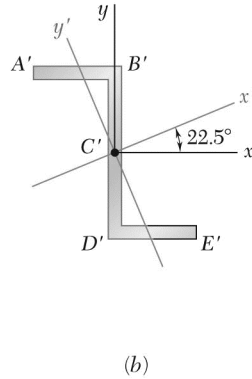
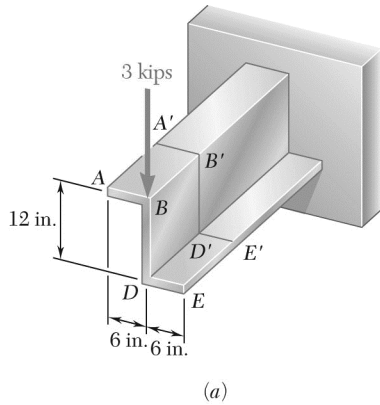
Data: $l = \pi a = \pi(1.25) = 3.927 \text{ in.}$ $t = 0.250 \text{ in.}$ $c_1 = 0.31996$

$$\tau_M = \frac{397.9}{(0.31996)(3.927)(0.250)^2} = 5067 \text{ psi}$$

By superposition $\tau = \tau_V + \tau_M = 1019 \text{ psi} + 5067 \text{ psi}$

$$\tau = 6090 \text{ psi} \quad \blacktriangleleft$$

PROBLEM 6.85*



The cantilever beam shown consists of a Z-shape of $\frac{1}{4}$ -in. thickness. For the given loading, determine the distribution of the shearing stresses along line $A'B'$ in the upper horizontal leg of the Z-shape. The x' and y' axes are the principal centroidal axes of the cross section and the corresponding moments of inertia are $I_{x'} = 166.3 \text{ in}^4$ and $I_{y'} = 13.61 \text{ in}^4$.

SOLUTION

$$V = 3 \text{ kips} \quad \beta = 22.5^\circ$$

$$V_{x'} = V \sin \beta \quad V_{y'} = V \cos \beta$$

In upper horizontal leg, use coordinate x : $(-6 \text{ in} \leq x \leq 0)$

$$A = \frac{1}{4}(6 + x) \text{ in.}$$

$$\bar{x} = \frac{1}{2}(-6 + x) \text{ in.}$$

$$\bar{y} = 6 \text{ in.}$$

$$\bar{x}' = \bar{x} \cos \beta + \bar{y} \sin \beta$$

$$\bar{y}' = \bar{y} \cos \beta - \bar{x} \sin \beta$$

Due to $V_{x'}$:

$$\tau_1 = \frac{V_{x'} A \bar{x}'}{I_{y'} t}$$

$$\tau_1 = \frac{(V \sin \beta) \left(\frac{1}{4} \right) (6 + x) \left[\frac{1}{2} (-6 + x) \cos \beta + 6 \sin \beta \right]}{(13.61) \left(\frac{1}{4} \right)}$$

$$= 0.084353(6 + x)(-0.47554 + 0.46194x)$$

Due to $V_{y'}$:

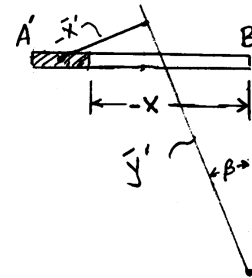
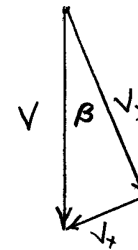
$$\tau_2 = \frac{V_{y'} A \bar{y}'}{I_{x'} t} = \frac{(V \cos \beta) \left(\frac{1}{4} \right) (6 + x) \left[6 \cos \beta - \frac{1}{2} (-6 + x) \sin \beta \right]}{(166.3) \left(\frac{1}{4} \right)}$$

$$= 0.0166665(6 + x)[6.69132 - 0.19134x]$$

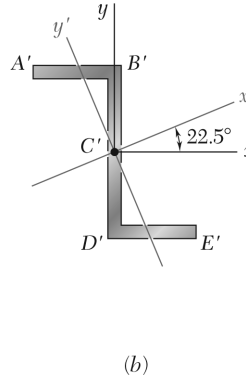
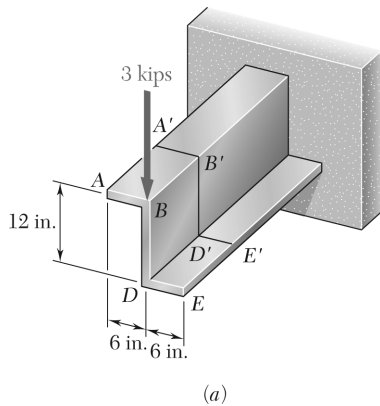
Total:

$$\tau_1 + \tau_2 = (6 + x)[-0.07141 + 0.035396x]$$

x (in)	-6	-5	-4	-3	-2	-1	0
τ (ksi)	0	-0.105	-0.140	-0.104	0.003	0.180	0.428



PROBLEM 6.86*



For the cantilever beam and loading of Prob. 6.85, determine the distribution of the shearing stress along line $B'D'$ in the vertical web of the Z-shape.

PROBLEM 6.85* The cantilever beam shown consists of a Z-shape of $\frac{1}{4}$ -in. thickness. For the given loading, determine the distribution of the shearing stresses along line $A'B'$ in the upper horizontal leg of the Z-shape. The x' and y' axes are the principal centroidal axes of the cross section and the corresponding moments of inertia are $I_{x'} = 166.3 \text{ in}^4$ and $I_{y'} = 13.61 \text{ in}^4$.

SOLUTION

$$V = 3 \text{ kips} \quad \beta = 22.5^\circ$$

$$V_{x'} = V \sin \beta \quad V_{y'} = V \cos \beta$$

For part AB'

$$A = \left(\frac{1}{4}\right)(6) = 1.5 \text{ in}^2$$

$$\bar{x} = -3 \text{ in.}, \quad y = 6 \text{ in.}$$

For part $B'Y$,

$$A = \frac{1}{4}(6 - y)$$

$$\bar{x} = 0 \quad \bar{y} = \frac{1}{2}(6 + y)$$

$$x' = x \cos \beta + y \sin \beta$$

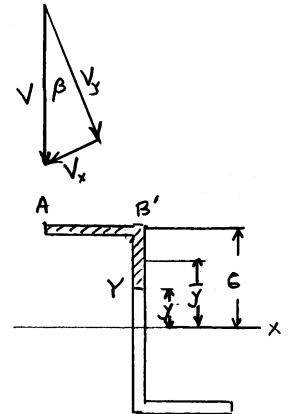
$$y' = y \cos \beta - x \sin \beta$$

Due to $V_{x'}$:

$$\tau_1 = \frac{V_{x'}(A_{AB}\bar{x}_{AB} + A_{BY}\bar{x}_{BY})}{I_{y'}t}$$

$$\tau_1 = \frac{(V \sin \beta)[(1.5)(-3 \cos \beta + 6 \sin \beta) + \frac{1}{4}(6 - y)\frac{1}{2}(6 + y) \sin \beta]}{(13.61)\left(\frac{1}{4}\right)}$$

$$= \frac{(V \sin \beta)[-0.7133 + 1.7221 - 0.047835 y^2]}{3.4025} = 0.3404 - 0.01614 y^2$$



PROBLEM 6.86* (Continued)

Due to $V_{y'}$:

$$\tau_2 = \frac{V_{y'}(A_{AB}\bar{y}_{AB}' + A_{BY}\bar{y}')}{I_{x'}t}$$

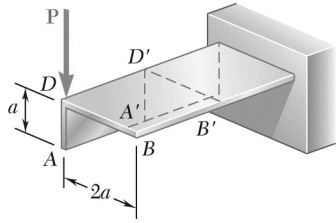
$$\tau_2 = \frac{(V \cos \beta)[(1.5)(6 \cos \beta + 3 \sin \beta) + \frac{1}{4}(6 - y)\frac{1}{2}(6 + y) \cos \beta]}{(166.3)\left(\frac{1}{4}\right)}$$

$$= \frac{(V \cos \beta)[10.037 + 4.1575 - 0.11548 y^2]}{(166.3)\left(\frac{1}{4}\right)} = 0.9463 - 0.00770 y^2$$

Total: $\tau_1 + \tau_2 = 1.2867 - 0.02384 y^2$

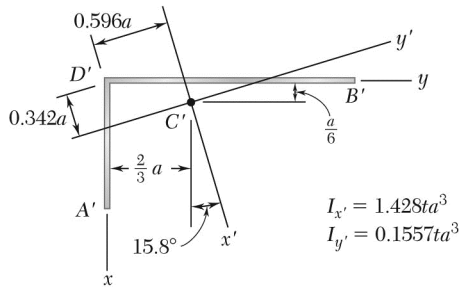
y (in.)	0	± 2	± 4	± 6
τ (ksi)	1.287	1.191	0.905	0.428





PROBLEM 6.87*

Determine the distribution of the shearing stresses along line $D'B'$ in the horizontal leg of the angle shape for the loading shown. The x' and y' axes are the principal centroidal axes of the cross section.



SOLUTION

$$\beta = 15.8^\circ \quad V'_x = P \cos \beta \quad V'_y = -P \sin \beta$$

$$A(y) = (2a - y)t \quad \bar{y} = \frac{1}{2}(2a + y), \quad \bar{x} = 0$$

Coordinate transformation.

$$y' = \left(y - \frac{2}{3}a \right) \cos \beta - \left(x - \frac{1}{6}a \right) \sin \beta$$

$$x' = \left(x - \frac{1}{6}a \right) \cos \beta + \left(y - \frac{2}{3}a \right) \sin \beta$$

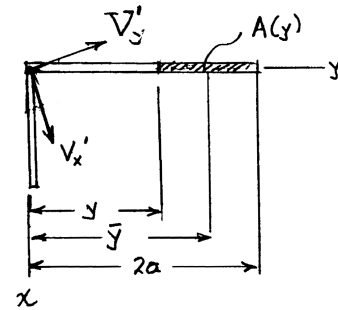
In particular,

$$\begin{aligned} \bar{y}' &= \left(\bar{y} - \frac{2}{3}a \right) \cos \beta - \left(\bar{x} - \frac{1}{6}a \right) \sin \beta \\ &= \left(\frac{1}{2}y + \frac{1}{3}a \right) \cos \beta - \left(-\frac{1}{6}a \right) \sin \beta \\ &= 0.48111y + 0.36612a \end{aligned}$$

$$\begin{aligned} \bar{x}' &= \left(\bar{x} - \frac{1}{6}a \right) \cos \beta + \left(\bar{y} - \frac{2}{3}a \right) \sin \beta \\ &= \left(-\frac{1}{6}a \right) \cos \beta + \left(\frac{1}{2}y + \frac{1}{3}a \right) \sin \beta \\ &= 0.13614y - 0.06961a \end{aligned}$$

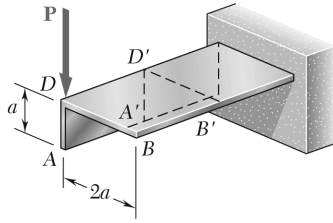
PROBLEM 6.87* (Continued)

$$\begin{aligned}
 + \rightarrow \tau &= \frac{V_{x'} A \bar{x}'}{I_{y'} t} + \frac{V_{y'} A \bar{y}'}{I_{x'} t} \\
 &= \frac{(P \cos \beta)(2a - y)(t)(0.13614y - 0.06961a)}{(0.1557 ta^3)(t)} \\
 &\quad + \frac{(-P \sin \beta)(2a - y)(0.48111y + 0.36612a)}{(1.428 a^3 t)(t)} \\
 &= \frac{P(2a - y)(0.750y - 0.500a)}{ta^3}
 \end{aligned}$$



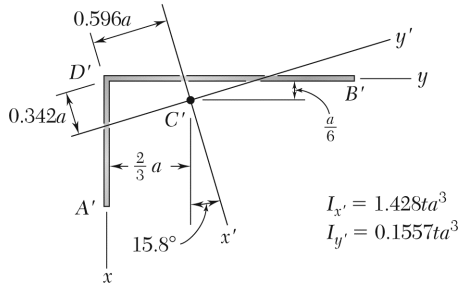
$y(a)$	0	$\frac{1}{3}$	$\frac{2}{3}$	1	$\frac{4}{3}$	$\frac{5}{3}$	2
$-\tau \left(\frac{P}{at} \right)$	-1.000	-0.417	0	0.250	0.333	0.250	0





PROBLEM 6.88*

For the angle shape and loading of Prob. 6.87, determine the distribution of the shearing stresses along line $D'A'$ in the vertical leg.



PROBLEM 6.87* Determine the distribution of the shearing stresses along line $D'B'$ in the horizontal leg of the angle shape for the loading shown. The x' and y' axes are the principal centroidal axes of the cross section.

SOLUTION

$$\begin{aligned}\beta &= 15.8^\circ & V_{x'} &= P \cos \beta \\ V_{x'} &= -P \sin \beta & A(x) &= (a-x)t \\ \bar{x} &= \frac{1}{2}(a+x), & \bar{y} &= 0\end{aligned}$$

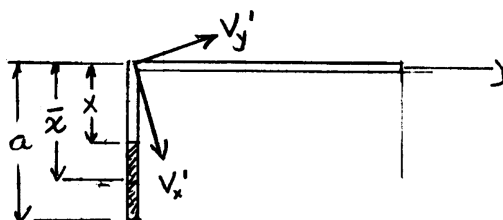
Coordinate transformation.

$$\begin{aligned}y' &= \left(y - \frac{2}{3}a\right) \cos \beta - \left(x - \frac{1}{6}a\right) \sin \beta \\ x' &= \left(x - \frac{1}{6}a\right) \cos \beta + \left(y - \frac{2}{3}a\right) \sin \beta\end{aligned}$$

In particular,

$$\begin{aligned}\bar{y}' &= \left(\bar{y} - \frac{2}{3}a\right) \cos \beta - \left(\bar{x} - \frac{1}{6}a\right) \sin \beta \\ &= \left(-\frac{2}{3}a\right) \cos \beta - \left(\frac{1}{2}x + \frac{1}{3}a\right) \sin \beta \\ &= -0.13614x - 0.73224a \\ \bar{x}' &= \left(\bar{x} - \frac{1}{6}a\right) \cos \beta + \left(\bar{y} - \frac{2}{3}a\right) \sin \beta \\ &= \left(\frac{1}{2}x + \frac{1}{3}a\right) \cos \beta + \left(-\frac{2}{3}a\right) \sin \beta \\ &= 0.48111x + 0.13922a\end{aligned}$$

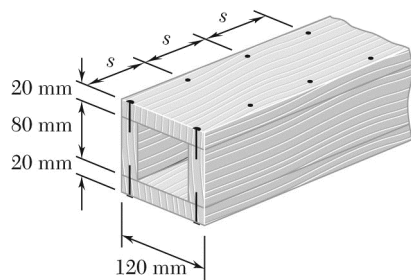
PROBLEM 6.88* (Continued)



$$\begin{aligned}
 +\downarrow \tau &= \frac{V_{x'} A(x) \bar{x}'}{I_{y'} t} - \frac{V_{y'} A(x) \bar{y}'}{I_{x'} t} \\
 &= \frac{(P \cos \beta)(a-x)(t)(0.48111x + 0.13922a)}{(0.1557 ta^3)(t)} \\
 &\quad + \frac{(-P \sin \beta)(a-x)(t)(-0.13614x - 0.73224a)}{(1.428 ta^3)(t)} \\
 &= \frac{P(a-x)(3.00x + 1.000a)}{ta^3}
 \end{aligned}$$

$x(a)$	0	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{2}$	$\frac{2}{3}$	$\frac{5}{6}$	1
$\tau \downarrow \left(\frac{P}{at} \right)$	1.000	1.250	1.333	1.250	1.000	0.583	0





PROBLEM 6.89

A square box beam is made of two 20×80-mm planks and two 20×120-mm planks nailed together as shown. Knowing that the spacing between the nails is $s = 30$ mm and that the vertical shear in the beam is $V = 1200$ N, determine (a) the shearing force in each nail, (b) the maximum shearing stress in the beam.

SOLUTION

$$\begin{aligned}
 I &= \frac{1}{12} b_2 h_2^3 - \frac{1}{12} b_1 h_1^3 \\
 &= \frac{1}{12} (120)(120)^3 - \frac{1}{12} (80)(80)^3 = 13.8667 \times 10^6 \text{ mm}^4 \\
 &= 13.8667 \times 10^{-6} \text{ m}^4
 \end{aligned}$$

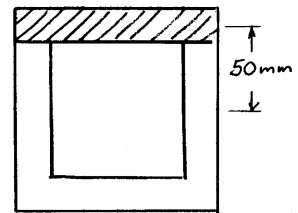
$$\begin{aligned}
 (a) \quad A_1 &= (120)(20) = 2400 \text{ mm}^2 \\
 \bar{y}_1 &= 50 \text{ mm}
 \end{aligned}$$

$$Q_1 = A_1 \bar{y}_1 = 120 \times 10^3 \text{ mm}^3 = 120 \times 10^{-6} \text{ m}^3$$

$$q = \frac{VQ}{I} = \frac{(1200)(120 \times 10^{-6})}{13.8667 \times 10^{-6}} = 10.385 \times 10^3 \text{ N/m}$$

$$qs = 2F_{\text{nail}}$$

$$F_{\text{nail}} = \frac{qs}{2} = \frac{(10.385 \times 10^3)(30 \times 10^{-3})}{2}$$

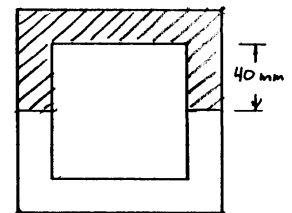


$$F_{\text{nail}} = 155.8 \text{ N} \quad \blacktriangleleft$$

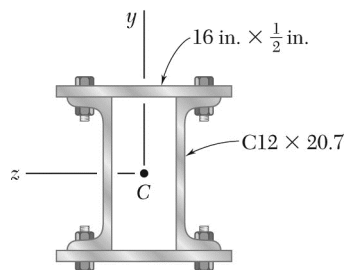
$$\begin{aligned}
 (b) \quad Q &= Q_1 + (2)(20)(40)(20) \\
 &= 120 \times 10^3 + 32 \times 10^3 = 152 \times 10^3 \text{ mm}^3 \\
 &= 152 \times 10^{-6} \text{ m}^3
 \end{aligned}$$

$$\tau_{\text{max}} = \frac{VQ}{It} = \frac{(1200)(152 \times 10^{-6})}{(13.8667 \times 10^{-6})(2 \times 20 \times 10^{-3})}$$

$$= 329 \times 10^3 \text{ Pa}$$



$$\tau_{\text{max}} = 329 \text{ kPa} \quad \blacktriangleleft$$



PROBLEM 6.90

The beam shown is fabricated by connecting two channel shapes and two plates, using bolts of $\frac{3}{4}$ -in. diameter spaced longitudinally every 7.5 in. Determine the average shearing stress in the bolts caused by a shearing force of 25 kips parallel to the y -axis.

SOLUTION

$$\text{C12} \times 20.7: \quad d = 12.00 \text{ in.}, \quad I_x = 129 \text{ in}^4$$

For top plate,

$$\bar{y} = \frac{12.00}{2} + \left(\frac{1}{2}\right)\left(\frac{1}{2}\right) = 6.25 \text{ in.}$$

$$I_t = \frac{1}{12}(16)\left(\frac{1}{2}\right)^3 + (16)\left(\frac{1}{2}\right)(6.25)^2 = 312.667 \text{ in}^4$$

For bottom plate,

$$I_b = 312.667 \text{ in}^4$$

Moment of inertia of fabricated beam:

$$\begin{aligned} I &= (2)(129) + 312.667 + 312.667 \\ &= 883.33 \text{ in}^4 \end{aligned}$$

$$Q = A_{\text{plate}} \bar{y}_{\text{plate}} = (16)\left(\frac{1}{2}\right)(6.25) = 50 \text{ in}^3$$

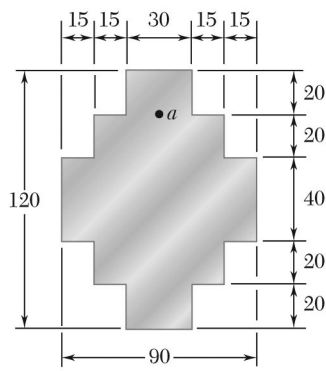
$$q = \frac{VQ}{I} = \frac{(25)(50)}{883.33} = 1.41510 \text{ kips/in}$$

$$F_{\text{bolt}} = \frac{1}{2}qs = \left(\frac{1}{2}\right)(1.41510)(7.5) = 5.3066 \text{ kips}$$

$$A_{\text{bolt}} = \frac{\pi}{4}(d_{\text{bolt}})^2 = \frac{\pi}{4}\left(\frac{3}{4}\right)^2 = 0.44179 \text{ in}^2$$

$$\tau_{\text{bolt}} = \frac{F_{\text{bolt}}}{A_{\text{bolt}}} = \frac{5.3066}{0.44179} = 12.01 \text{ ksi}$$

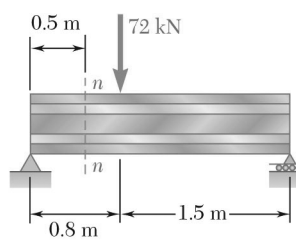
$$\tau_{\text{bolt}} = 12.01 \text{ ksi} \quad \blacktriangleleft$$



Dimensions in mm

PROBLEM 6.91

For the beam and loading shown, consider section $n-n$ and determine (a) the largest shearing stress in that section, (b) the shearing stress at point a .



SOLUTION

$$+\circlearrowleft \Sigma M_B = 0: -2.3A + (1.5)(72) = 0$$

$$A = 46.957 \text{ kN } \uparrow$$

At section $n-n$, $V = A = 46.957 \text{ kN}$

Calculate moment of inertia:

$$I = 2 \left[\frac{1}{12} (15)(40)^3 \right] + 2 \left[\frac{1}{12} (15)(80)^3 \right] + \frac{1}{12} (30)(120)^3$$

$$= 5.76 \times 10^6 \text{ mm}^4 = 5.76 \times 10^{-6} \text{ m}^4$$

At a ,

$$t_a = 30 \text{ mm} = 0.030 \text{ m}$$

$$Q_a = (30 \times 20)(50) = 30 \times 10^3 \text{ mm}^3$$

$$= 30 \times 10^{-6} \text{ m}^3$$

$$\tau_a = \frac{VQ_a}{It_a} = \frac{(46.957 \times 10^3)(30 \times 10^{-6})}{(5.76 \times 10^{-6})(0.030)}$$

$$= 8.15 \times 10^6 \text{ Pa} = 8.15 \text{ MPa}$$

At b ,

$$t_b = 60 \text{ mm} = 0.060 \text{ m}$$

$$Q_b = Q_a + (60 \times 20)(30) = 30 \times 10^3 + 36 \times 10^3 = 66 \times 10^3 \text{ mm}^3 = 66 \times 10^{-6} \text{ m}^3$$

$$\tau_b = \frac{VQ_b}{It_b} = \frac{(46.957 \times 10^3)(66 \times 10^{-6})}{(5.76 \times 10^{-6})(0.060)} = 8.97 \times 10^6 \text{ Pa} = 8.97 \text{ MPa}$$

At NA ,

$$t_{NA} = 90 \text{ mm} = 0.090 \text{ m}$$

$$Q_{NA} = Q_b + (90 \times 20)(10) = 66 \times 10^3 + 18 \times 10^3 = 84 \times 10^3 \text{ mm}^3 = 84 \times 10^{-6} \text{ m}^3$$

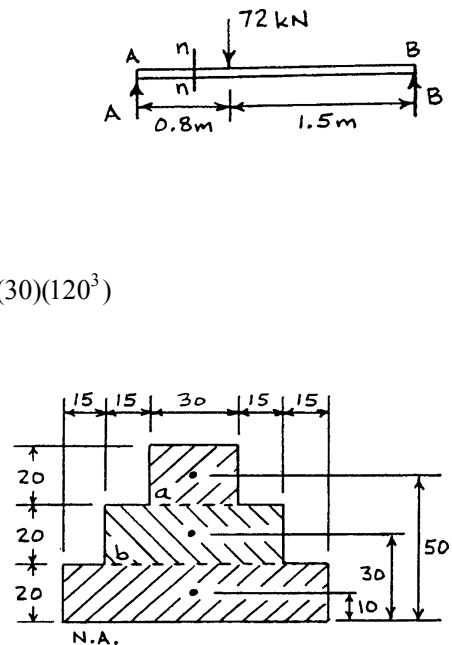
$$\tau_{NA} = \frac{VQ_{NA}}{It_{NA}} = \frac{(46.957 \times 10^3)(84 \times 10^{-6})}{(5.76 \times 10^{-6})(0.090)} = 7.61 \times 10^6 \text{ Pa} = 7.61 \text{ MPa}$$

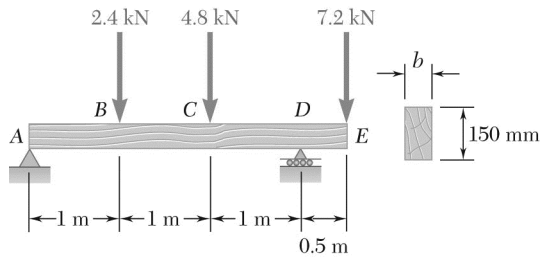
(a) $\tau_{\max}$ occurs at b .

$$\tau_{\max} = 8.97 \text{ MPa} \quad \blacktriangleleft$$

(b)

$$\tau_a = 8.15 \text{ MPa} \quad \blacktriangleleft$$





PROBLEM 6.92

For the beam and loading shown, determine the minimum required width b , knowing that for the grade of timber used, $\sigma_{\text{all}} = 12 \text{ MPa}$ and $\tau_{\text{all}} = 825 \text{ kPa}$.

SOLUTION

$$+\circlearrowleft \Sigma M_D = 0: -3A + (2)(2.4) + (1)(4.8) - (0.5)(7.2) = 0$$

$$A = 2 \text{ kN} \uparrow$$

Draw the shear and bending moment diagrams.

$$|V|_{\text{max}} = 7.2 \text{ kN} = 7.2 \times 10^3 \text{ N}$$

$$|M|_{\text{max}} = 3.6 \text{ kN} \cdot \text{m} = 3.6 \times 10^3 \text{ N} \cdot \text{m}$$

Bending:

$$\sigma = \frac{M}{S}$$

$$S_{\text{min}} = \frac{|M|_{\text{max}}}{\sigma}$$

$$= \frac{3.6 \times 10^3}{12 \times 10^6}$$

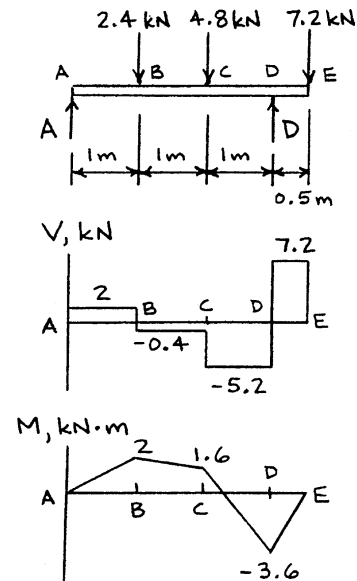
$$= 300 \times 10^{-6} \text{ m}^3$$

$$= 300 \times 10^3 \text{ mm}^3$$

For a rectangular section,

$$S = \frac{1}{6}bh^2$$

$$b = \frac{6S}{h^2} = \frac{(6)(300 \times 10^3)}{(150)^2} = 80 \text{ mm}$$



Shear: Maximum shearing stress occurs at the neutral axis of bending for a rectangular section.

$$A = \frac{1}{2}bh, \quad \bar{y} = \frac{1}{4}h, \quad Q = A\bar{y} = \frac{1}{8}bh^2$$

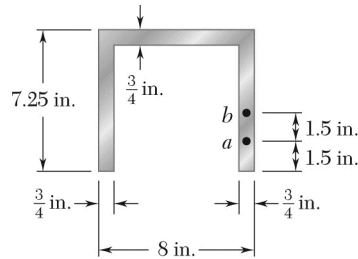
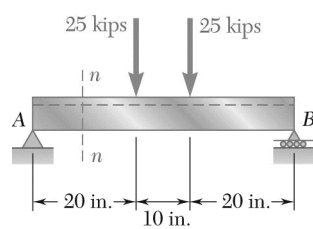
$$I = \frac{1}{12}bh^3 \quad t = b$$

$$\tau = \frac{VQ}{It} = \frac{V(\frac{1}{8}bh^2)}{(\frac{1}{12}bh^3)(b)} = \frac{3}{2} \frac{V}{bh}$$

$$b = \frac{3V}{2h\tau} = \frac{(3)(7.2 \times 10^3)}{(2)(150 \times 10^{-3})(825 \times 10^3)} = 87.3 \times 10^{-3} \text{ m}$$

The required value of b is the larger one.

$$b = 87.3 \text{ mm} \quad \blacktriangleleft$$



PROBLEM 6.93

For the beam and loading shown, consider section $n-n$ and determine the shearing stress at (a) point a , (b) point b .

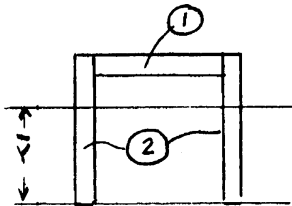
SOLUTION

$$R_A = R_B = 25 \text{ kips}$$

At section $n-n$,

$$V = 25 \text{ kips}$$

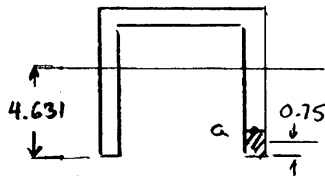
Locate centroid and calculate moment of inertia.



Part	$A(\text{in}^2)$	$\bar{y}(\text{in.})$	$A\bar{y}(\text{in}^3)$	$d(\text{in.})$	$Ad^2(\text{in}^4)$	$\bar{I}(\text{in}^4)$
①	4.875	6.875	33.52	2.244	24.55	0.23
②	10.875	3.625	39.42	1.006	11.01	47.68
Σ	15.75		72.94		35.56	47.86

$$\bar{Y} = \frac{\Sigma A\bar{y}}{\Sigma A} = \frac{72.94}{15.75} = 4.631 \text{ in.}$$

$$I = \Sigma Ad^2 + \Sigma \bar{I} = 35.56 + 47.86 = 83.42 \text{ in}^4$$



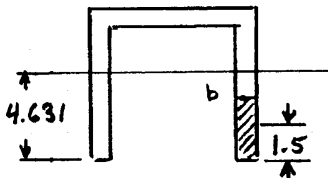
(a)

$$Q_a = A\bar{y} = \left(\frac{3}{4}\right)(1.5)(4.631 - 0.75) = 4.366 \text{ in}^3$$

$$t = \frac{3}{4} = 0.75 \text{ in.}$$

$$\tau_a = \frac{VQ}{It} = \frac{(25)(4.366)}{(83.42)(0.75)}$$

$$\tau_a = 1.745 \text{ ksi} \quad \blacktriangleleft$$



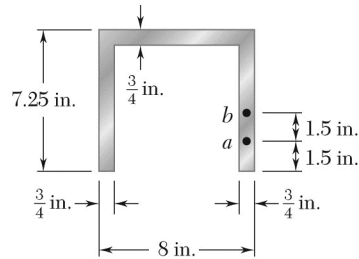
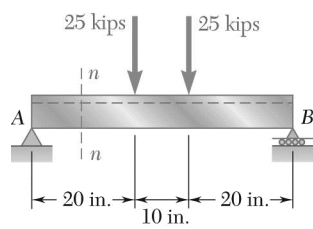
(b)

$$Q_b = A\bar{y} = \left(\frac{3}{4}\right)(3)(4.631 - 1.5) = 7.045 \text{ in}^3$$

$$t = 0.75 \text{ in.}$$

$$\tau_b = \frac{VQ}{It} = \frac{(25)(7.045)}{(83.42)(0.75)}$$

$$\tau_b = 2.82 \text{ ksi} \quad \blacktriangleleft$$



PROBLEM 6.94

For the beam and loading shown, determine the largest shearing stress in section $n-n$.

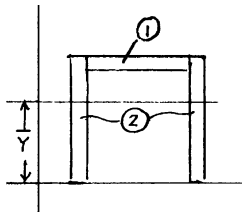
SOLUTION

$$R_A = R_B = 25 \text{ kips}$$

At section $n-n$,

$$V = 25 \text{ kips.}$$

Locate centroid and calculate moment of inertia.

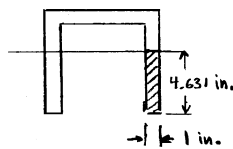


Part	$A(\text{in}^2)$	$\bar{y}(\text{in.})$	$A\bar{y}(\text{in}^3)$	$d(\text{in.})$	$Ad^2(\text{in}^4)$	$\bar{I}(\text{in}^4)$
①	4.875	6.875	33.52	2.244	24.55	0.23
②	10.875	3.625	39.42	1.006	11.01	47.68
Σ	15.75		72.94		35.56	47.86

$$\bar{Y} = \frac{\Sigma A\bar{y}}{\Sigma A} = \frac{72.94}{15.75} = 4.631 \text{ in.}$$

$$I = \Sigma Ad^2 + \Sigma \bar{I} = 35.56 + 47.86 = 83.42 \text{ in}^4$$

Largest shearing stress occurs on section through centroid of entire cross section.

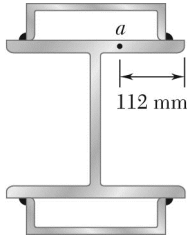


$$Q = Ay = \left(\frac{3}{4}\right)(4.631)\left(\frac{4.631}{2}\right) = 8.042 \text{ in}^3$$

$$t = \frac{3}{4} = 0.75 \text{ in.}$$

$$\tau = \frac{VQ}{It} = \frac{(25)(8.042)}{(83.42)(0.75)}$$

$$\tau_m = 3.21 \text{ ksi} \quad \blacktriangleleft$$



PROBLEM 6.95

The composite beam shown is made by welding C200×17.1 rolled-steel channels to the flanges of a W250×80 wide-flange rolled-steel shape. Knowing that the beam is subjected to a vertical shear of 200 kN, determine (a) the horizontal shearing force per meter at each weld, (b) the shearing stress at point *a* of the flange of the wide-flange shape.

SOLUTION

For W250 × 80, $d = 257 \text{ mm}$, $t_f = 15.6 \text{ mm}$, $I_x = 126 \times 10^6 \text{ mm}^4$

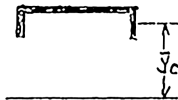
For C200 × 17.1, $A = 2170 \text{ mm}^2$, $b_f = 57.4 \text{ mm}$, $t_f = 9.91 \text{ mm}$

$$I_y = 0.545 \times 10^6 \text{ mm}^4, \quad \bar{x} = 14.5 \text{ mm}$$

For the channel in the composite beam,

$$\bar{y}_c = \frac{257}{2} + 57.4 - 14.5 = 171.4 \text{ mm}$$

For the composite beam,

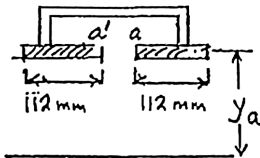


$$I = 126 \times 10^6 + 2 \left[0.545 \times 10^6 + (2170)(171.4)^2 \right] \\ = 254.59 \times 10^6 \text{ mm}^4 = 254.59 \times 10^{-6} \text{ m}^4$$

(a) For the two welds,

$$Q_w = A\bar{y}_c = (2170)(171.4) = 371.94 \times 10^3 \text{ mm}^3 = 371.94 \times 10^{-6} \text{ m}^3$$

$$q = \frac{VQ}{I} = \frac{(200 \times 10^3)(371.94 \times 10^{-6})}{254.59 \times 10^{-6}} = 292.2 \times 10^3 \text{ N/m}$$



For one weld, $\frac{q}{2} = 146.1 \times 10^3 \text{ N/m}$

Shearing force per meter of weld:

$$146.1 \text{ kN/m} \quad \blacktriangleleft$$

(b) For cuts at *a* and *a'* together,

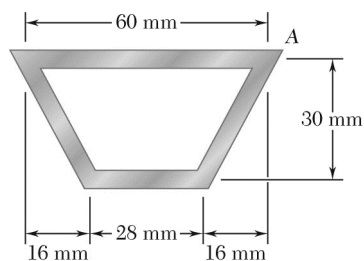
$$A_a = 2(112)(15.6) = 3494.4 \text{ mm}^2 \quad \bar{y}_a = \frac{257}{2} - \frac{15.6}{2} = 120.7 \text{ mm}$$

$$Q_a = 371.94 \times 10^3 + (3494.4)(120.7) = 793.71 \times 10^3 \text{ mm}^3 = 793.71 \times 10^{-6} \text{ m}^3$$

Since there are cuts at *a* and *a'*, $t = 2t_f = (2)(15.6) = 31.2 \text{ mm} = 0.0312 \text{ m}$.

$$\tau_a = \frac{VQ_a}{It} = \frac{(200 \times 10^3)(793.71 \times 10^{-6})}{(254.59 \times 10^6)(0.0312)} = 19.99 \times 10^6 \text{ Pa}$$

$$\tau_a = 19.99 \text{ MPa} \quad \blacktriangleleft$$



PROBLEM 6.96

An extruded beam has the cross section shown and a uniform wall thickness of 3 mm. For a vertical shear of 10 kN, determine (a) the shearing stress at point A, (b) the maximum shearing stress in the beam. Also, sketch the shear flow in the cross section.

SOLUTION

$$\tan \alpha = \frac{16}{30} \quad \alpha = 28.07^\circ$$

Side:

$$A = (3 \sec \alpha)(30) = 102 \text{ mm}^2$$

$$\bar{I} = \frac{1}{12} (3 \sec \alpha)(30)^3 = 7.6498 \times 10^3 \text{ mm}^4$$

Part	$A(\text{mm}^2)$	$\bar{y}_0(\text{mm})$	$A\bar{y}(10^3 \text{ mm}^3)$	$d(\text{mm})$	$Ad^2(10^3 \text{ mm}^4)$	$\bar{I}(10^3 \text{ mm}^4)$
Top	180	30	5.4	11.932	25.627	neglect
Side	102	15	1.53	3.077	0.966	7.6498
Side	102	15	1.53	3.077	0.966	7.6498
Bot	84	0	0	18.077	27.449	neglect
Σ	468		8.46		55.008	15.2996

$$\bar{Y}_0 = \frac{\Sigma A\bar{y}}{\Sigma A} = \frac{8.46 \times 10^3}{468} = 18.077 \text{ mm}$$

$$I = \Sigma Ad^2 + \Sigma \bar{I} = 70.31 \times 10^3 \text{ mm}^4 = 70.31 \times 10^{-9} \text{ m}^4$$

$$(a) \quad Q_A = (180)(11.932) = 2.14776 \times 10^3 \text{ mm}^3 = 2.14776 \times 10^{-6} \text{ m}^3$$

$$t = (2)(3 \times 10^{-3}) = 6 \times 10^{-3} \text{ m}$$

$$\tau_A = \frac{VQ}{It} = \frac{(10 \times 10^3)(2.14776 \times 10^{-6})}{(70.31 \times 10^{-9})(6 \times 10^{-3})} = 50.9 \times 10^6 \text{ Pa}$$

$$\tau_A = 50.9 \text{ MPa} \quad \blacktriangleleft$$

$$(b) \quad Q_m = Q_A + (2)(3 \sec \alpha)(11.932) \left(\frac{1}{2} \times 11.932 \right)$$

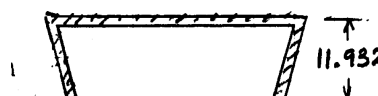
$$= 2.14776 \times 10^3 + 484.06 = 2.6318 \times 10^3 \text{ mm}^3$$

$$= 2.6318 \times 10^{-6} \text{ m}^3$$

$$t = 6 \times 10^{-3} \text{ m}$$

$$\tau_m = \frac{VQ_m}{It} = \frac{(10 \times 10^3)(2.6318 \times 10^{-6})}{(70.31 \times 10^{-9})(6 \times 10^{-3})} = 62.4 \times 10^6 \text{ Pa}$$

$$\tau_m = 62.4 \text{ MPa} \quad \blacktriangleleft$$

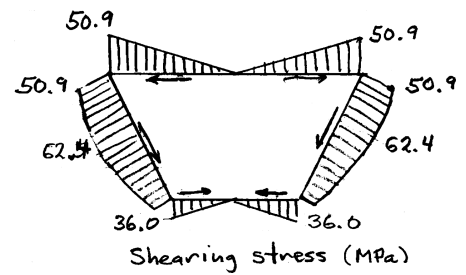


PROBLEM 6.96 (Continued)

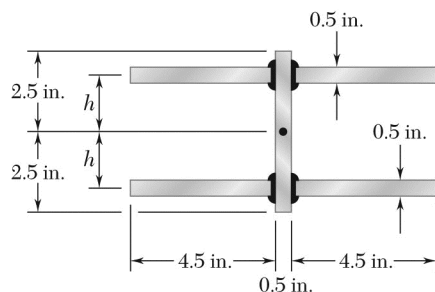
$$Q_B = (28)(3)(18.077) = 1.51847 \times 10^3 \text{ mm}^3$$

$$\tau_B = \frac{Q_B}{Q_A} \tau_A = \frac{1.51847 \times 10^3}{2.14776 \times 10^3} (50.9)$$

$$= 36.0 \text{ MPa}$$



Multiply shearing stresses by $t(3 \text{ mm} = 0.003 \text{ m})$ to get shear flow.



PROBLEM 6.97

The design of a beam requires welding four horizontal plates to a vertical 0.5×5 -in. plate as shown. For a vertical shear V , determine the dimension h for which the shear flow through the welded surface is maximum.

SOLUTION

Horizontal plate:

$$I_h = \frac{1}{12}(4.5)(0.5)^3 + (4.5)(0.5)h^2$$

$$= 0.046875 + 2.25h^2$$

Vertical plate:

$$I_v = \frac{1}{12}(0.5)(5)^3 = 5.2083 \text{ in}^4$$

Whole section:

$$I = 4I_h + I_v = 9h^2 + 5.39583 \text{ in}^4$$

For one horizontal plate,

$$Q = (4.5)(0.5)h = 2.25h \text{ in}^3$$

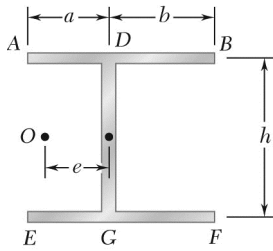
$$q = \frac{VQ}{I} = \frac{2.25Vh}{9h^2 + 5.39583}$$

To maximize q , set

$$\frac{dq}{dh} = 0.$$

$$2.25V \frac{(9h^2 + 5.39583) - 18h^2}{(9h^2 + 5.39583)^2} = 0$$

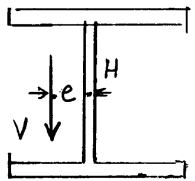
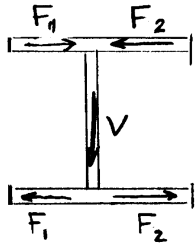
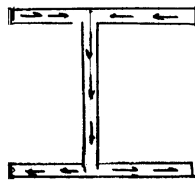
$$h = 0.774 \text{ in.} \quad \blacktriangleleft$$



PROBLEM 6.98

Determine the location of the shear center O of a thin-walled beam of uniform thickness having the cross section shown.

SOLUTION



Part AD:

$$I_{AB} = I_{EF} = (a+b)t \left(\frac{h}{2} \right)^2 + \frac{1}{12} (a+b)t^3 \approx \frac{1}{4} t(a+b)h^2$$

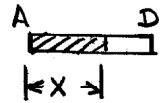
$$I_{DG} = \frac{1}{12} th^3 \quad I = \Sigma I = \frac{1}{12} t(6a+6b+h)h^2$$

$$Q = tx \frac{h}{2} = \frac{1}{2} thx$$

$$\tau = \frac{VQ}{It} = \frac{Vhx}{2I}$$

$$F_1 = \int \tau dA = \int_0^a \frac{Vhx}{2I} t dx = \frac{Vht}{2I} \int_0^a x dx$$

$$= \frac{Vht}{2I} \frac{x^2}{2} \bigg|_0^a = \frac{Vhta^2}{4I}$$



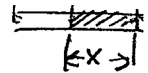
Part BD:

$$Q = tx \frac{h}{2} = \frac{1}{2} thx$$

$$\tau = \frac{VQ}{It} = \frac{Vhx}{2I}$$

$$F_2 = \int \tau dA = \int_0^b \frac{Vhx}{2I} t dx = \frac{Vht}{2I} \int_0^b x dx$$

$$= \frac{Vht}{2I} \frac{x^2}{2} \bigg|_0^b = \frac{Vhtb^2}{4I}$$

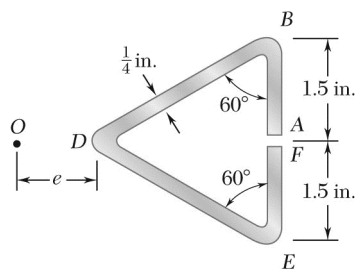


$$+\circlearrowleft \Sigma M_H = +\circlearrowleft \Sigma M_H:$$

$$Ve = F_2 h - F_1 h = \frac{Vh^2 t(b^2 - a^2)}{4I}$$

$$= \frac{Vh^2 t(b^2 - a^2)}{4 \cdot \frac{1}{12} t(6a+6b+h)h^2} = \frac{3V(b^2 - a^2)}{6a+6b+h}$$

$$e = \frac{3(b^2 - a^2)}{6(a+b)+h} \quad \blacktriangleleft$$



PROBLEM 6.99

Determine the location of the shear center O of a thin-walled beam of uniform thickness having the cross section shown.

SOLUTION

$$I_{AB} = \frac{1}{3} \left(\frac{1}{4} \right) (1.5)^3 = 0.28125 \text{ in}^4$$

$$L_{BD} = 3 \text{ in.} \quad A_{BD} = (3) \left(\frac{1}{4} \right) = 0.75 \text{ in}^2$$

$$I_{BD} = \frac{1}{3} A_{BD} h^2 = \frac{1}{3} (0.75) (1.5)^2 = 0.5625 \text{ in}^4$$

$$I = (2)(0.28125) + (2)(0.5625) = 1.6875 \text{ in}^4$$

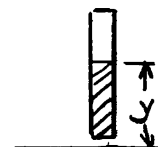
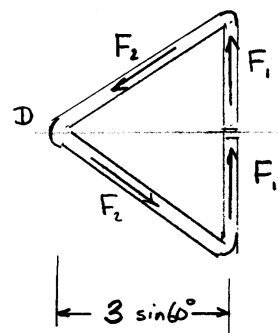
Part AB:

$$A = \frac{1}{4} y \quad \bar{y} = \frac{1}{2} y \quad Q = A\bar{y} = \frac{1}{8} y^2$$

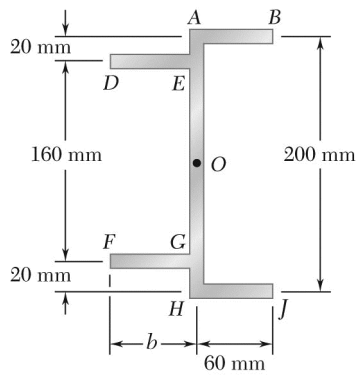
$$\tau = \frac{VQ}{It} = \frac{Vy^2}{(8)(1.6875)(0.25)} = \frac{Vy^2}{3.375}$$

$$\begin{aligned} F_1 &= \int \tau dA = \int_0^{1.5} \frac{Vy^2}{3.375} (0.25 dy) \\ &= \frac{(0.25)V}{3.375} \frac{y^3}{3} \bigg|_0^{1.5} = \frac{(0.25)(1.5)^3}{(3.375)(3)} \\ &= 0.08333V \end{aligned}$$

$$\begin{aligned} +\curvearrowright M_D = +\curvearrowright M_D: \quad Ve &= 2F_1(3 \sin 60^\circ) \\ Ve &= (2)(0.08333)V(3 \sin 60^\circ) \\ e &= (2)(0.08333)(3 \sin 60^\circ) \end{aligned}$$



$$e = 0.433 \text{ in.} \quad \blacktriangleleft$$



PROBLEM 6.100

A thin-walled beam of uniform thickness has the cross section shown. Determine the dimension b for which the shear center O of the cross section is located at the point indicated.

SOLUTION

Part AB :

$$A = tx, \quad \bar{y} = 100 \text{ mm}, \quad Q = 100 tx$$

$$q = \frac{VQ}{I} = \frac{100Vtx}{I}$$

$$\begin{aligned} F_1 &= \int_{x_A}^{x_B} q dx = 100 \frac{Vt}{I} \int_0^{60} x dx \\ &= \frac{(100)(60)^2}{2} \frac{Vt}{I} = 180 \times 10^3 \frac{Vt}{I} \end{aligned}$$

Part DE :

$$A = tx, \quad \bar{y} = 80 \text{ mm}, \quad Q = 80 tx$$

$$q = \frac{VQ}{I} = \frac{80 Vtx}{I}$$

$$\begin{aligned} F_2 &= \int_{x_D}^x q dx = 80 \frac{Vt}{I} \int_0^b x dx \\ &= (40b^2) \frac{Vt}{I} \end{aligned}$$

$$+\circlearrowleft \Sigma M_O = 0: \quad (200)(180 \times 10^3) \frac{Vt}{I} - (160)(40b^2) \frac{Vt}{I} = 0$$

$$b^2 = \frac{(200)(180 \times 10^3)}{(160)(40)} = 5625 \text{ mm}^2$$

$$b = 75.0 \text{ mm} \quad \blacktriangleleft$$

